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National Schools of Excellence
Chemistry Teacher's' Resource Book
Grade 12
Term 1

DOMAIN **Phylum**

SCIENCE **Chemistry**

Technology

Engineering

Mathematics

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THEMA: BIOTECHNOLOGY

Introduction

Biotechnology is a branch of biology and chemistry that employs mathematics and physics to develop tools and engineering insights for modern biology and biomedical research. It entails the study and application of living cells and cellular materials in the development of pharmacological, diagnostic, agricultural, environmental, and other products that promote human welfare and society. It's also utilized to examine and manipulate genetic information in animals in order to imitate and study human disease. Biotechnology is a science that is both fundamental and applied. Agriculture, health care and medical sciences, biomedical engineering, cell biology, immunology, seed technology, virology, plant physiology, forensics, industrial processing, and environmental management all benefit from the technologies it develops.

There are a number of career paths linked to the field of Biotechnology, such as: working in research labs, graduate training in biology and chemistry, and professional careers in the health sciences. If you are someone who enjoys STEM (particularly biology and chemistry), below are some best biotechnology careers that may be of interest to you (research on them for more information)

- Biomedical Engineer
- Biochemistry
- Medical Scientist
- Clinical Technician
- Microbiologist
- Process Development Scientist
- Biomanufacturing Specialist
- Business Development Manager
- Product Strategist
- Biopharma Sales Representative
- Medical Scientist
- Biotechnological Technician
- Epidemiologist
- Microbiologist
- Medical and Clinical Lab Technologist
- Biomanufacturing Specialist
- Bioproduction Specialist

MODULE 12.1: PROTEIN SEQUENCING

4-5 Weeks

Protein sequencing is the technique of determining the amino acid sequence of a protein or peptide in its entirety or in part.¹ This module is designed to introduced students to the process or steps involved in Protein sequencing. The process involves:

1. Separating proteins into subunits
2. Producing Smaller Peptide Fragments
3. Sequencing the Protein Fragments
4. Repeating steps 2 and 3 using a different cleavage/hydrolysis method
5. Assembling the overlapping fragments for the full protein sequence

The entire practical learning experience will be covered within the following topics:

Topic 1: Amino Acids

Topic 2: Protein – Structure and Function

Topic 3: Protein Purification

Topic 4: Protein Sequencing

¹ (Protein Sequencing, 2021)

Topic 1: Amino Acids and Proteins

1.1 Amino Acid - Chemical Structure

An Amino Acid is an organic compound that contains:

- A carbon atom - typically known as alpha (α) carbon atom
- An Amino functional group ($-\text{NH}_3^+$)
- A Carboxylate functional group ($-\text{COO}^-$)
- A Hydrogen atom (H)
- R group - side chain alkyl group that is specific to each amino acid.

(See Figure SC-BT-TRB-1)

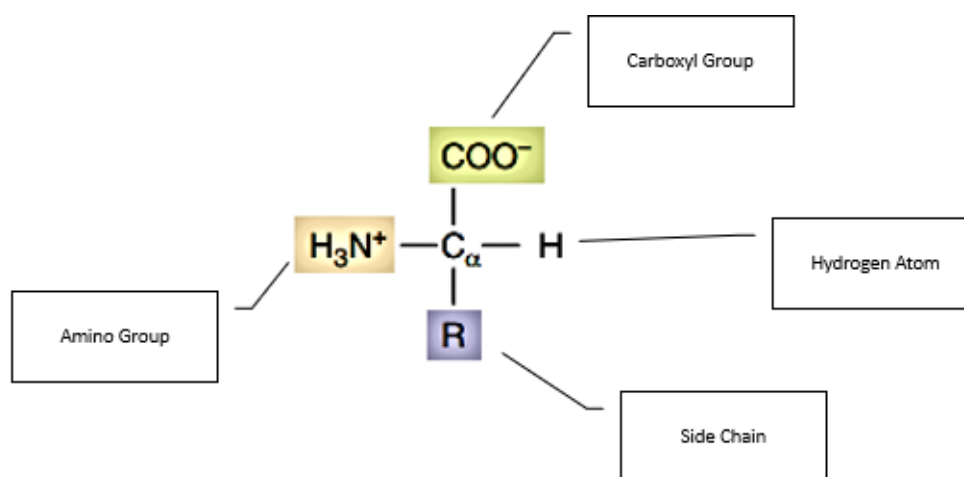


Figure SC-BT-TRB- 1: An alpha amino acid's structure. The amine group is protonated (i.e., the addition of a proton) to form ($-\text{NH}_3^+$) and the carboxyl group is deprotonated to form ($-\text{COO}^-$). (Source: (htt

The atoms in a hydrocarbon chain connected to a carboxyl ($-\text{COOH}$) group are denoted by Greek letters in the pre-IUPAC nomenclature system, hence these amino acids are called α -amino acids. The carbon atom closest to the carboxyl group is designated α , the next β , and so forth.

1.2 20 Common Amino Acids

There are about 20 common Amino Acids. It's inconvenient to write the complete names of the amino acids, especially for polypeptide chains containing numerous amino acids. *Table 1* shows their Three-Letter Code and one Letter Codes.

Table 1: 20 Common Amino Acids with their Three-Letter and One-Letter Code

#	Amino Acid	Three-Letter Code	One-Letter Code
1	Arginine	Arg	R
2	Lysine	Lys	K
3	Aspartic Acid	Asp	D
4	Glutamic Acid	Glu	E
5	Glutamine	Gln	Q
6	Asparagine	Asn	N
7	Histidine	His	H
8	Serine	Ser	S
9	Threonine	Thr	T

10	Tyrosine	Tyr	Y
11	Cysteine	Cys	C
12	Tryptophan	Trp	W
13	Methionine	Met	M
14	Alanine	Ala	A
15	Isoleucine	Ile	I
16	Leucine	Leu	L
17	Phenylalanine	Phe	F
18	Valine	Val	V
19	Proline	Pro	P
20	Glycine	Gly	G

The physical and chemical properties of each amino acid are determined by its unique side chain. The properties of amino acids with same side chains are frequently identical. When comparing amino acid sequences of two different polypeptides or determining the influence of an amino acid substitution on protein function, this relationship is critical to consider.

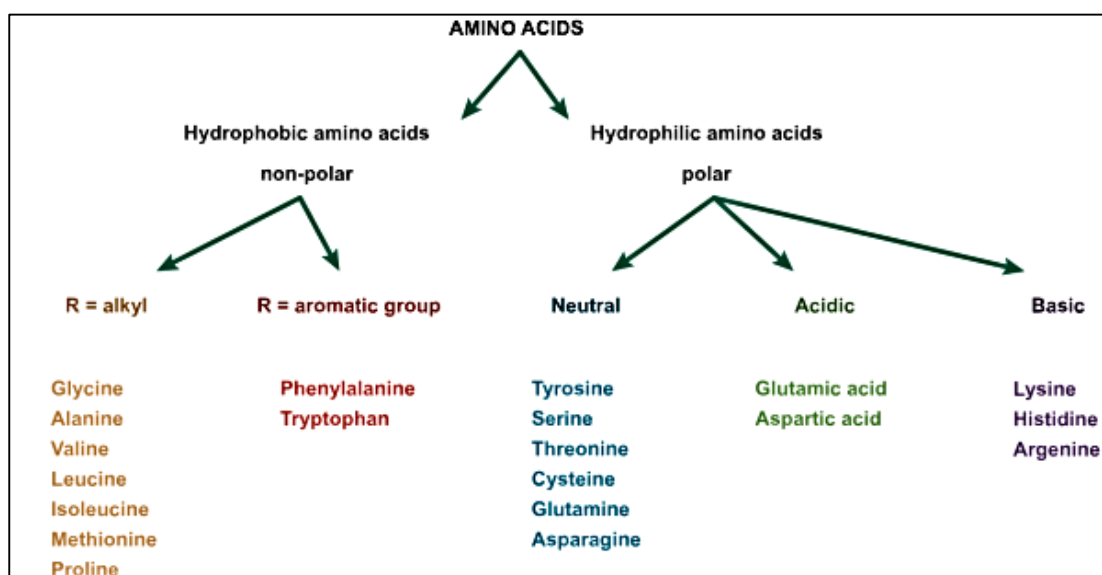


Figure SC-BT-TRB- 2:: Classification of Amino Acids based on Hydrophobic (water fearing) and Hydrophilic (water-loving) property

- **Alkyl group** – a functional group that contains only carbon and hydrogen atoms, which are arranged in a chain.
- **Aromatic Group** – chemical compounds that contain one or more rings with pi electrons delocalized all the way around them.
- **Neutral group** – includes functional groups such as the hydrocarbons (alkanes, alkenes, alkynes, aromatics) alcohols, aldehydes, ketones, esters, amides, nitro compounds.
- **Acidic group** – includes functional groups that can donate (or lose) a proton (H^+)
- **Basic group** – includes functional groups that can accept a proton (H^+). Carboxyl, Hydroxyl, Carbonyl, phosphate, sulfhydryl, amino, etc.

1.3 20 Amino Acids into Four Groups

Based on side chain structure, the 20 amino acids can be divided into four groups.

1. Side Chains with Basic Groups
2. Side Chains with Acidic Groups
3. Side Chains with Polar but Uncharged Groups
4. Side Chains with Nonpolar Groups

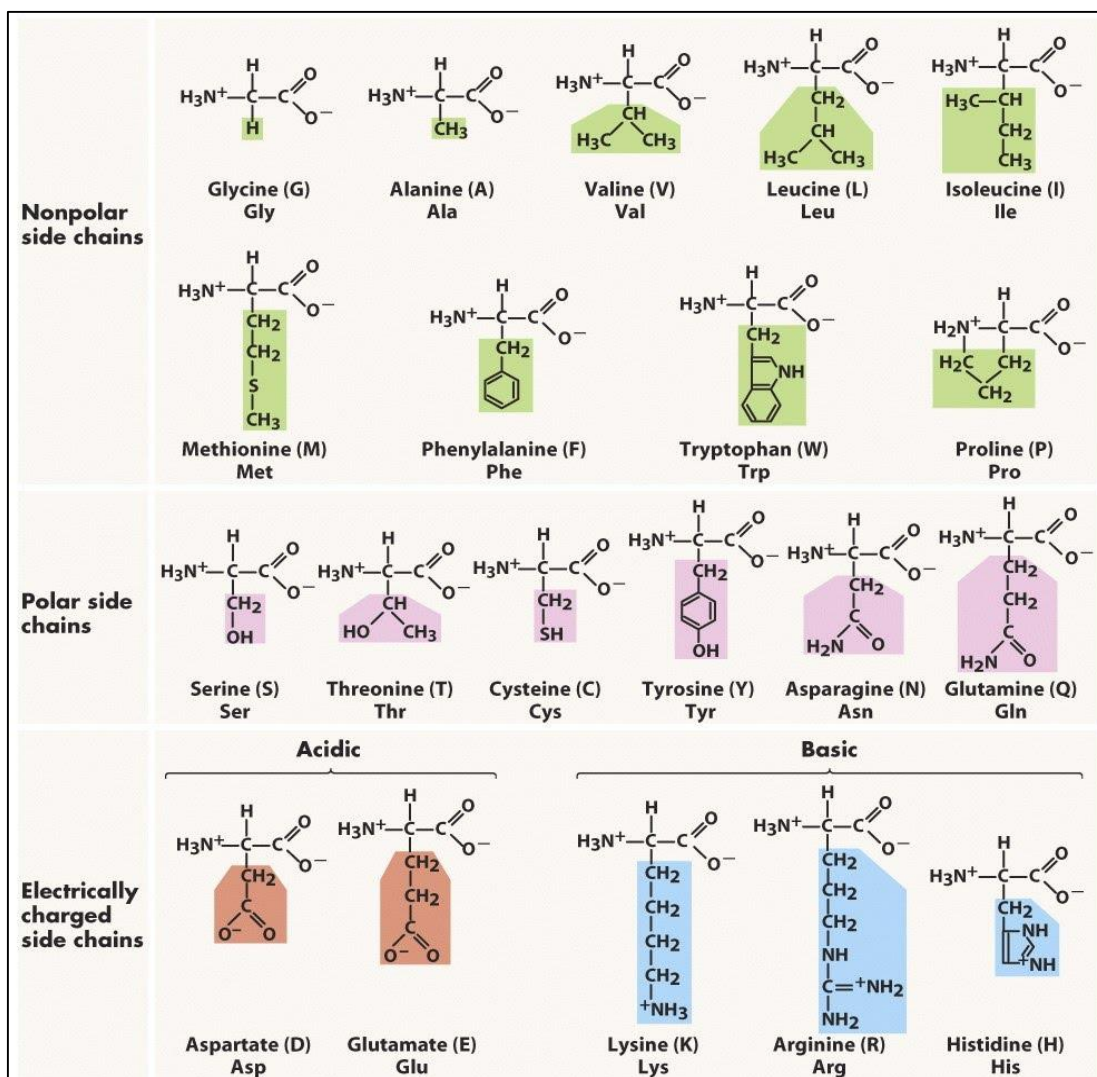


Figure SC-BT-TRB- 3: The chemical structures of the 20 amino acids classified accordingly (Nonpolar, Polar, Acidic (Negatively Charged) and Basic (Positively Charged) to their R side chains.

(Source: <https://sites.google.com/a/providenceday.org/apbiology/> /rsrc/1360033313223/codon-charts-periodic-table/aminoacids01.jpg)

1.3.1 Side Chains with Basic Groups

Three out of the 20 amino acids are basic amino acids:

1. Arginine
2. Lysine
3. Histidine

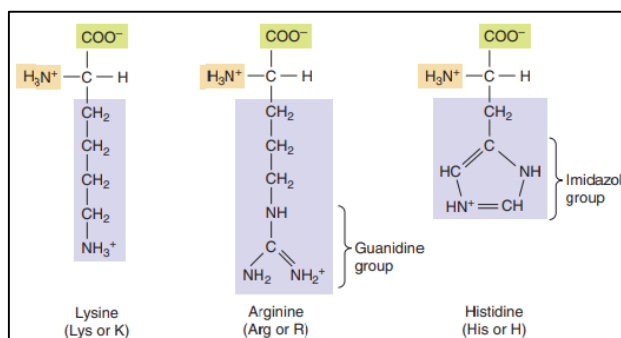


Figure SC-BT-TRB- 4: Amino Acids with Basic Side Chains (Source: (htt

They classified as basic amino acids because they contain side chains that are **Proton Acceptors**.

- The guanidino group in the side chain of arginine is a fairly strong base.
- The amine group in the side chain of lysine is a slightly weaker base, while
- the imidazole group in the side chain of histidine is the weakest of the three bases.

As a result, at pH7, arginine and lysine side chains are very likely to contain positive charges, but histidine side chains have a 10% chance of being positive.

1.3.2 Side Chains with Acidic Groups

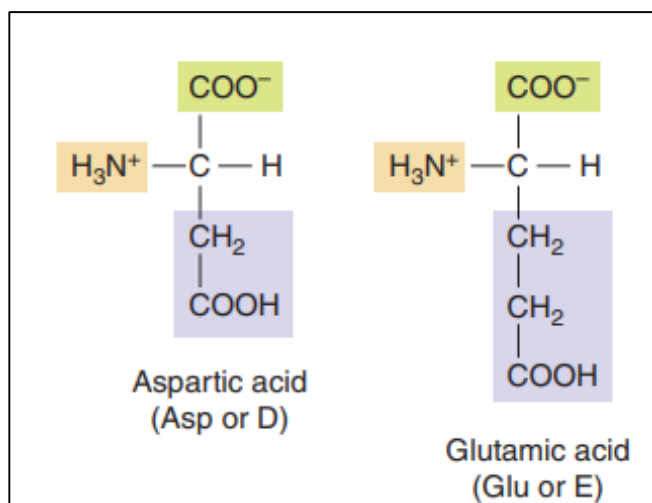


Figure SC-BT-TRB- 5: Amino Acids with Acidic Side Chains (Source: (htt

Two out of 20 amino acids contain acidic side chains:

1. Aspartic Acid
2. Glutamic Acid

These is because each of their side chains are carboxyl groups (See *Figure SC- BT-TRB-5*).

At pH7, both the α -carboxyl and side chain carboxyl groups are deprotonated, resulting in negative charges. Because the α -carbon is additionally connected to a positively charged amino group, the α -carboxyl group is a significantly stronger acid. Aspartic and glutamic acids are more accurately called aspartates and glutamate, respectively, when the side chain is deprotonated. The terms aspartic acid and aspartate are interchangeable since they refer to the same amino acid at various pH levels. Glutamic acid and glutamate are in the same boat.

1.3.3 Side Chains with Polar but Uncharged Groups

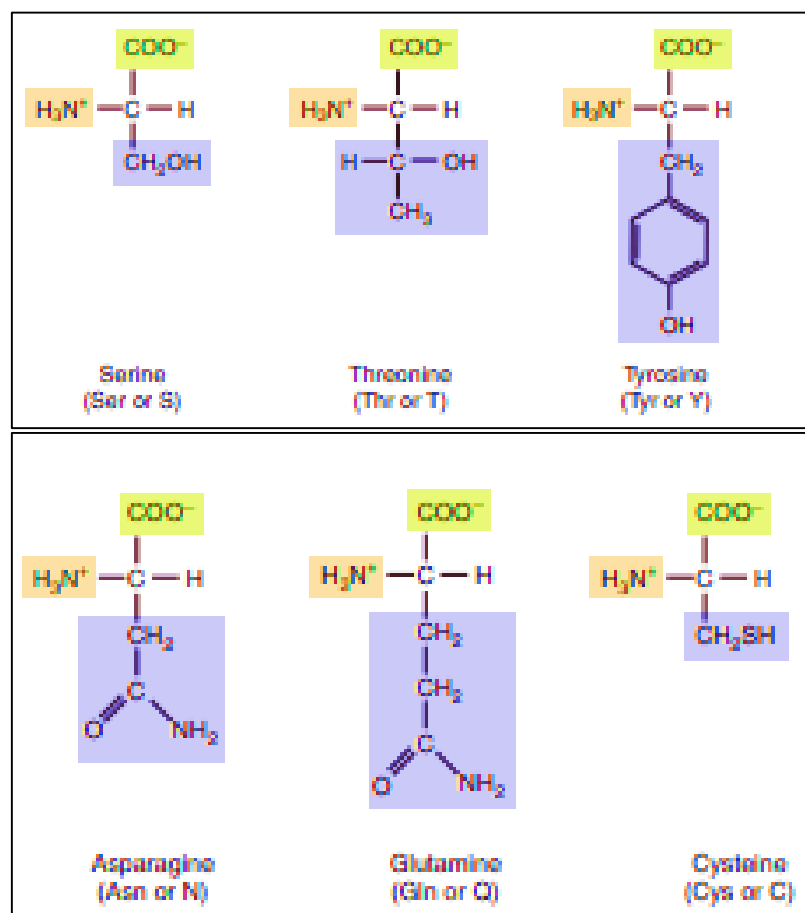


Figure SC-BT-TRB- 6:: Amino acids with polar but uncharged side chains at pH 7.0 (Source: (htt

Six amino acids have side chains with polar groups: Serine, Threonine, Tyrosine, Asparagine, Glutamine, and Cysteine.

Below are some characteristics of these amino acids.

- Asparagine is an amide derivative of Aspartate
- Glutamine is an amide derivative of Glutamate
- Serine, Threonine, and Tyrosine have side chains that contains hydroxyl (-OH) groups.
- Tyrosine's side chains also contain aromatic properties.
- Serine and Cysteine are similar but differ in that sulfhydryl (-SH) group is the place of the hydroxyl group.

When two cysteine molecules are exposed to oxygen or other oxidizing agents, the sulfhydryl groups react to create a disulfide (-S-S-) bond, resulting in the formation of Cystine.

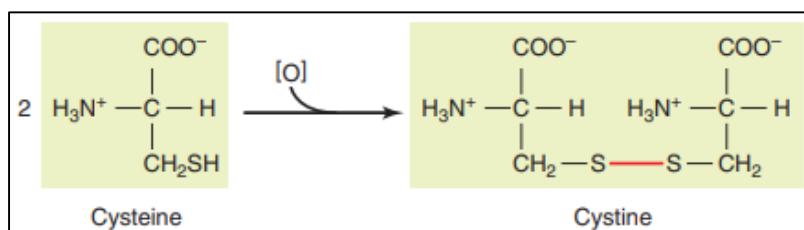


Figure SC-BT-TRB- 7: Oxidation of Cysteine to form Cystine. (Source: (htt

1.3.4 Side Chains with Nonpolar Groups

Nine amino acids have side chains with nonpolar groups:

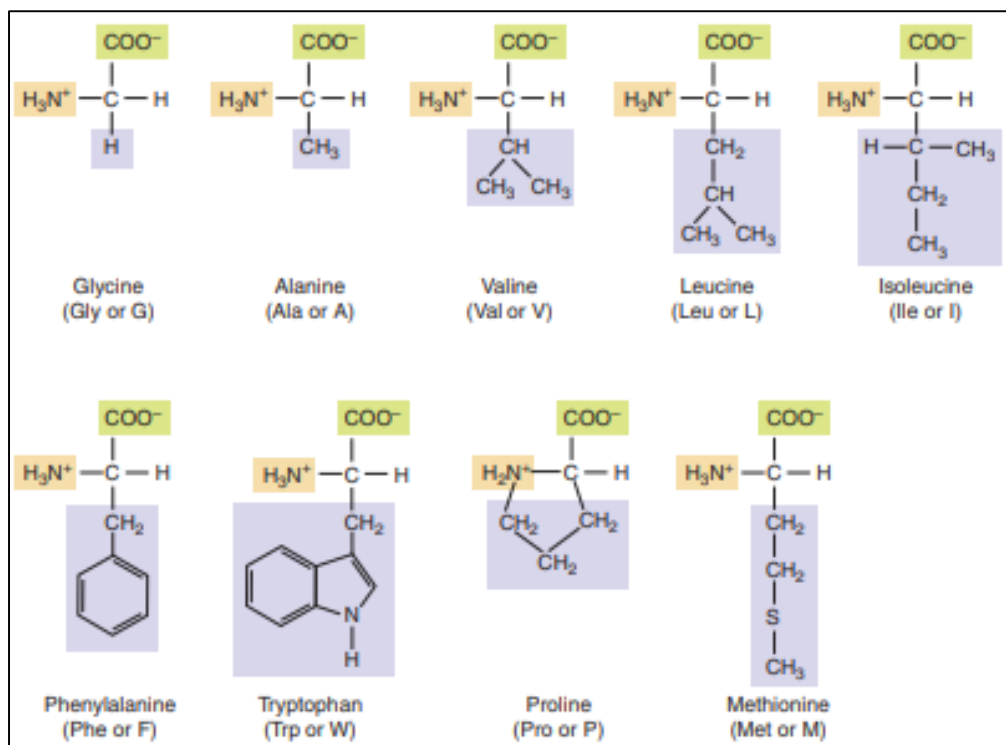


Figure SC-BT-TRB- 8: Amino acids with nonpolar side chains (Source: [htt](#))

Nonpolar amino acids are split into two groups:

- aliphatic amino acids
- aromatic amino acids.

Glycine, Alanine, Valine, Leucine, Isoleucine, and Proline are aliphatic amino acids that have branched hydrocarbon chains, ranging from the simplest glycine to the more complex structures of leucine and valine. Proline is an aliphatic amino acid having distinctive features due to the cyclization of the hydrocarbon chain with the terminal amine, resulting in a unique 5-membered ring structure. Proline can significantly alter the 3-dimensional structure of the due to the structural rigidity of the ring structure when it is incorporated into the polypeptide chain, as we will see in the next section on primary structure, and is commonly found in regions of the protein where folds or turns occur.

Because of their high carbon/hydrogen content, aromatic amino acids (phenylalanine, tyrosine, and tryptophan) include aromatic functional groups within their structure, making them predominantly nonpolar and hydrophobic. It should be remembered, however, that hydrophobicity and hydrophilicity are on a sliding scale, and each amino acid might have distinct physical and chemical properties depending on its structure. When compared to phenylalanine, the hydroxyl group present in tyrosine increases its reactivity and solubility. *Because the terminal methyl group generates a thioether functional group, which cannot form a stable dipole inside the molecule and retains poor solubility, methionine, one of the sulfur-containing amino acids, is commonly categorized as a nonpolar, hydrophobic amino acid.*

1.4 Proteins: What Are They?

Proteins are molecules that are called Polymers. A polymer is any molecule that is made up of individual building blocks that are linked together in repeated units. Those individual building blocks are called “Monomers” (See *Figure SC-TRB-9*).

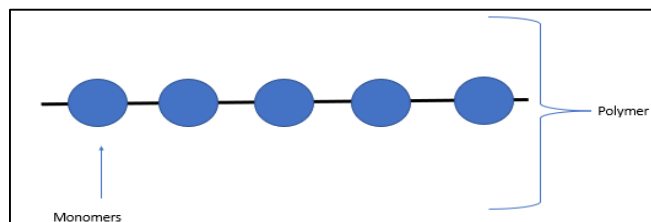


Figure SC-BT-TRB- 9: Simple illustration of a polymer, which contains repeated chemical units called Monomers

Monomers that make up proteins are the Amino Acids. Each of the individual circles in *Figure SC- BT-TRB-9* represents Amino Acids. When Multiple amino acids are linked or bonded together, it creates a “Polypeptide” or “Protein” They can be linked together through a reaction known as **Condensation Reaction**. Condensation reaction involves the removal of a water molecule (H_2O) in order to form a bond between the amino acids.

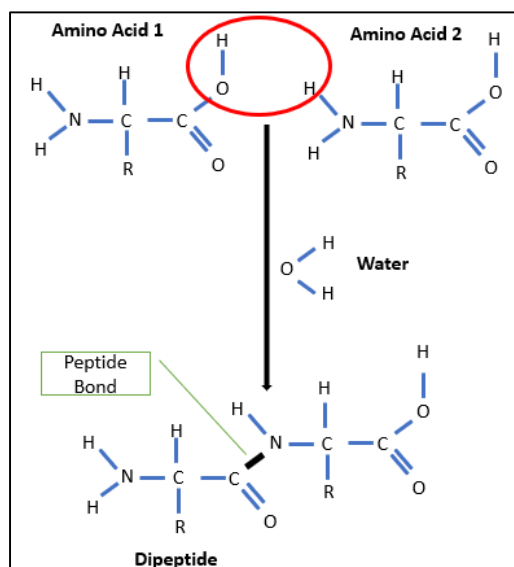


Figure SC-BT-TRB- 10: Condensation reaction of two amino acids (1 and 2) to form a dipeptide

As you can see in *Figure SC- BT-TRB-10*, the condensation reaction to forming polymers of amino acids is possible due to the α -carboxyl end of Amino acid 1 and the α -amino group end of Amino acid 2. The hydroxyl group of the carboxyl group in Amino acid 1 reacts with the Hydrogen atom of the Amino acid 2, create water which then detaches from both the Amino acids 1 and 2. When Amino Acid 1 and Amino Acid 2 are linked together due to the creation of a **peptide bond**.

Peptide bonds, which connect amino acids, are known as peptide bonds, and the compounds that result are known as peptides. Dipeptides are peptides containing two amino acids, tripeptides are peptides with three amino acids, and so on.

As a result, a polypeptide is made up of three or more amino acids joined together by a peptide bond formed by the carboxyl group of the amino acids and the amino group of the amino acids. Only 20 amino acids can be used to make any protein found in the human body.

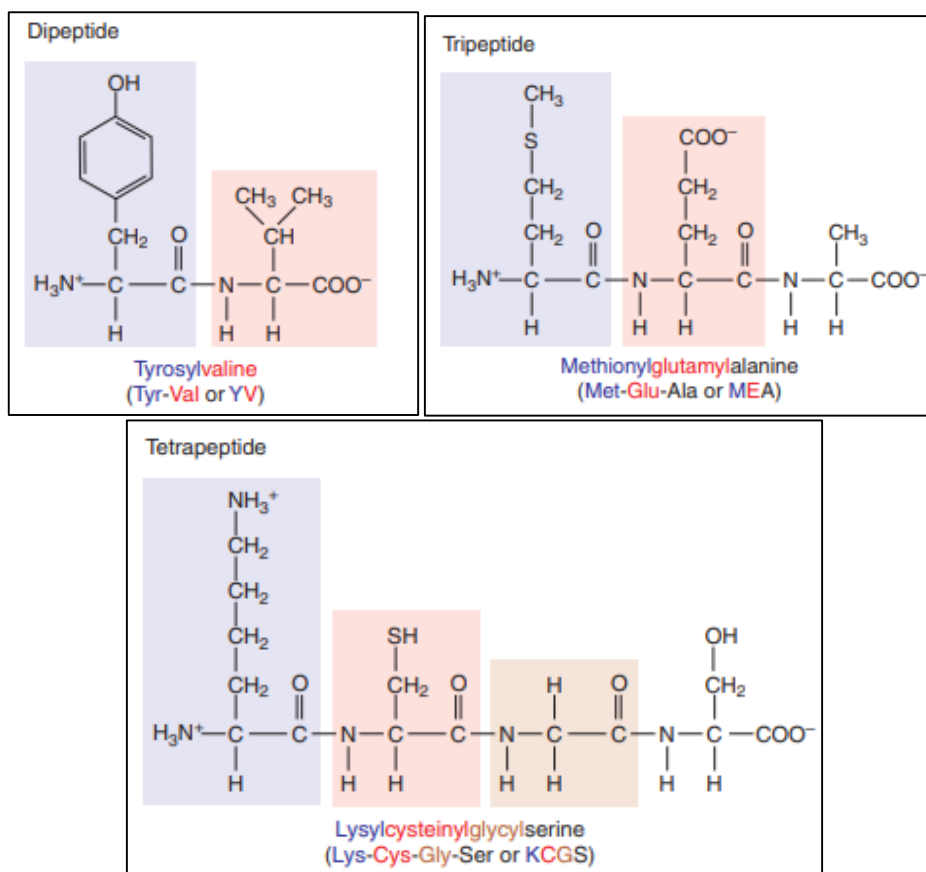


Figure SC-BT-TRB- 11: Conventions for drawing peptides. By convention, the amino acid terminus (N-terminus) is on the left and the carboxyl terminus (C-terminus) is on the right. peptides are named as derivatives of the carboxyl terminal amino acid.

Peptide chains, like polynucleotide chains, exhibit directionality. The peptide chain has a free amino group on one end and a free carboxyl group on the other. The free amino group is drawn on the left by convention. Amino acids are referred to be **amino acid residues** after they have been bonded together in a peptide chain (or **residues** for short). Figure SC- BT-TRB-11 summarizes some significant peptide nomenclature conventions.

1.5 N-Terminus and C-Terminus

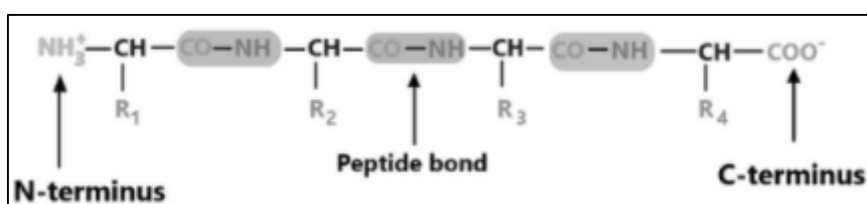


Figure SC-BT-TRB- 12: N-terminus and C-terminus of polypeptide chain (source: (ashajyoti, 2021))

A Peptide bond connects the carboxyl group (-COOH) of one amino acid to the amino group of another amino acid. A water molecule is lost when a peptide bond is formed. The amino residue is the name given to each amino acid unit in a peptide.

Peptides contain two ends:

- an amino-terminal or N terminus
- carboxyl-terminal or C terminus.

Protein residues at both the N and C termini play a crucial function in the peptide chain. One group is basic, while the other is acidic.

1.5.1 N-Terminus

In peptide bonding, there are two residues. One of these is the N terminus. The N-terminus of a peptide is the residue that has an amino group (-NH₂).

The amino group of one amino acid is connected to the carboxyl group (-COOH) of another amino acid in a peptide bond. The Carboxyl group (-COOH) is only linked to the terminal amino group of a polypeptide. The N-terminal amino group is the last surviving amino group.

1.5.1.1 Features

The N-Terminus contains the following features:

1. It's at the end of a polypeptide or protein, for starters.
2. The free alpha-amino groups are labeled at the N-terminus.
3. When drawing a peptide structure, the N-terminus should always be on the left side.
4. It indicates the free amino group of amino acids and describes the commencement of a polypeptide or protein.
5. The color of protein is blue.
6. During protein synthesis, it is the major component of a protein that emerges from a ribosome.
7. There is a free amino group at the N-terminus.
8. The N-terminus may contain a signal peptide that delivers protein to various cellular compartments (2). (3).

1.5.1.2 Analyzing N-Terminus

There are three methods that are generally used for sequencing peptides from the N-terminus.

- Sanger method or DNP method.
- Dansyl method.
- The Edman degradation

NOTE: We will discuss some of these methods in detail in Topic 3 of this Module.

1.5.1.3 Structure of N-Terminus

A peptide is a polymer of amino acids linked by amide bonds between the amino groups of one amino acid and the carboxyl groups of the neighboring amino acid (-COOH). They are primarily made up of short amino acid chains. Each amino acid unit of a peptide that is formed by the peptide bond is referred to as a residue.

The Carboxyl group (-COOH) of one amino acid is linked to the amino group of another amino acid. A dehydration reaction results in the formation of this connection. The end amino group of a protein is then exclusively connected to the end carboxyl group of the protein (-COOH).

1.5.1.4 Function of N-Terminus

- It is the protein's initial component. As a result, it plays a crucial role in protein synthesis.
- The N-terminus guides protein distribution to the correct cell organelles.
- The signal peptide sequence is frequently present.
- The protein's terminus is a significant determinant.
- Many of the structural features of a cell and attached cells are supplied by the amino-terminal or N-terminus.

1.5.2 C-Terminus

The C-terminus is one of the two terminal groups of a peptide chain. C-terminus refers to the end of a polypeptide having a free Carboxyl group (-COOH). Carboxyl-terminal, c-terminal, or carboxy-terminus are all terms for this terminal group.

The peptide bond is produced when one amino acid's -amino group and another amino acid's -Carboxyl group (-COOH) come together. The C-terminus or carboxyl-terminal residues have the residual Carboxyl group (-COOH).

1.5.2.1 Features

- The N-terminus and C-terminus are the same for all proteins.
- When drawing a peptide structure, the C-terminus is on the right side of the peptide chain, according to tradition.
- Its teeth are formed by the carboxyl-terminal or c-terminal residue.
- The color of protein is red.
- It has a Carboxyl group that is free (-COOH).
- C-terminals refer to the polypeptide chain's terminus and the presence of a free carboxyl group (-COOH).
- It's a positively charged residue, after all.
- The C-terminus is linked to a variety of functions, including protein degradation, translation termination efficiency, and so on. (2) and (5)

1.5.2.2 Analysis of C-Terminus

There are two ways for determining the C-terminus.

- Method based on enzymes
- Methods involving chemicals
- LiBH₄ treatment of the peptide, followed by acidic hydrolysis and analysis
- Hydrazine treatment of the peptide and examination of the results.

NOTE: We will discuss some of these methods in detail in Topic 3 of this Module)

1.5.2.3 Structure of C-Terminus

A peptide has two residues in a polypeptide chain: one at the N-terminus and the other at the C-terminus. A peptide chain is created when an amino acids carboxyl group (-COOH) is bonded to another amino acids amino group.

The alpha-carbon atoms connected to the amino group and the carboxyl group (-COOH) are attached alpha-carbon atoms. The initial amino group is then connected to the terminal Carboxyl group (-COOH). The remaining Carboxyl group (-COOH) is located at the C-terminus of the residues. The C-terminus is on the right side of a protein structure when shown according to convention (3) & (4).

1.5.2.4 Function of C-Terminus

- It could be delivering protein sorting signals. The protein is kept in the endoplasmic reticulum by this terminal.
- The protein's C-terminus connects lipid anchors to it, allowing it to pass across the cell membrane without becoming a transmembrane domain.
- It assists RNA polymerase in binding to other proteins in order to initiate polymerase activity.
- Because it is the end of the polypeptide chain, this terminal is also important in the creation of peptide bonds.

SAMPLE TOPIC ACTIVITIES

Activity 1: Exercises

1. Connect the following properties with amino acids

- | | |
|-----------------------|---------------|
| a) Hydrophobic | • Tryptophane |
| b) Positively Charged | • Serine |
| c) Small | • Lysine |
| d) Polar | • Glycine |
| e) Aromatic | • Glutamine |

2. You are working on a research project to elucidate the reaction mechanism of an enzyme. You think that a particular serine residue in the protein is required for catalytic activity. To test this hypothesis, you want to genetically replace this amino acid by another, and then test whether the enzyme is still active. Which amino acid should you choose to replace the Ser?

- a) Threonine
- b) Alanine
- c) Tryptophan
- d) Glutamic Acid
- e) Histidine

Multiple Choice Questions

1. What atoms are connected by a peptide bond?
 - a. Amine nitrogen and another amine nitrogen
 - b. Carbonyl carbon and carbonyl oxygen
 - c. Side chain groups
 - d. Carbonyl carbon and amino group nitrogen
2. Proteins are biological molecules made of:
 - a. Glycerol
 - b. Nucleotides
 - c. Amino acids
 - d. Sugars
3. Which of the following is a function performed by proteins?
 - a. Regulation
 - b. Muscle Contraction
 - c. All of the answers are correct.
 - d. Transportation

Short Answer Questions

1. _____: Building blocks for creating proteins
2. _____ different amino acids, but common structure.
3. Amino functional group: _____
4. _____ functional group – NH₂
5. Carboxyl functional group: _____
6. _____ functional group – COOH
7. _____: How small organic molecules link together to form larger units
8. Small subunits, or _____, link to form polymers.
9. Small subunits, or monomers, link to form _____.
10. Polymerization is the process of linking _____ together
11. Bond forms between _____ group of one amino acid and amino group of another.
12. Bond forms between carboxyl group of one _____ and amino group of another.
13. _____ bond are the C-N bonds between amino acids.

14. Electron sharing creates unusually _____ bond.
15. Peptide bonds stiff and unable to _____.
16. Linked amino acids create _____.
17. _____: Required for many cell functions.
18. Specialized proteins _____, or speed up, chemical reactions.
19. _____: protein that functions as catalyst
20. _____ antibodies attack and destroy viruses and bacteria
21. _____ proteins responsible for cellular movement.
22. _____: carry and receive chemical messages from cell to cell
23. Protein function is based upon _____.
24. Four basic levels of organization: _____, _____, _____, and _____.
25. Primary structure: Unique sequence of _____.

Long Answer Questions

1. How to determine n and c terminus?
2. What are the n terminus and c terminus of a protein?
3. How to identify c and n terminus?
4. What is the c and n terminus?
5. How to determine n terminus and c terminus?

SAMPLE VIDEO LINKS

Below are some sample videos links that you may use to understand the content.

1. Basic Structure of Amino Acids (Link: <https://youtu.be/amdAgCbwlSE>)
2. 20 Amino Acids (Link: <https://youtu.be/PmbcA1Sav7s>)
3. Proteins formed from Amino Acids (Link: <https://youtu.be/JQZQiEdOPJY>)
4. N-Terminus and C-Terminus (Link: <https://youtu.be/Rn9zldxtZko>)

Topic 2: Proteins – Structures and Functions

2.1 Four Structural Levels of Protein

Proteins are divided into four structural levels:

1. The **Primary Structure** of a protein refers to the order or "sequence" of amino acids in the polypeptide chain as well as the placement of any disulfide links.
2. **Secondary Structure** refers to the steric connection between amino acids that are near together.
3. **Tertiary Structure** refers to the overall organization and interrelationship of a single polypeptide chain's numerous sections or domains.
4. Proteins with **Quaternary Structure** are made up of two or more polypeptide chains bound together by noncovalent forces.

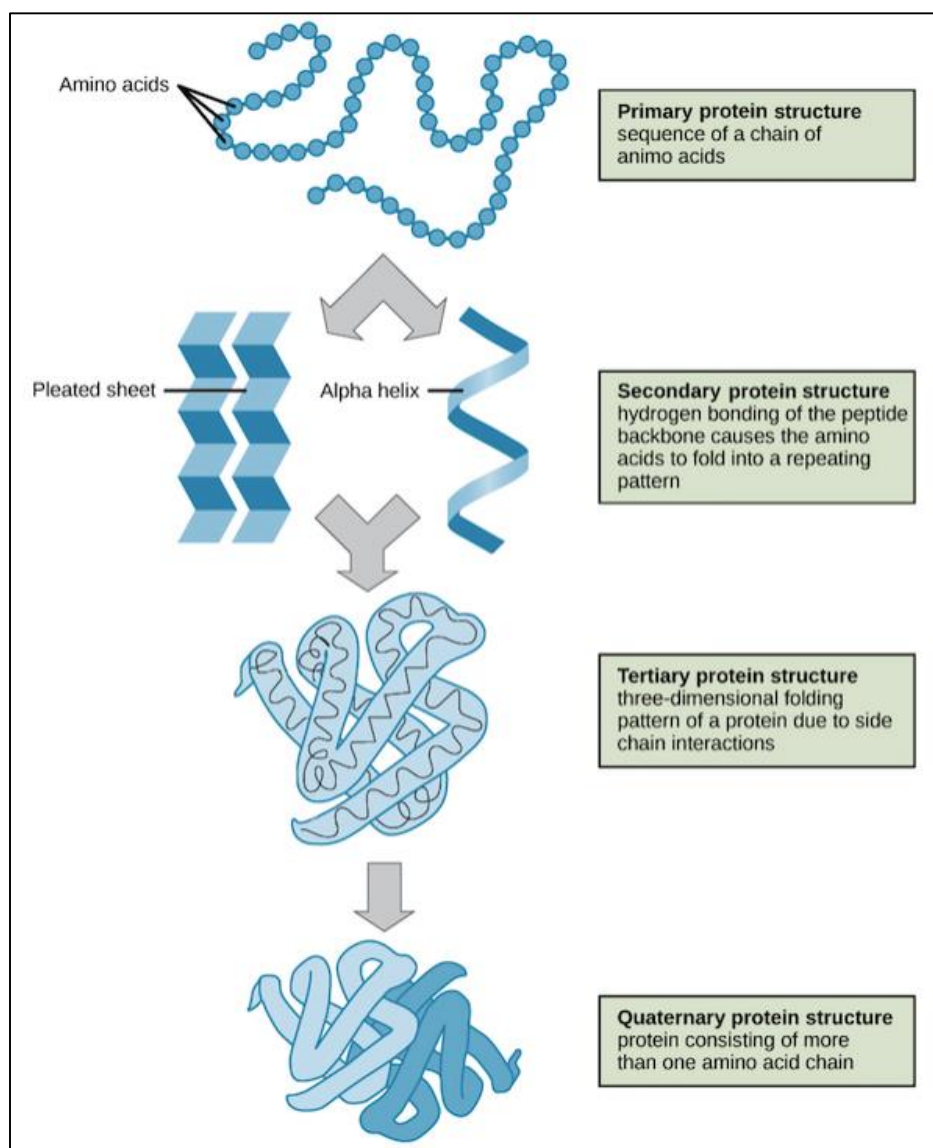


Figure SC-BT-TRB- 13: Four Levels of Protein Structure

2.2 Primary Structure

The amount and sequence of amino acids in a protein are referred to as **Primary Structure**. The primary structure determines the upper levels of organization. The genes determine the amino acid sequence of each polypeptide chain. The primary structure is maintained by the peptide connections' covalent bonding.

The following example can help students understand what is meant by the term "Sequence":

Gly-Ala-Val (1)
Gly-Val-Ala (2)

The amino acids in both of the tripeptides illustrated above are the same, but their sequences are different. When the sequence is altered, the peptide changes as well.

2.2.1 Numbering of Amino Acids in Proteins

One free alpha amino group will exist at one end of a polypeptide chain. The amino terminal (N-terminal) end is referred to as such, and the amino acid that contributes the alpha-amino group is referred to as the first amino acid. (See Figure SC-BT-TRB-14.)

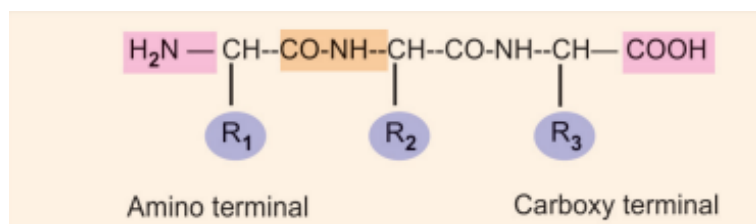


Figure SC-BT-TRB- 14: End groups of polypeptide chain (source: [file:///C:/Users/UTL%2007/Downloads/Ch-4Protein%20\(1\).pdf](file:///C:/Users/UTL%2007/Downloads/Ch-4Protein%20(1).pdf))

When writing the sequence of a protein, the N-terminal amino acid is usually written on the left-hand side. The production of the protein, by the way, begins at the amino terminal end.

The carboxy terminal end (C-terminal) of the polypeptide chain has a free alpha carboxyl group that is provided by the final amino acid (Figure SC-TRB-26). The creation of peptide bonds involves all other alpha amino and alpha carboxyl groups. Change the suffix "-ine" to "-yl" to designate amino acid residues in polypeptides, e.g., Glycine to Glycyl.



The amino group of glycine is free in the preceding example; nevertheless, the carboxyl group of glycine is bound to the amino group of alanine; the carboxyl group of alanine is then connected to the amino group of Valines; while the carboxyl group of Valine is free. As a result, the peptide is called glycyl-alanyl-valine. Gly-Ala-Val, or GAV, is the abbreviation for Gly-Ala-Val.

2.2.1.1 Branched and Circular Proteins

Polypeptide chains are usually linear. Interchain disulphide bridges, on the other hand, may form branching points in the chains. The basic structure also includes covalent disulphide linkages between separate polypeptide chains in the same protein (interchain) or between parts of the same polypeptide chain (intrachain). Glutathione (gamma-glutamyl-cysteinyl-glycine) is an example of a peptide bond formed by the gamma carboxyl group of glutamic acid instead of the alpha COOH group. A peptide bond produced by a carboxyl group different than that found in the

alpha position is referred to as a pseudopeptide. Gramicidin is an example of a protein that is in a circular configuration.

2.2.1.2 Example: Primary Structure of Insulin

The primary structure of insulin is depicted in Figure SC-TRB-15 as an example of a protein's primary structure. Sanger, who won the Nobel Prize in 1958, first documented this phenomenon in 1955. Insulin is made up of two chains of polypeptides. The A (Glycine) chain comprises 21 amino acids, whereas the B (Phenylalanine) chain has 30. Two interchain disulfide bonds hold them together (Figure SC- BT-TRB-15).

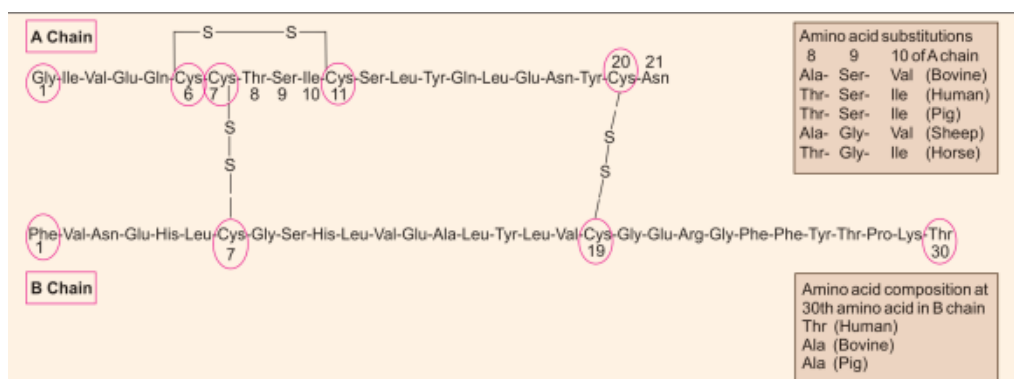


Figure SC-BT-TRB- 15: Primary Structure of Human Insulin (source: [file:///C:/Users/UTL%2007/Downloads/Ch-4Protein%20\(1\).pdf](file:///C:/Users/UTL%2007/Downloads/Ch-4Protein%20(1).pdf))

The 7th cysteine in the A chain is linked to the 7th cysteine in the B chain. Similarly, the 20th cysteine of A chain and the 19th cysteine of B chain are linked. Another intrachain disulfide link exists between A chain's 6th and 11th cysteine residues. The amino acids in positions 8, 9, and 10 in the A chain, as well as the C-terminal of the B chain, are the only ones that differ across species (Figure SC- BT-TRB-15). The amino acid sequence has remained remarkably consistent throughout evolution.

2.2.1.3 Primary Structure Determine Biological Activity

The sequence of the protein is determined by the DNA of the gene that produces it (or that encodes a portion of the protein, for multi-subunit proteins). A change in a gene's DNA sequence can result in a change in the amino acid sequence of the protein. Changing just one amino acid in a protein's sequence can change its entire structure and function.

A single amino acid mutation, for example, is associated to sickle cell anemia, an inherited disease affecting red blood cells. In sickle cell anemia, one of the polypeptide chains that make up hemoglobin (the protein that transports oxygen in the blood) has a slight sequence change. In the hemoglobin β -chain, a valine replaces glutamic acid, which is normally the sixth amino acid (one of two types of protein chains that make up hemoglobin). This substitution is depicted in the diagram below for a section of the β chain.

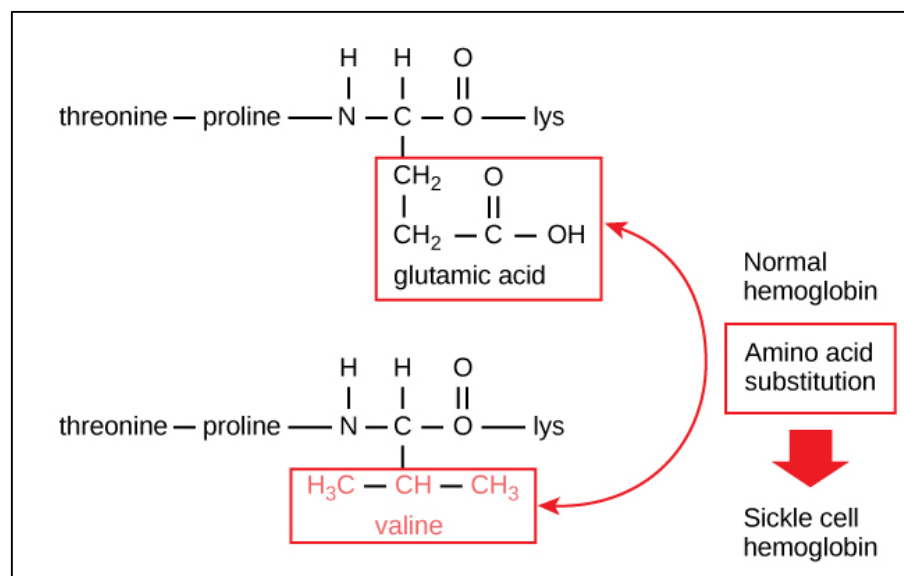


Figure SC-BT-TRB- 16: substitution of glutamic acid with valine leads to sickle cell hemoglobin

It's incredible that a hemoglobin molecule consists of two chains and two chains, each with about 150 amino acids, for a total of about 600 amino acids in the whole protein. A normal hemoglobin molecule differs from a sickle cell molecule by only two amino acids out of roughly 600.

Sickle cell anemia occurs when a person's body only generates sickle cell hemoglobin. The glutamic acid-to-valine amino acid transition causes hemoglobin molecules to assemble into long filaments. The filaments twist disc-shaped red blood cells into crescent shapes. "Sickled" cells can be seen mixed in with regular, disc-like cells in the blood sample below.

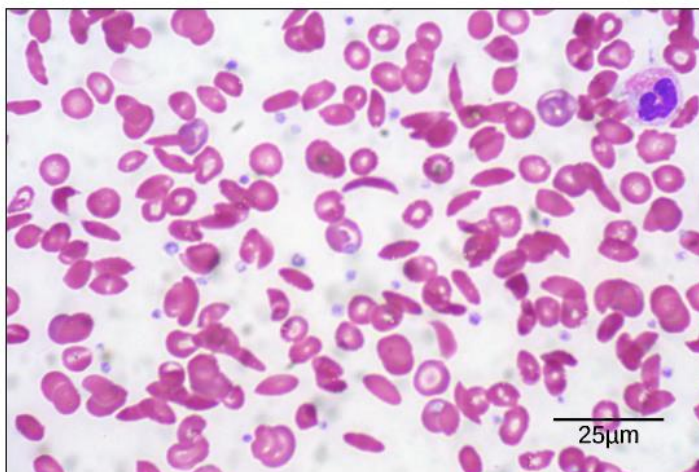


Image credit: OpenStax Biology modification of work by Ed Uthman;
scale-bar data from Matt Russell.

Figure SC-BT-TRB- 17: Disc-like cells mixed with Sickle Cell (Source: <https://cdn.kastatic.org/ka-perseus-images/ce17dc976b7e2692d4a664c804653b4bd95e1440.png>)

Sickled cells become trapped when they try to travel through blood vessels. As a result of the imprisoned cells impeding blood flow, people with sickle cell anemia may experience dyspnea, disorientation, headaches, and stomach pain.

2.3 Secondary Structure

The secondary structure of a polypeptide chain relates to the shape that a segment takes. It isn't always the complete polypeptide's three-dimensional shape, but rather a portion of it. Shapes are divided into two categories:

1. Alpha (α) Helix
2. β Pleated Sheet

Local folded structures that emerge within a polypeptide as a result of interactions between backbone atoms are referred to as secondary structure, the next level of protein structure. (The backbone simply refers to the polypeptide chain away from the R groups, and secondary structure does not contain R group atoms.) The two most common secondary structures are the helix and the pleated sheet. Between the carbonyl O of one amino acid and the amino H of another, hydrogen bonds form, binding both structures together.

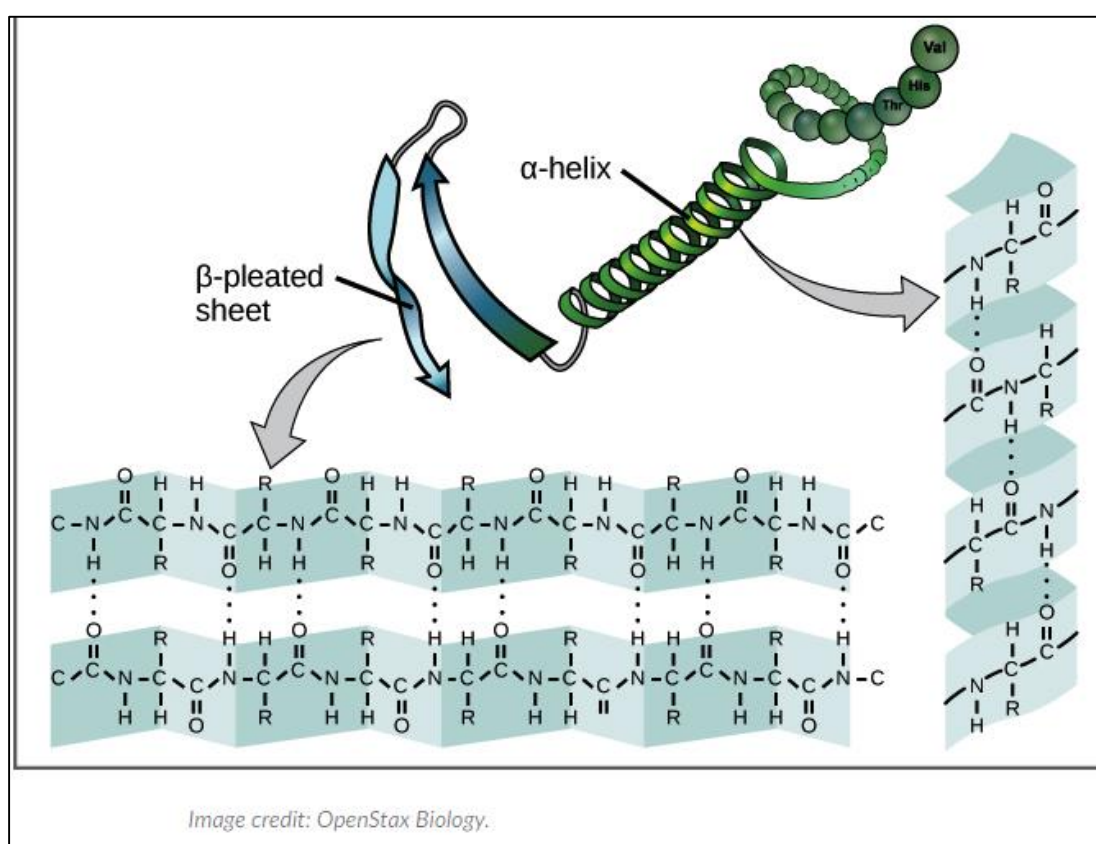


Figure SC-BT-TRB- 18: Secondary Protein Structure (Source: <https://cdn.kastatic.org/ka-perseus-images/cd59edaf690af9b30fad410a48d4a8003f3cda53.png>)

2.3.1 Alpha (α) Helix

Carbonyl (C=O) of one amino acid is hydrogen linked to amino H (N-H) of an amino acid four along the chain in a helix. (For example, the carbonyl of amino acid 1 would form a hydrogen bond with the N-H of amino acid 5.) The polypeptide chain is pulled into a helical structure that resembles a curled ribbon by this pattern of bonding, which has 3.6 amino acids per turn of the helix. The R groups of amino acids protrude from the helix, allowing them to interact freely.

2.3.2 β pleated sheet

In a β pleated sheet, two or more polypeptide chain segments line up next to each other, forming a sheet-like structure held together by hydrogen bonds. The backbone's carbonyl and amino groups form hydrogen connections, while the R groups stretch above and below the sheet's plane. The strands of a pleated sheet might be parallel, pointing in the same direction (meaning their N- and C-termini are aligned), or antiparallel, pointing in opposite directions (meaning their N- and C-termini are not aligned) (meaning that the N-terminus of one strand is positioned next to the C-terminus of the other).

Certain amino acids are more or less likely to be found in α -helices or β -pleated sheets. Because its distinctive R group (which binds to the amino group to form a ring) induces a bend in the chain and hinders helix formation, the amino acid proline is known as a "helix breaker." Proline is commonly found in bends and unstructured sections between secondary structures. Similarly, amino acids having large ring structures in their R groups, such as tryptophan, tyrosine, and phenylalanine, are typically found in pleated sheets, presumably because the pleated sheet shape gives the side chains plenty of room.

Despite the fact that many proteins have both helices and pleated sheets, some have only one type of secondary structure (or do not form either type).

2.4 Tertiary Structure

The overall three-dimensional structure of a polypeptide is referred to as its tertiary structure. The tertiary structure is mostly determined by interactions between the R groups of the amino acids that make up the protein.

R group interactions that contribute to tertiary structure include hydrogen bonding, ionic bonding, dipole-dipole interactions, and London dispersion forces – basically, all non-covalent connections. Like-charged R groups, for example, repel one other, but opposite-charged R groups can form an ionic bond. Polar R groups can also form hydrogen bonds and other dipole-dipole interactions. **Hydrophobic Interactions** are also important for tertiary structure, in which nonpolar, hydrophobic R groups cluster together on the inside of the protein, leaving hydrophilic amino acids on the outside to interact with surrounding water molecules.

Finally, a disulfide bond is a sort of covalent bond that can aid in the formation of tertiary structure. Disulfide bonds are stronger than the other types of tertiary structure bonding because they develop between the Sulphur-containing side chains of cysteines. They serve as molecular "safety pins," ensuring that polypeptide components are held together firmly.

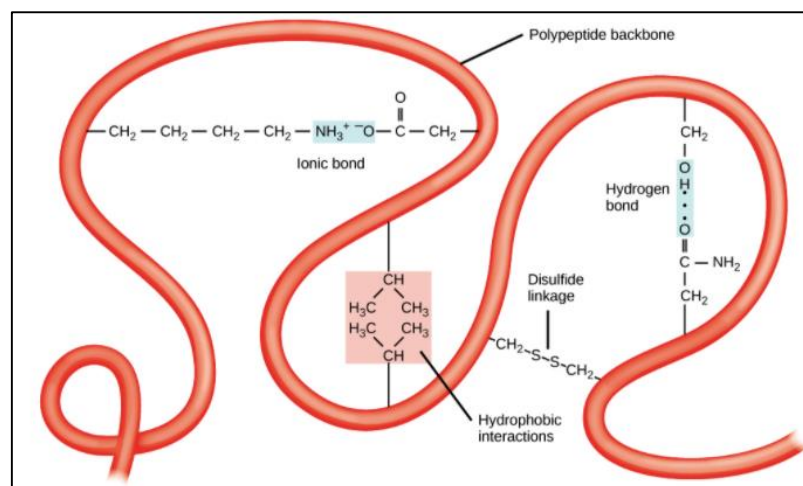


Figure SC-BT-TRB- 19: Tertiary Protein Structure (Source: <https://cdn.kastatic.org/ka-perseus-images/453bb048ca337d2042268fed53be2cab6df7d089.png>)

2.5 Quaternary Structure

When a protein has more than one polypeptide chain, it is classified as this level. Many proteins are made up of a single polypeptide chain and have only three levels of structure (the ones we just spoke about). Some proteins, on the other hand, are formed up of subunits, which are polypeptide chains. When these subunits come together, the protein's **Quaternary Structure** is produced.

We've already seen hemoglobin as an example of a quaternary structure protein. As previously stated, hemoglobin transports oxygen in the blood and is made up of four subunits, two of each kind. DNA polymerase, for example, is a ten-subunit enzyme that synthesizes new DNA strands.

The weak interactions that contribute to tertiary structure (such as hydrogen bonding and London dispersion forces) keep the subunits together to form quaternary structure.

2.6 The bonds that hold Protein Structures together

- Primary Structures are held together with peptide bonds
- Secondary Structures such as the alpha helix and the
- Tertiary Structures are held together with many interactions such as
 - hydrogen bonds between the different variable groups of the amino acids.
 - Ionic bonds or (acid-base interactions) when certain amino acids have either acidic or basic variable groups.
 - Van der Waals interactions hold the protein shape together
 - Disulfide bridges – a special type of covalent bond between two sulfurs that are both contained inside amino acids that make up a polypeptide chain. The chain folds in on itself in a way that the two sulfurs are across from each other to form this disulfide bridge.
- Quaternary structure is also as similar to Tertiary Structures.

2.7 Polypeptide vs. Protein

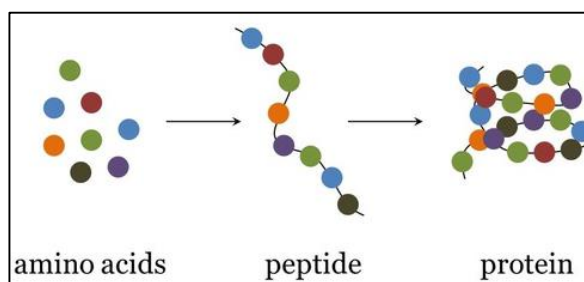


Figure SC-BT-TRB- 20: amino acids vs. peptide vs. protein (Source: <https://www.peptidesciences.com/glossary/wp-content/uploads/2016/05/peptides-vs-proteins.png>)

A polypeptide is a three- or more-amino-acid linkable chain that has not yet been folded. One or more polypeptides have been folded into their whole three-dimensional structure to form a protein. Another distinction is that a protein can be entirely made up of one folded polypeptide chain. Other proteins, especially those with a quaternary structure, are composed of two or more folded interacting polypeptide chains. Protein is either a single folded polypeptide chain or a collection of folded polypeptide chains as a result of this.

Consider a polypeptide to be a straight strand of yarn, and a protein to be a kneaded sweater that has taken shape.

A normal protein has 300 or more amino acids, with the number of amino acids and their sequence varying from one protein to the next. Like the alphabet, the amino acid 'letters' can be arranged in a million different ways to create 'words' and a full protein 'language.' Depending on the number and sequence of amino acids, the final protein will fold into a specific shape. This shape is critical since it influences the protein's function (e.g., muscle or enzyme). Humans, like all other species, have unique protein characteristics.

2.8 Chaperonins

Chaperonins are chemical molecules that aid in protein folding and are found in living cells. It's an enormously complex structure made completely of proteins. The product is seen in both front and side views in the image below.

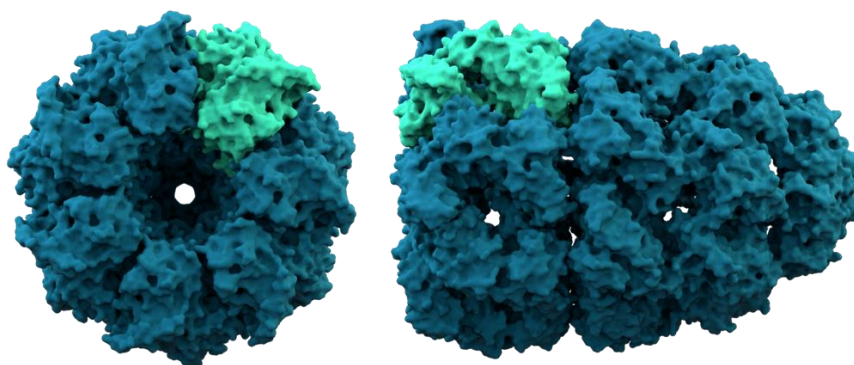


Figure SC-BT-TRB- 21: Chaperonins front view (left) and side view (right)(Source: https://kids.kiddle.co/images/thumb/6/6c/Chaperonin_1AON.png/1036px-Chaperonin_1AON.png)

When a polypeptide needs to be folded, it enters the cavity of a chemical structure like the Chaperonin (Figure SC- BT-TRB-21, Left side view). This cavity provides an ideal environment for all of the amino acids to create the necessary interactions.

2.9 Denaturation

Proteins can sometimes really unfold, a process known as denaturation. As a result, when a protein unfolds, it becomes Denatured. This can occur as a result of high temperatures, which can cause the hydrogen bonds that hold the protein together to break. Denaturation can also occur in extremely basic or acidic conditions, where the ionic interactions that hold the protein's acidic and basic amino acids together are disturbed.

As a result, if the acidic or basic contacts or hydrogen bonds are disrupted, the protein's three-dimensional form can be lost owing to denaturation. Because the form of the protein is so important to its function, this can be quite harmful.

2.10 Structure-Function Relationship of Proteins

Proteins are able to detect and interact with a wide range of molecules, which allows them to continue their roles. The functional properties are provided and maintained by the three-dimensional structural conformation. The primary structure, in turn, is dependent on the three-dimensional structure. As a result, any changes in the primary structure may result in a protein that is unable to perform its job.

The primary types and functions of proteins are listed in Table 2 below.

Table 2: Types, Examples and Functions of Proteins

Type of Protein	Example	Functions
Digestive Enzymes	Amylase, Lipase, Pepsin, Trypsin	Help in digestion of food by catabolizing nutrients into monomeric units
Transport	Hemoglobin, albumin	Carry substances in the blood or lymph throughout the body
Structural	Acting, Tubulin, Keratin	Construct different structures, like the cytoskeleton
Hormone	Insulin, Thyroxine	Coordinate the activity for different body systems
Defense	Immunoglobulins	Protect the body from foreign pathogens
Contractile	Actin, Myosin	Effect muscle contraction
Storage	Legume storage proteins, egg white (albumin)	Provide nourishment in early development of embryo and the seedling.

Examples of what proteins do are listed below:

- **Antibody** - Antibodies help the body protect itself by binding to certain foreign particles such as viruses and bacteria. Consider the following scenario: Immunoglobulin G (IgG) is a kind of immunoglobulin (IgG)
- **Enzyme** - Enzymes are responsible for practically all of the chemical reactions that occur in cells. They also help to make new molecules by interpreting the genetic information stored in DNA. Phenylalanine Hydroxylase, for example.
- **Messenger** - Messenger proteins, such as certain hormones, send signals between cells, tissues, and organs to coordinate biological activities. For instance, Growth Hormone.
- **Structural Component** - These proteins give cells their structure and support. They also allow the body to move on a bigger scale.
- **Transport/Storage** - These proteins bind and transport atoms and tiny molecules across the body and within cells. Actin, for example.²

² [What are proteins and what they do?: MedlinePlus Genetics](#)

SAMPLE TOPIC ACTIVITIES

Activity 1: Self-Test Exercise

1. Which of the following describes the structure of albumin?
 - a. A molecule containing carbon, hydrogen, and oxygen in a 1:2:1 ration
 - b. A chain of amino acids folded and twisted into a molecule.
 - c. Three fatty acids attached to one glycerol molecule
 - d. A series of nitrogenous bases attached to a sugar-phosphate backbone.
2. Which of the following are functions of proteins?
 - a. Cell signaling
 - b. Encoding Genetic Information
 - c. Catalyzing Chemical Reactions
 - d. Energy Storage
3. Which of the following is an example of protein denaturation?
 - a. Several amino acids are joined together via peptide bonds
 - b. A protein binds with a substrate, lowering the activation energy of reaction.
 - c. Amino acids fold into repeating patterns due to hydrogen bonding of the peptide backbone.
 - d. A protein is exposed to extremely high heat, causing it to lose its secondary structure and be left with only its primary structure.
4. Which of the following statements is accurate regarding these protein structures?
 - a. Secondary structures are formed by multiple polypeptide chains.
 - b. Interactions between the R groups in amino acids form tertiary structure.
 - c. The two types of primary structure are α -helices and β -pleated sheets.
 - d. Proteins in a quaternary structure consist of a simple polypeptide chain.

SAMPLE VIDEO LINKS

Below are some sample videos links that you may use to understand the content.

1. Protein Structure Levels (Link: <https://youtu.be/YhTpnRzp-eQ>)
2. Chaperonins: Protein Folding Mechanism (Link: <https://youtu.be/Syw-m4ipsO4>)
3. Denaturation (Link: <https://youtu.be/qEycDKQn93Y>)

Activity 2: Homework Questions

1. During denaturation, proteins will not lose their structure, with regard to:
 - A. Primary structure
 - B. Secondary structure
 - C. Tertiary structure
 - D. Quaternary structure
2. The characteristic features of the peptide bond include all the following, except:
 - A. Does not allow freedom of rotation
 - B. It is a partial double bond
 - C. Always has cis configuration
 - D. Absorbs UV light at 280 nm
3. The force maintaining the primary structure of a protein:
 - A. Peptide bonds
 - B. Hydrophobic forces
 - C. Hydrogen bonds
 - D. Electrostatic (ionic) bonds
4. The forces maintaining the secondary, tertiary and quaternary structures of a protein are the following, except:
 - A. Electrostatic (ionic) bonds
 - B. Hydrophobic forces
 - C. Van der Waals forces
 - D. Peptide bonds
5. The amino acid which did not allow formation of alpha-helix is:
 - A. Glutamate
 - B. Proline
 - C. Tyrosine
 - D. Histidine
6. Tertiary structure of a protein describes:
 - A. The sequence of amino acids
 - B. Location of disulphide bonds
 - C. Amino terminal end amino acid
 - D. The nature of protein folding
7. One of the following proteins does not have a quaternary structure:
 - A. Albumin
 - B. Hemoglobin
 - C. Lactate dehydrogenase
 - D. Immunoglobulin G
8. All the following reagents are used for identifying the first amino acid in a protein, except:
 - A. Cyanogen bromide
 - B. Fluorodinitrobenzene

- C. Dansyl chloride
 - D. Phenyl isothiocyanate
9. Proteins can be precipitated by the following methods, except:
- A. Adding alcohol and acetone
 - B. Saturating with ammonium sulphate
 - C. Using salts of heavy metals
 - D. Shifting the pH away from the iso electric point
10. Denatured proteins:
- A. Are soluble
 - B. Are difficult to digest
 - C. Are biologically inactive
 - D. Peptide bonds are broken
11. Which of the following is a simple protein?
- A. Casein
 - B. Insulin
 - C. Hemoglobin
 - D. Tyrosinase
12. In glycoproteins, the carbohydrate chains are combined through glycosidic linkages with:
- A. Hydroxyl groups of serine or threonine residues of proteins
 - B. Epsilon amino nitrogen of lysine residues of proteins
 - C. Guanidium group of arginine residues of proteins
 - D. Phenol group of tyrosine residues of proteins
13. The protein which does not answer the aldehyde test is:
- A. Hemoglobin
 - B. Albumin
 - C. Casein
 - D. Gelatin
14. Proteins may be estimated by the following methods, except:
- A. Biuret method
 - B. Heat coagulation
 - C. Kjeldahl's digestion
 - D. Nephelometry
15. All the following are examples of tertiary structure of proteins, except:
- A. Alpha helix
 - B. Beta pleated sheet
 - C. Triple stranded helix
 - D. Peptide bonds
16. Tertiary structure of a protein describes:
- A. Sequence of amino acids
 - B. Location of disulphide bonds

- C. Amino terminal end amino acid
 - D. The nature of protein folding
- 17.** Proteins may be denatured irreversibly by:
- A. Adding urea
 - B. Bringing to iso electric pH
 - C. Heat coagulation
 - D. Reduction with mercaptoethanol
- 18.** Lectins are:
- A. Animal proteins having specific amino acid binding site
 - B. Antibody molecules acting against cells
 - C. Plant proteins having specific carbohydrate binding site
 - D. Blood proteins having a lecithin group
- 19.** Ultraviolet light at 280 nm is absorbed by which component of proteins?
- A. Peptide bonds
 - B. Sulfhydryl group of cysteine
 - C. Indole ring of tryptophan
 - D. Imidazole ring of proline
- 20.** The nature of the bond linking amino acids to each other is:
- A. Covalent
 - B. Co-ordinate
 - C. Ionic
 - D. Hydrophobic
- 21.** How many peptide bonds are present in glutathione?
- A. 1
 - B. 2
 - C. 3
 - D. 4
- 22.** Basic difference between two polypeptides is in the —
- A. Structural conformation
 - B. Primary sequence of amino acids
 - C. Number of side chains
 - D. Number of hydrophobic bonds
- 23.** Human insulin differs from bovine insulin in:
- A. Biological activity
 - B. Number of amino acids
 - C. Position of disulfide bonds
 - D. Sequence of amino acids
- 24.** A covalent bond between the alpha carboxyl group of one amino acid and alpha amino group of the neighboring amino acid is called
- A. Cis double bond

- B. Isopeptide bond
 - C. Pseudopeptide bond
 - D. D. Peptide bond
- 25.** Study of linear sequence of amino acids is done by all techniques listed except:
- A. End group analysis
 - B. Hydrolysis by proteolytic enzymes
 - C. Analyzing the content of each amino acid
 - D. Denaturing the protein
- 26.** Different polypeptide chains are held together by:
- A. Peptide bonds
 - B. Disulphide bonds
 - C. Glycosidic bonds
 - D. Ester bonds
- 27.** Primary structure decides:
- A. Rate of synthesis of protein
 - B. Biological activity of the protein
 - C. Rate of degradation of the protein
 - D. Effect of proteolytic enzymes on protein
- 28.** Secondary and tertiary levels of protein structure are dependent on:
- A. Presence of disulfide bonds
 - B. Primary structure
 - C. PH of the medium
 - D. PK value of component amino acids
- 29.** The protein having predominantly alpha helical structure is
- A. Collagen
 - B. Keratin
 - C. Fibroin
 - D. Myoglobin
- 30.** Which of the following is NOT true regarding the tertiary structure of proteins?
- A. It is a random coil structure
 - B. Disulfide bonds are formed between any two cysteine residues
 - C. Position of disulfide bonds are predetermined and fixed
 - D. Denaturation using reducing agents does not affect the disulfide bonds
- 31.** Protein having a large number of disulfide bonds is:
- A. collagen
 - B. keratin
 - C. hemoglobin
 - D. albumin
- 32.** Which of the following proteins does not possess a quaternary structure?
- A. Myoglobin

- B. Lactate dehydrogenase
- C. Hemoglobin
- D. Immunoglobulin M

33. Which of the following is NOT true regarding hemoglobin?

- A. Has 4 independent subunits
- B. Each subunit has one heme residue
- C. Each subunit can bind one molecule of oxygen
- D. All four subunits are similar

Activity 2: Assignment Questions

1. How proteins are made up of?
2. What is a peptide bond?
3. What is a dipeptide?
4. How many peptide bonds are present in a tripeptide?
5. What is a polypeptide?
6. What are the levels of organizations of proteins?
7. What is meant by primary structure of a protein?
8. What force maintains the primary structure?
9. How are the end amino acids of proteins called?
10. What is a pseudopeptide?
11. Can you give an example of a pseudopeptide?
12. What are the salient structural features of insulin?
13. What is proinsulin?
14. What is a mutation?
15. Can you give an example?
16. What are the forces that maintain the secondary, tertiary and quaternary structures of a protein?
17. What are the salient features of alpha structure of proteins?
18. What is secondary structure of a protein?
19. What is meant by tertiary structure of a protein?
20. What is quaternary structure of a protein?
21. Name proteins having quaternary structure.
22. What are the reagents that are used for identifying the first amino acid in a protein?
23. What is isoelectric point of a protein?
24. What are the features of isoelectric point?
25. What is the isoelectric pH of human albumin?
26. How proteins are precipitated from solution?
27. How is albumin precipitated?
28. What about globulins?
29. Give example of isoelectric precipitation.
30. What is the isoelectric pH of casein?
31. What are the features of denaturation?
32. What is heat coagulation?
33. Give examples of proteins that coagulate.

Activity 3: Practical Experiment 2

Title: Laboratory Test of Proteins

Source: [Test of Proteins Experiment - Chemistry Practicals Class 12 \(byjus.com\)](https://www.byjus.com/chemistry/test-of-proteins-experiment/)

Proteins are a class of biologically important compounds having high molecular weight. Proteins are perhaps the most complex organic materials produced in nature. Plants build up their proteins from carbon dioxide, water and minerals in the presence of sunlight.

Animals derive proteins from plants.

Aim: To perform some simple tests to identify the presence of proteins in the given sample.

Theory:

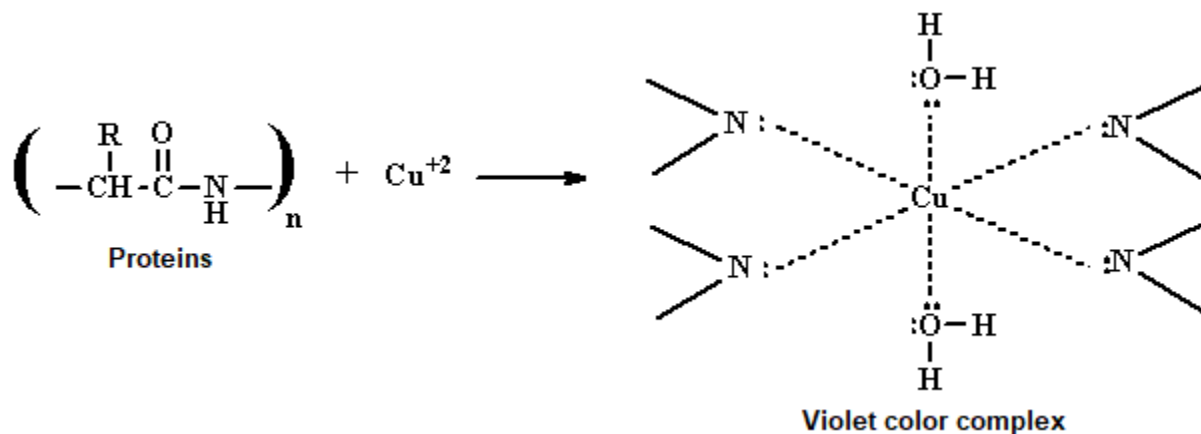
Protein is a long chain polymer with a high molecular mass made up of α -amino acids. Proteins are found in living organisms since they are components of cells. Carbon, hydrogen, nitrogen, oxygen, and phosphorus and sulphur are all found in proteins.

The following are the tests carried to find the presence of proteins in a given sample.

1. Biuret test
2. Xanthoproteic test
3. Million's test
4. Ninhydrin test

(a) Biuret Test:

This test is performed on peptide-linked molecules. Polypeptides of amino acids are joined together by peptide bonds to form proteins. When a bluish violet color is achieved from an alkaline protein solution, it is treated with a drop of aqueous copper sulfate.



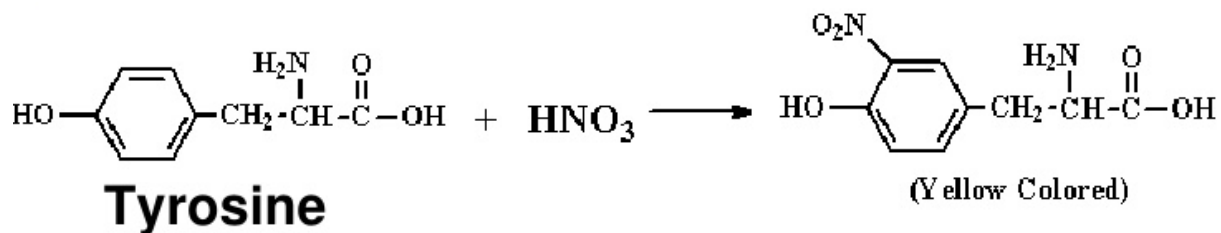
This test can be used to identify proteins as well as estimate protein amounts. When urea is heated to 180°C , a chemical called biuret is created. Two urea molecules condense to form a bi-urea or biuret molecule in this process. In the presence of copper ions, the biuret reagent creates a violet color complex.

Note: Formation of violet colouration confirms the presence of Proteins.

(b) Xanthoproteic Test:

Proteins that have been treated with nitric acid turn yellow or orange. Nitration is done with concentrated nitric acid. Proteins produce a yellow precipitate when exposed to nitric acid, which turns

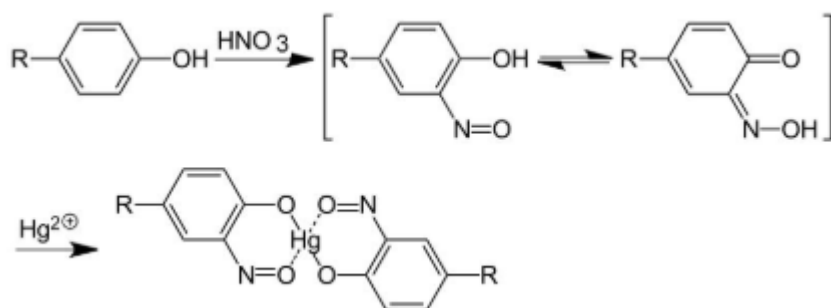
orange when exposed to alkali.



Note: The appearance of yellow colour solution confirms the presence of proteins.

(c) Millon's Test:

In the presence of sodium nitrite and sulfuric acid, the phenolic group of tyrosine in proteins reacts with mercuric sulfate to produce a red color. Proteins containing phenolic amino acids are used in Millon's test. This test is not performed on gelatin. When proteins are treated with millon reagent, a white precipitate forms that turns brick red when boiled, indicating the presence of proteins.

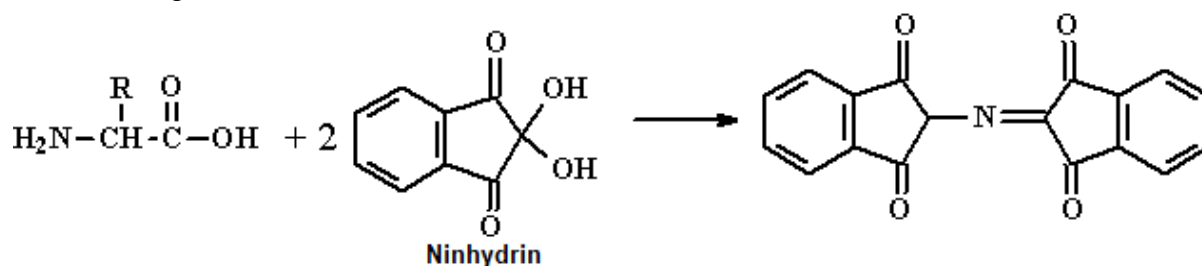


Note: The appearance of brick red colour solution confirms the presence of proteins.

(d) Ninhydrin Test:

Proteins react with the pyridine solution of ninhydrin to produce a coloured solution ranging from deep blue to violet pink, and in rare cases even red. Ninhydrin solution is made by dissolving 0.1 grams of ninhydrin in 100 milliliters of distilled water. However, this ninhydrin solution is stable and can be maintained for two days.

The following is a chemical reaction.



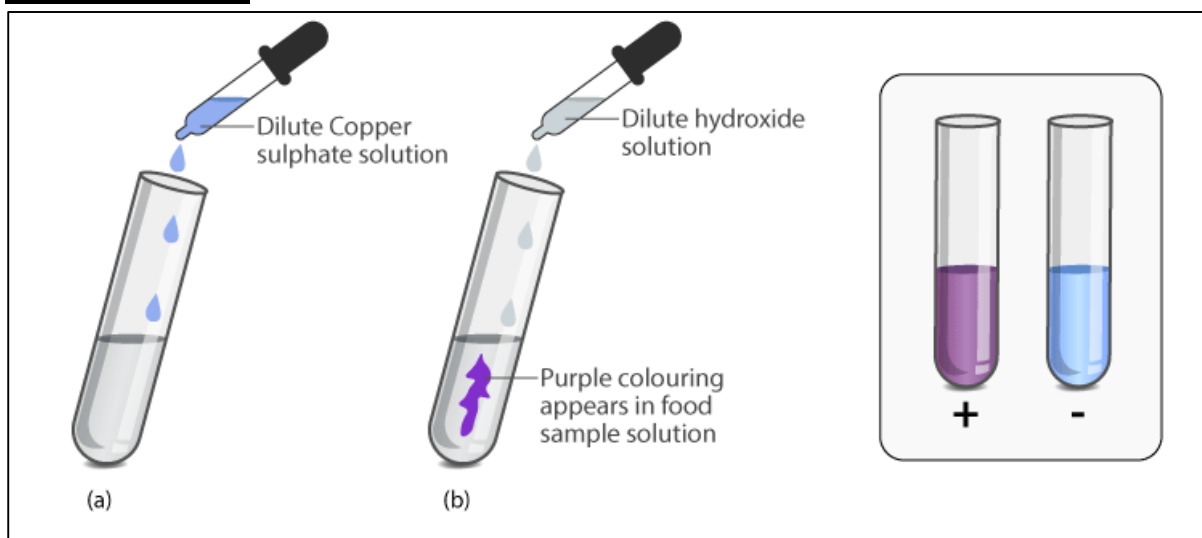
Note: The appearance of violet colour solution confirms the presence of proteins.

Materials Required:

1. Copper sulfate solution
2. Sodium hydroxide
3. Nitric acid
4. Mercuric sulfate
5. Sodium nitrite

6. Sulfuric acid
7. pyridine solution
8. Ninhydrin reagent
9. Distilled water
10. Test tubes
11. Test tube holder
12. Dropper
13. Water bath
14. Stirrer

Apparatus Setup:



Procedure:

(a) Biuret Test:

1. Take the given sample to be tested in a clean test tube.
2. Add 2ml of sodium hydroxide solution to it.
3. To that add 5 to 6 drops of copper sulfate solution to it.
4. If there is the appearance of bluish violet colour indicates the presence of protein.

(b) Xanthoproteic Test:

1. Take 2ml of given sample compound in a test tube.
2. Add a few drops of **concentrated sulfuric acid** and heat.
3. If there is formation of yellow precipitate then the presence of protein is confirmed.

(c) Millions Test:

1. Take 2ml of given sample solution in a clean test tube.
2. Add 2-3 drops of Millon's reagent and shake well.
3. Observe the change.
4. If there is formation of white precipitate and changes to brick red on heating than the presence of protein is confirmed.

(d) Ninhydrin Test:

1. Take the sample solution to be tested in a clean test tube.
2. Add 1-2ml of ninhydrin solution to it.
3. Boil the mixture and observe the change.
4. If there is the appearance of blue colouration then the presence of protein is confirmed.

Observation and Inference:

Biuret test	Formation of violet colouration confirms the presence of Proteins.
Xanthoproteic test	The appearance of yellow colour solution confirms the presence of proteins.
Millions test	The appearance of brick red colour solution confirms the presence of proteins.
Ninhydrin test	The appearance of violet colour solution confirms the presence of proteins.

Results and Discussions:

The given sample contains _____ (proteins).

Precautions:

1. While performing the experiment handle chemicals with care.
 2. Use lab aprons and hand gloves during the experiment.
 3. Use droppers to take the reagents from the bottle.
-

Activity 4: Practical Experiment

Title: Protein Gel Electrophoresis

Source: [Microsoft Word - Protein Gel Electrophoresis.doc \(cpb-us-e1.wpmucdn.com\)](https://cpb-us-e1.wpmucdn.com)

STUDENTS PRACTICAL ON PROTEIN GEL ELECTROPHORESIS

Objectives

In this lab you will:

1. Design an experiment to investigate a question about evolutionary relatedness between organisms.
2. Prepare protein samples from grocery store meats.
3. Do protein gel electrophoresis.
4. Analyze your results in relation to the questions you chose to investigate.

Introduction

Gel electrophoresis is a powerful research tool widely used by scientists today. Using this tool, scientists can separate a complex mixture of large molecules (macromolecules) according to their size. The two types of macromolecules most often separated by gel electrophoresis are nucleic acids and proteins. Information about the size of a molecule provides important clues as to its identity.

In this experiment, the detergent sodium dodecyl sulfate (SDS) and heat are used to denature proteins in a muscle tissue taken from different grocery store meat samples. The SDS coats the resulting polypeptides with a negative charge (see Figure 1).

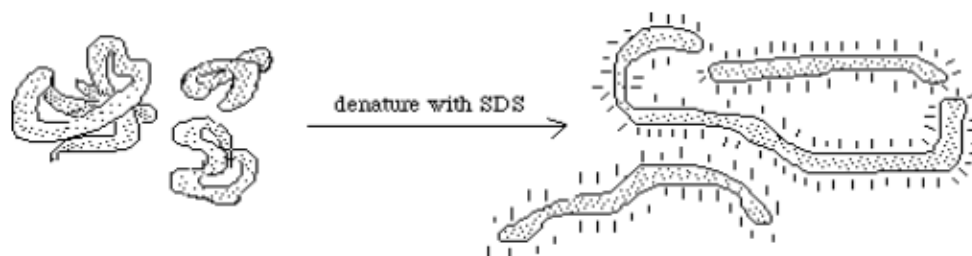


Figure 1: When three functional polypeptides (on the left) are coated with SDS and heated they lose their three-dimensional structure and take on a net negative charge (on the right). Bigger polypeptides are coated with more molecules of SDS; the ratio of molecular weight to charge is approximately the same for all proteins.

The polyacrylamide gel is placed in a gel box between two electrodes, the sample is then placed into a well at the top of the gel, and the electrodes are connected to a power supply that produces a voltage gradient across the gel. This causes the negatively-charged, SDS-coated polypeptide molecules to move

towards the positive electrode (the anode). The polyacrylamide gel is composed of long molecules that form a tangled mesh (see Figure 2). Simply put, to get through the gel, the macromolecules must make their way through a maze-like network of pores. The largest molecules move more slowly because they have a harder time negotiating the pores. The smallest molecules move fastest because they can quickly slip through the pores. As soon as the current is applied, the SDS-treated polypeptides begin their race towards the positive electrode. As time passes the smaller polypeptides begin to “pull away from” the larger ones. As a result the proteins from muscle tissue will be separated according to size. The size of a macromolecule is commonly expressed by its molecular weight which is measured in daltons.

The polypeptide molecules cannot be seen while the gel is running. If the current is left on for too long, all the proteins will run off the gel at the bottom. For this reason, a blue dye is added to the SDS solution. This blue dye is also negatively charged and drawn toward the positive electrode. Since it is a relatively small molecule (smaller than any of the polypeptides), it moves quickly through the gel. When this dye reaches the bottom of the gel it is time to turn off the current.

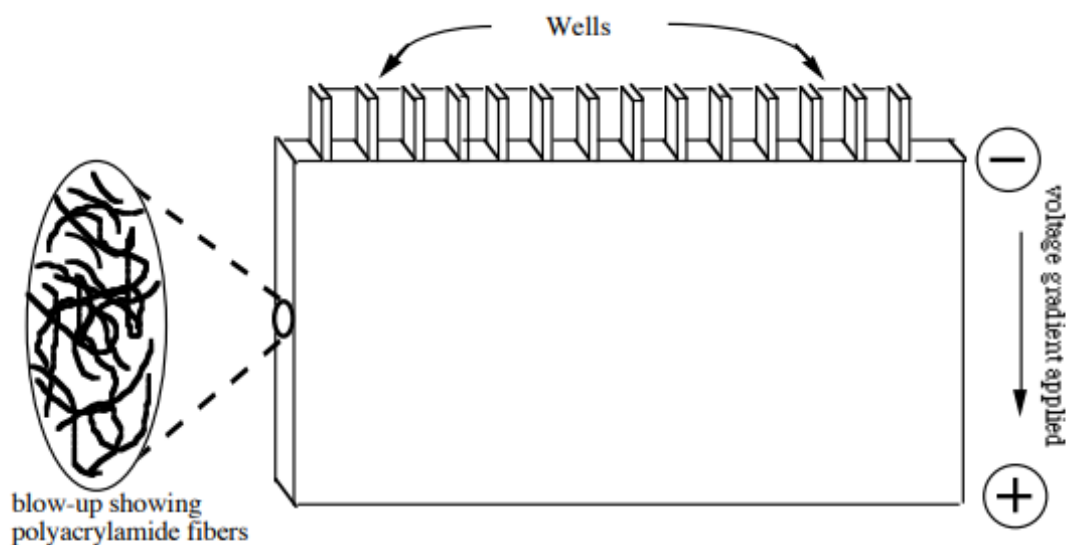


Figure 2: A simplified diagram of a polyacrylamide gel. The glass plates and supports which hold the gel upright are not shown. Mixtures of proteins can be loaded in any of the “wells” at the top of the gel, then the electric current is turned on.

Materials

- plastic cutting boards
- transfer pipettes
- 500 ml or 1 liter flask or beaker
- gel box and power supply
- 2 –20 μ l micropipettors and tips
- marker and tape
- stain
- staining tray with lid
- glass or plastic rods
- razor blades or scalpels
- 1X Tris-Glycine-SDS running buffer
- blue solubilization (Laemmli) buffer
- 15 ml plastic tubes and rack
- microcentrifuge tubes and rack
- gloves
- distilled or deionized water
- safety glasses
- clinical centrifuge or funnel with cheesecloth

Safety Precautions

Good laboratory practice should be followed while carrying out all aspects of this lab. Wear gloves and safety glasses while handling the meats, the polyacrylamide gels, the protein stain and the other chemicals. Be sure to wash your hands and bench tops after working with the raw meats.

Procedure

Experimental Design

1. Your teacher will list the samples available to you. In the process of designing your experiment you must:
 - i. Determine what question you would like to investigate.
 - ii. Determine which protein sources you are going to use.
 - iii. Determine the number of replicates you will include.
 - iv. Determine which sample will go in which well.
 - v. Determine how you will interpret your results.
2. Consult with your group members and fill out the **Student Experimental Design Sheet** (page 9).
3. Your **Student Experimental Design Sheet** must be turned in to your teacher and approved before moving on to the rest of the lab.

Sample preparation--extraction of muscle protein from animal tissues

1. Your teacher will assign your group a protein sample to prepare. You will make a protein extract for the whole class to share. This has no relation to the samples you choose to use in your experimental design.
2. Obtain a pea-sized sample of muscle tissue. Remove any skin, bone or other non-muscle tissue.
3. Mince the tissue into **very tiny** pieces on the plastic cutting board using scissors, a razor blade, or a scalpel.
4. Put the muscle into a 15 ml **labeled** conical tube and, using a transfer pipet, add 1 ml of blue solubilization buffer (Laemmli Buffer). Mash the tissue with a rod for 1 min. With a clean transfer pipet, add 3 ml of deionized or distilled water and mash again with the rod for 1 min.
5. Place the plastic tube with the cap loosened in a dry bath set at 95°C for 5 minutes.
6. Remove your test tube from the dry bath and let it cool 2 to 5 minutes (if the protein has congealed, remash it with the rod).
7. Centrifuge the sample with the cap tightened for 5 to 10 minutes at high speed or allow the sample to settle by gravity for 1/2 hour.

*(During steps 5-7, begin steps 11&12 and start the next part of the lab, **Gel electrophoresis.**)*

8. Use a transfer pipet to draw off 0.5 ml of the supernatant and place into a 15 ml **labeled** conical tube. Some meat samples will have a whitish layer of fat or clear oil floating on top. Avoid this layer! Insert the tip of your transfer pipette below this layer when drawing off the supernatant. Avoid the pellet, also!
9. Use a clean transfer pipette to add 1 ml blue solubilization buffer (Laemmli Buffer) and 1 ml distilled water to your new tube. Mix gently. Your original protein sample is now one fifth of the final solution.
10. Give this labeled **Master Supply Tube** to your teacher who will place all the test tubes in a rack at the front of the room. All students will obtain their samples from these tubes.
11. Obtain a microcentrifuge tube rack and microcentrifuge tubes. Label the tubes with the name of the protein samples you are going to use in your experiment. Take your rack with labeled tubes to the **Master Supply Tubes**. Add **one drop** of each protein sample to the appropriate tube.
12. Label the diagram on your **Lab Report Sheet** (Page 10) with the sequence of samples that you will use. This should be the same as your labeled diagram on your **Student Experimental Design Sheet** (Page 9).

Gel Electrophoresis

Two gels can be run in one gel box, so you may be sharing a gel box with another group. Referring to the addendum “Gel Box Assembly,” disassemble the gel box into its four parts: Lid, tank, clamping frame, and electrode assembly.

1. First, remove the lid and, using two hands, lift the clamping frame out of the tank. Then make sure to open the cam levers on the clamping frame before removing the electrode assembly.
2. Notice that one gel will fit against the green U-shaped gasket on each side of the electrode assembly. If you find a buffer dam on one side remove it, unless you are only planning to run one gel in the box. During electrophoresis, the gels will be clamped against the electrode assembly, forming an inner buffer chamber sealed off from the outer chamber in the tank. Note the location of the red anode (positive electrode) and black cathode (negative electrode). When the buffer chambers are sealed, the top of the gel will only be exposed to buffer from the inner chamber, and the bottom of the gel will only be exposed to buffer from the outer chamber.
3. **You must wear gloves** when handling the polyacrylamide gel. Obtain a gel from your teacher. Use a marking pen to indicate the location of each well on the large outside plate. This will help you locate the lanes. The actual gel itself is sandwiched between the two glass plates.
4. Insert the gels into the slots at the bottom of each side of the electrode assembly, so that the shorter glass plate on each gel faces inward, toward the green U-shaped gaskets. They will rest at an angle, with only the bottom of the gels against the gaskets.
5. Gently press the gels up against the gaskets and fit the electrode assembly into the clamping frame. Be sure to press down on the Electrode Assembly while closing the two cam levers on the clamping frame. This ensures a tight seal so that no buffer will leak between the inner and outer buffer chambers.



Figure 3. Sealing the Inner Chamber

From Bio Rad's *Ready Gel Cell Instruction Manual*, p.5.

6. Finally, set the clamping frame and electrode assembly inside the tank, so that the notches on the electrode assembly fit into the clear tabs on the tank.
7. To make 1X running buffer for each gel box, combine 40 ml of 10X Tris-Glycine-SDS Running Buffer with 360 ml of deionized H₂O and mix well. Fill the inner chamber so that the buffer reaches a level between the tops of the short and long plates of the gel. Don't overfill the inner chamber, or

buffer will spill over the top of the long plate, interrupting electrophoresis. Fill the outer chamber to a level just above the bottom of the two cam levers (approximately 200 ml).

8. Remove the comb carefully using two hands and your index fingers. Check for bubbles in the top of the gel; these may be dislodged by a gentle stream of buffer from a transfer pipet.
9. Set the micropipettor to 10 μ l, put on a yellow tip, and draw up a muscle sample as shown by your instructor. Line up the tip with the middle of the well. **Rest** the tip between the two gel plates. Gently dispense the sample into the well. The sample will sink to the bottom of the well. Be certain your diagram correctly shows the lane each sample was loaded into.
10. You also need to add 10 μ l of the protein standard (molecular weight marker) to the appropriate lanes. The protein standard is a mixture of specific polypeptides of known size. By including them on the gel, you can estimate the size of the polypeptides in your sample. The molecular weight marker used here has three proteins:

protein	molecular weight (in daltons)
Bovine serum albumin	66,000 Da
Carbonic anhydrase	29,000 Da
Soybean trypsin inhibitor	20,000 Da

11. Slide the lid into position, making sure the black plug fits on the black electrode and the red plug fits on the red electrode. Attach the black and red cords to the power supply in the proper orientation, plug in the power supply, and turn the power on at 200 volts. Bubbles rising from the bottom of the tank indicate the current is running.
12. Describe or sketch the initial appearance of the gel and the appearance after 5 min. as indicated on your **Lab Report Sheet** (Post Lab Questions).
13. Run the gel for at least 45 minutes. It's important to let the bands of blue dye run off the bottom of the gel before turning off the power supply. Notice that the bottom of the gel plates extend below the green gasket on the electrode assembly. The bands of dye will be hard to see when they move in front of the gasket, but will reappear a few minutes later near the bottom of the gel. **Turn off the power supply, unplug the power cord, and remove the lid.**
14. **Wearing gloves and goggles**, pour out the used running buffer from both gel chambers. With two hands, remove the clamping frame from the tank. Make sure the inner chamber is emptied before opening the cam levers on the clamping frame, or the buffer will leak out. Then lift the electrode assembly carefully out of the clamping frame and remove the gels. To expose the gels, slice the plastic along the outside of the white spacer on one side of the gel plates. Then use the razor blade to pry the plates apart. Open the glass plates as if you were "opening a book." **Be careful!** The gel is fragile and tears easily.
15. Carefully place the gel into a dish of deionized water and allow the gel to soak for 10-15 minutes. Using a spatula to hold the gel in place, pour out the water and add enough of protein stain so the gel is completely covered and floating in the stain). This solution precipitates the proteins, "fixing"

them into place so that they cannot diffuse through the gel. In addition, it contains a dye which will stain the proteins blue.

16. Allow the gel to soak in staining solution for no more than 5 minutes. The gel should be uniformly blue, but the bands will not yet be visible. However, there is no need to stain the gel for a longer period of time.
17. **Wearing gloves**, discard the stain, rinse the gel once with water, and place the gel in water to remove stain which is not bound to proteins. The water should be changed several times during this destaining process. The proteins will remain blue while the background becomes clear. The gel will probably need to destain overnight.

Observations and Analysis

1. Remove your gel from the Tupperware container and put it on a light box or white surface. Draw a picture of your gel and answer the post-lab questions.
2. With your teacher's assistance, photograph your gel. The shutter speed for the camera should be set at 1/125 of a second. Start with an f stop of f 11. If the photograph is too dark change to f 8. If the photograph is too light change to f 16.

Name: _____ Date: _____

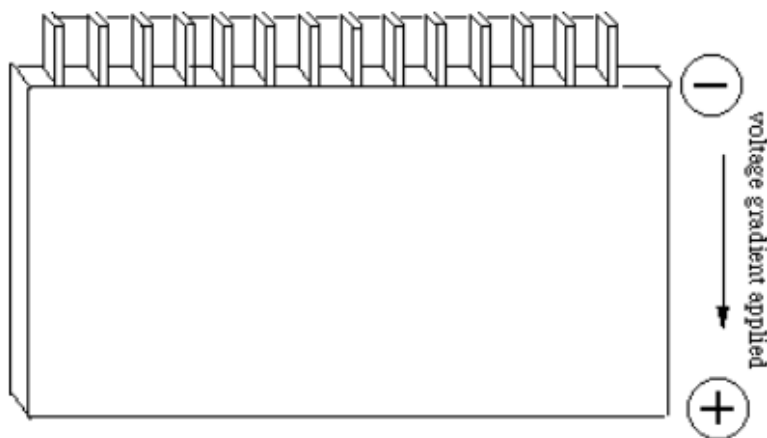
Student Experimental Design Sheet

What question is your group going to investigate?

What protein sources will you use?

How many replicates will you have of each sample?

Label the following diagram with the position of each protein you will use. Only 12 wells will fit on the photograph. Your 6th and 12th lanes should be the molecular weight standard. This will prevent confusion if the gel becomes flipped over.



How will you analyze your results?

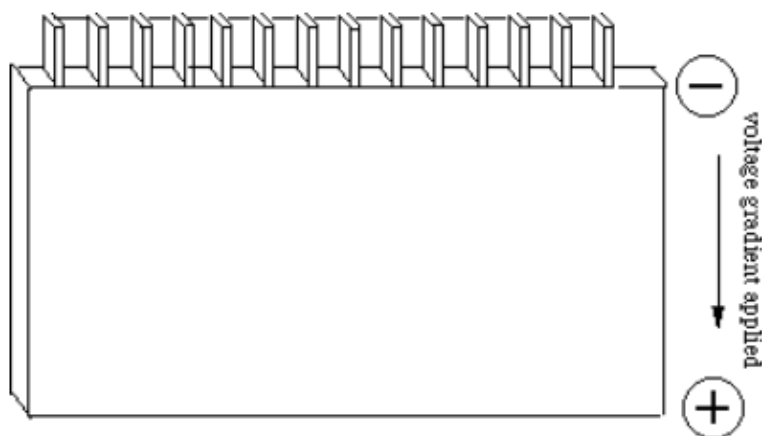
Name: _____ Date: _____

Lab Report Sheet

Pre-Lab Questions

Write all of your answers on a separate page.

1. Why do SDS-coated polypeptides separate when placed in an electrical field?
- 2a. On the diagram below, sketch the predicted results if lane #1 was loaded with a sample composed of 4 different types of polypeptides (having 100 amino acids, 175 amino acids, 250 amino acids, 300 amino acids) while lane #2 was loaded with a sample composed of 3 types of polypeptides (200 amino acids, 250 amino acids, and 350 amino acids) and lane 3 was loaded with a sample of 5 different types (100 amino acids, 150 amino acids, 175 amino acids, 250 amino acids, 300 amino acids).



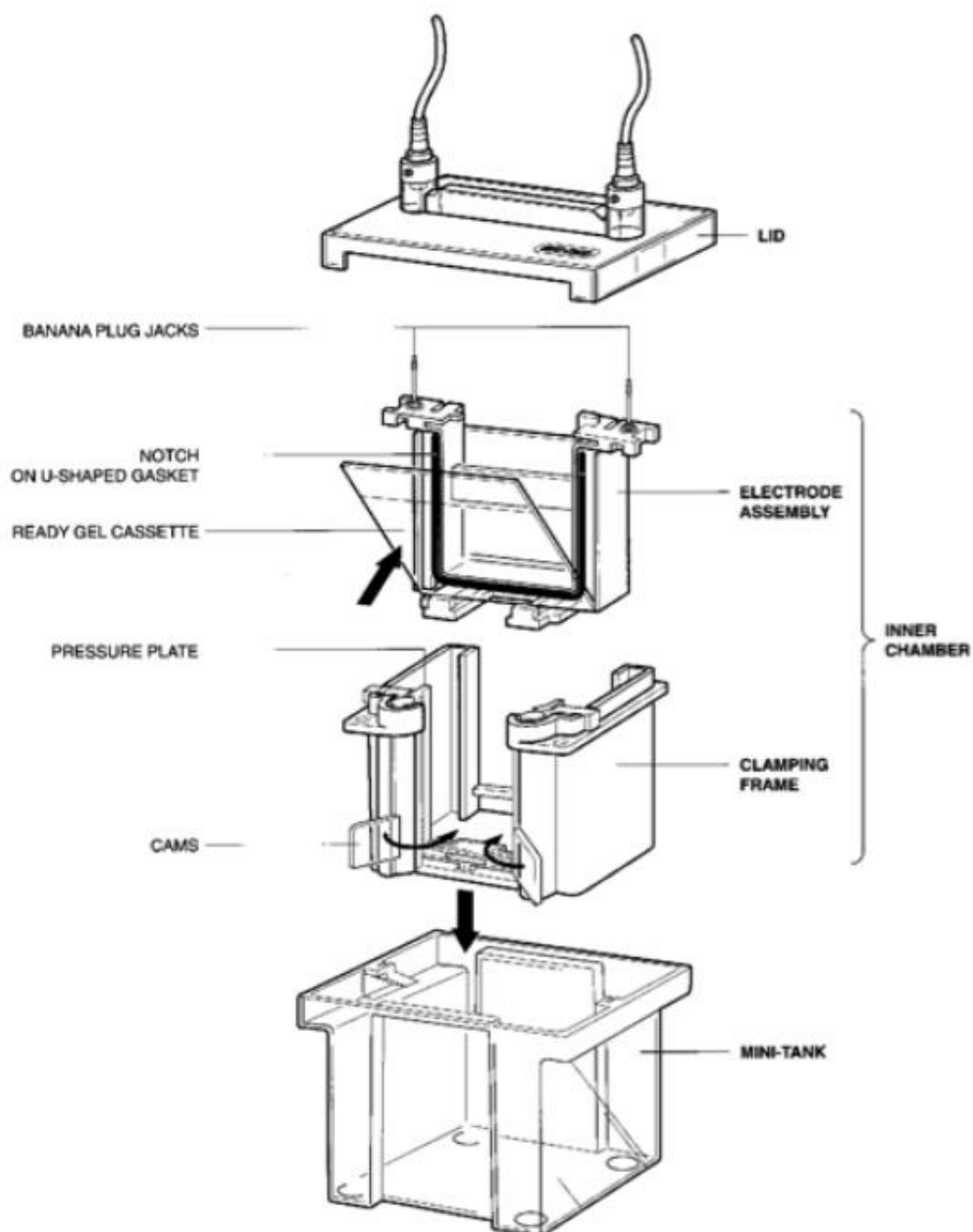
- b. If each lane represents the proteins from a different species of animal, which two would you say are the most closely related? Why?
3. If two proteins have the same molecular weight are they necessarily the same protein? Explain.

Name: _____ Date: _____

Lab Report Sheet

Post-Lab Questions

1. Restate the question your group was asking.
2. Describe or sketch the appearance of the gel initially.
3. Describe or sketch the appearance of the gel after 5 minutes. Explain the difference between the “just loaded” gel and the gel after 5 minutes.
4. After the staining and destaining procedure is done, observe the bands on the gel and make a sketch of your gel. Be sure to indicate the positive and negative sides of the gel and label each lane, indicating which sample was loaded into it.
- 5a. Based on your analysis of the gel, which organisms are closely related?
- b. Which organisms are distantly related? How do you know?
6. Discuss the results of your experiment in light of the question you asked.
7. (optional) It is possible to determine the size of individual proteins by comparing their distance migrated on a gel to the distances moved by proteins of known size. The distance migrated is inversely related to the log of the size.
 - a. From your gel photo (or from the gel itself) measure the distance from the well to each of the bands in the lane corresponding to molecular weight markers.
 - b. Obtain a piece of 2 cycle semi-log paper. For each band, plot the distance migrated from the well on the linear axis and the size (molecular weight) on the log scale. Connect the points with a smooth curve. This is a standard curve and can be used to determine the size of any of the proteins separated from your muscle tissues. Conditions vary from gel to gel, this is why it is very important to include molecular weight markers on every gel that you run. You cannot use a set of molecular weight markers from another gel in your class to determine the size of proteins on your gel. Determine the size of several of the most abundant proteins in each of your samples.
8. An amino acid is approximately 120 daltons. Use this information to calculate the approximate number of amino acids in the proteins you sized in part b.



Gel Box Assembly

From Bio Rad's Ready Gel Cell Instruction Manual, p.4.

Link Here: [Microsoft Word - Protein Gel Electrophoresis.doc \(cpb-us-e1.wpmucdn.com\)](https://cpb-us-e1.wpmucdn.com/cpb-us-e1.wpmucdn.com/microsoft-word-protein-gel-electrophoresis.doc/cpb-us-e1.wpmucdn.com/)

Topic 3: Protein Purification

3.1 What is Protein Purification?

A protein is a polymer of amino acids organized in the form of polypeptide chains in biochemistry. Proteins are essential for the structure and function of cells.

However, before studying proteins, it is necessary to first identify and isolate them. Biochemists use a technique called as **High-Throughput Protein Purification** to do this. Individual proteins are isolated first, then classified according to their molecular weight, solubility, charge, and particular binding affinity. Researchers can use this method to examine individual proteins as well as study protein characterization and relationships.

Protein purification is a process that involves a number of stages to extract and study a desired protein from a complicated mixture. The following are some of the benefits of isolating a protein:

- Size and structure
- Binding affinity
- Biological activity
- Physio-chemical properties
- Protein-protein interactions

Many experimental applications exist for this procedure, particularly in the large-scale manufacturing of recombinant proteins. (However, native protein purification is frequently difficult.) Polyhistidine, Glutathione-S-transferase, and His-tags are employed to isolate and immobilize them in certain circumstances.)

3.2 How does Protein Purification Work?

Protein purification is required in order to isolate a protein successfully. The process's general steps are listed below.³

- **Cell lysis** - Cell lysis is the process of disrupting cells and denaturing proteins utilizing a variety of enzymatic, chemical (such as protease inhibitors and detergents) or physical means (e.g. sonication).
- As a result, both soluble and insoluble proteins, nucleic acids, cell debris, organelles, and membrane remnants will be present in the cell lysate.
- **Clarification** - On the other hand, the goal is to acquire just pure proteins. To do so, cells must undergo a purification procedure to remove any contaminants. This is accomplished through filtration or centrifugation. The protein extract can then be concentrated once this is done.
- **Protein binding** - The protein must then be isolated from the extract. This period is also known as the recuperation stage. The protein is separated here by attaching the protein molecule to a matrix using techniques such as affinity chromatography and magnetic bead separation.
- **Elution** - Elution is the process of washing one item that has been adsorbed to another to extract it. All non-specific binding on the column is washed away with a suitable elution buffer. The number and length of washes will be determined by the column type. Changing the pH

³ [What is Protein Purification? Definition, Methods, & More for 2021 \(excedr.com\)](https://www.excedr.com/what-is-protein-purification-definition-methods-more-for-2021/)

of the column causes the charged functional groups of proteins to neutralize, allowing the required protein to be extracted. An elution buffer and reagents like imidazole (used in His-tagged proteins) are utilized in the procedure, which successfully elute the target protein at high concentrations. This method aids researchers in obtaining high protein concentrations for future study.

3.3 Protein Purification Methods and Strategies

Below are different methods of carrying out the purification of proteins.

3.3.1 Column Chromatography

Column chromatography is a typical approach for protein purification that separates proteins in a mixture by partitioning them between a mobile aqueous solution and an immobile solid matrix. The protein mixture solution is percolated via a column that contains an immobile solid matrix made up of hundreds of small beads.

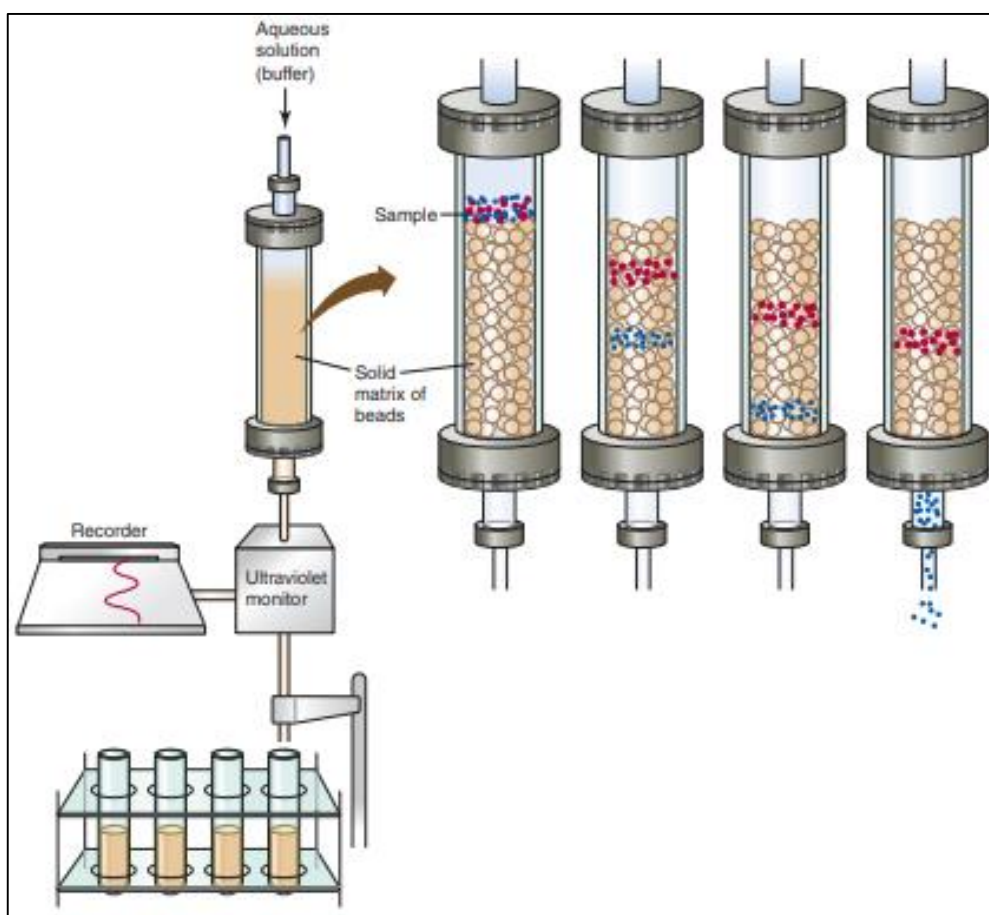


Figure SC-BT-TRB- 22: Schematic column chromatography.

Proteins interact with the immobile matrix and are slowed as the solution goes through the column, as described below. Proteins with differing migration speeds can be separated if the column is long enough. An UV monitor can detect proteins released from the column, which are subsequently collected in tubes by a fraction collector.

Protein separation can be increased by using a longer column or finer beads to increase the surface area of the solid matrix. However, all strategies for increasing the surface area reduce

the flow of solvent through the column. The first attempt at increasing the flow rate failed because the pressure squashed the soft beads, preventing solvent flow.

The flow rate issue was handled by inventing hard chromatographic beads that allow the aqueous solution to be driven through the column under high pressure, reducing separation time and boosting resolution. **High Performance Liquid Chromatography (HPLC)**, a modified method, is currently frequently employed in protein purification. A protein's migration through the solid matrix can be slowed by a variety of favorable interactions.

3.3.2 Ion-exchange Chromatography

Ion-exchange Chromatography uses electrostatic interactions between the protein and the solid matrix to fractionate proteins (*Figure SC- BT-TRB-23*).

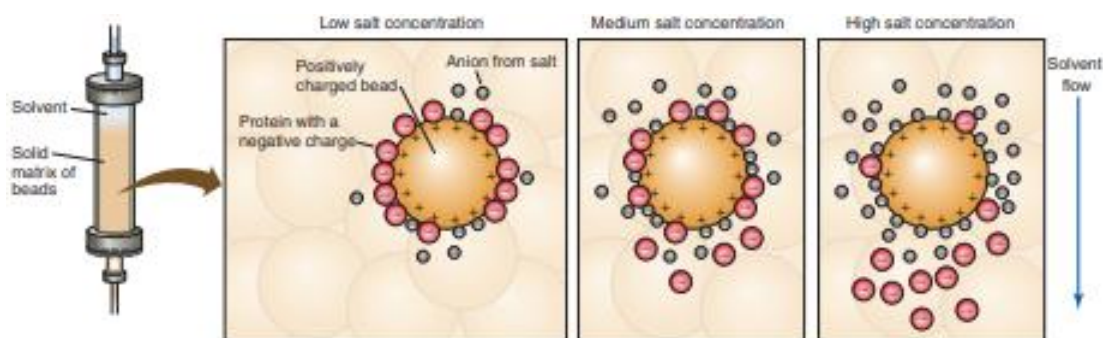


Figure SC-BT-TRB- 23: Anion Exchange Chromatography. Negatively charged groups on the proteins bind to positively charged groups on the anion exchange resin. Increasing salt concentrations produce anions that displace the proteins. Cation exchange resins work in a similar way except in this case, positively charged groups on the proteins bind to the negatively charged groups on the resin and cations displace the proteins.

A sample containing a mix of proteins is allowed to percolate through an immobile matrix, such as polysaccharide beads covered with positively (or negatively) charged groups, in a column. The charged groups on the beads interact with the protein's charged amino acid side chains.

Negatively charged aspartate, glutamate, and carboxyl terminal residues interact with positively charged resins at pH 7. (anion-exchange chromatography). Positively charged lysine, arginine, and amino terminal residues will interact with negatively charged resins (Cation exchange resins).

Aqueous solutions with gradually higher salt concentrations are passed through the column to release proteins. The charged side chains are displaced from the ion exchange beads by the salt ions. Proteins with the weakest interactions with the column will migrate across the column more quickly. Because proteins have both positively and negatively charged side chains on their surfaces, anion and cation exchange chromatography can be used to separate them.

3.3.3 Reverse Phase Chromatography

To prevent protein migration, **Reverse Phase Chromatography** (also known as **Hydrophobic Chromatography**) relies on weak attractive interactions between nonpolar amino acid side chains and nonpolar groups attached to polysaccharide beads, such as phenyl or octyl groups. (See Figure SC- BT-TRB-24.)

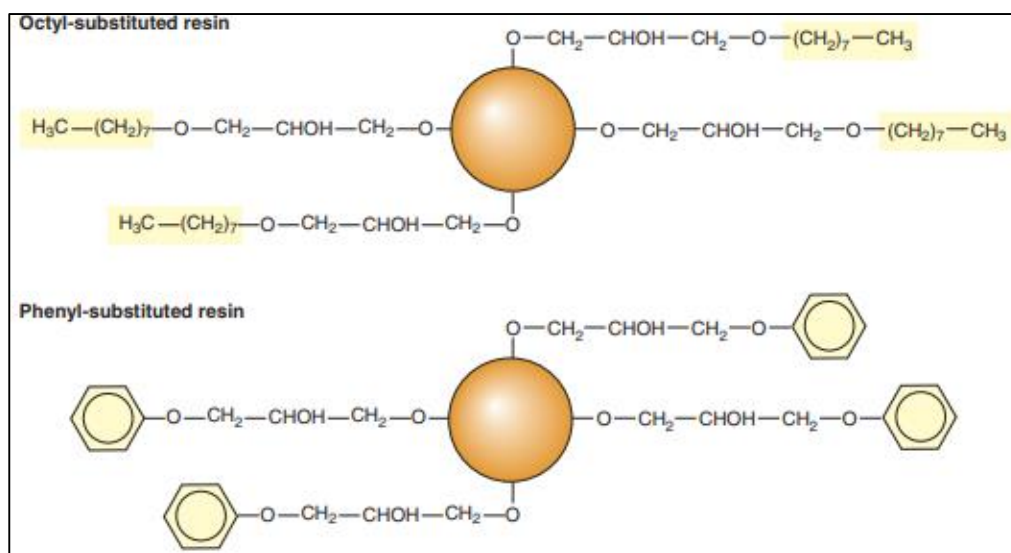


Figure SC-BT-TRB- 24: Resins used for reverse phase chromatography

Proteins are applied to the column filled with reverse phase chromatography beads after being dissolved in an aqueous solution with a high salt content. When the salt concentration is high, proteins cling to the beads more strongly. The quantity and location of nonpolar amino acid residues in a protein determine how well it interacts with reverse phase resin. Aqueous solutions with progressively reduced salt concentrations are passed through the column to release proteins. The proteins with the lowest affinity for the beads will be the first to be released. In reverse phase chromatography, the conditions for binding and eluting are thus the polar opposite of those utilized in ion exchange chromatography.

3.3.4 Gel Filtration

Molecular Exclusion or **Gel Filtration Chromatography** is a technique for separating molecules based on their size. This approach relies on unique beads that allow small proteins to access their core while keeping large proteins out of these areas (See Figure SC- BT-TRB-25).

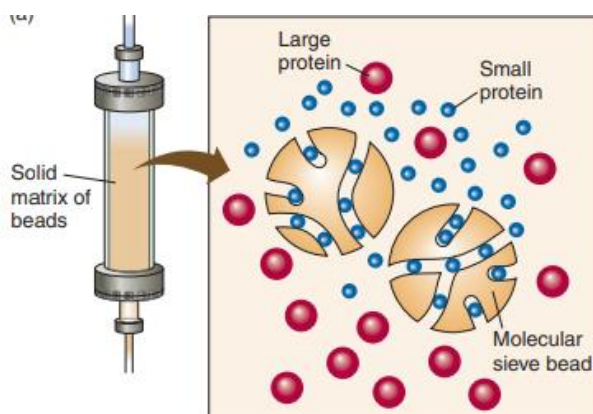


Figure SC-BT-TRB- 25: Gel Filtration Chromatography. The gel filtration column has an internal compartment consisting of the aqueous solutions inside specially designed beads and an external compartment consisting of the aqueous solutions outside the beads. The beads permit small proteins to penetrate into their matrix while excluding large proteins from the region; the large proteins can only get into the external compartment.

A gel filtration column has two different water compartments:

1. Internal Compartment – consist of the aqueous solution inside the beads
2. External Compartment – consist of the aqueous solution outside the beads.

Large protein molecules have access to only the exterior compartment, while small protein molecules have access to both. As a result, large proteins appear in fractions earlier than small proteins.

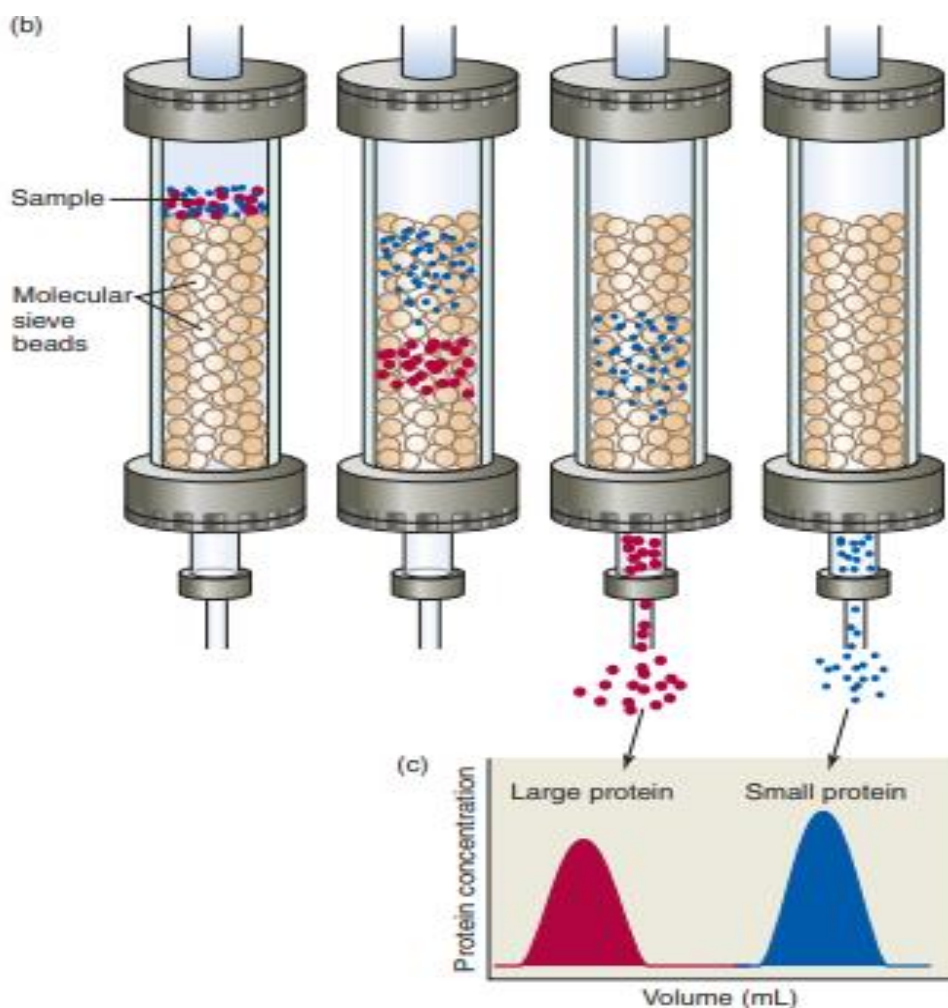


Figure SC-BT-TRB- 26: Gel Filtration Chromatography. (b) Proteins are separated by size with the larger proteins appearing in earlier fractions and smaller proteins in later fractions. (c) Protein elution profile.

3.3.5 Affinity Chromatography

The fact that many proteins can bind particular tiny molecules known as Ligands is used in affinity chromatography. To purify the protein, affinity chromatography takes advantage of this selectivity. Affinity beads are made by attaching ligands to small beads, which are then suspended in an aqueous buffer and put into a column. The deoxythymidylic acid derivative displayed (Figure SC-BT-TRB-27) was coupled to an insoluble polysaccharide to create affinity beads that bind nucleases in one experiment (enzymes that catalyze nucleic acid hydrolysis.)

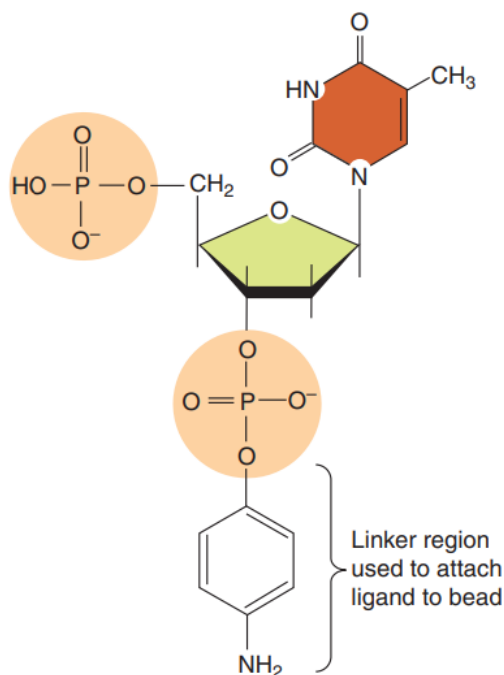


Figure SC-BT-TRB- 27: **Affinity Chromatography.** Affinity chromatography exploits a protein's ability to bind to ligands. Nucleases bind to deoxyribonucleotide derivatives such as the one shown. Affinity resins are prepared by attaching the ligand to tiny water-insoluble beads.

Aqueous solution containing a mix of bacterial proteins was passed through an affinity bead column. Because the majority of proteins were unable to attach to the ligand, they passed through the column. The nuclease, on the other hand, binds to the ligand and is captured by the column. Active nuclease was recovered by washing the beads in a buffer solution with a low pH. Enzymes can also be removed from a column by washing it with a free ligand solution.

Because of the selectivity of the binding process, affinity chromatography can give a high degree of protein purification. If the DNA that codes for a protein has been isolated, a variation of affinity chromatography can be used to purify practically any protein using recombinant DNA technology. The basic idea is to change the DNA so that the gene of interest now codes for a protein with six more histidine residues at the amino or carboxyl end. Specific resins with a high affinity for proteins having a (His)₆ sequence have been developed. The recombinant protein can be isolated from the other proteins in the mixture because only the protein encoded by the modified DNA has a (His)₆ sequence. Although the inclusion of a (His)₆ tag normally has no effect on a protein's biological properties, it must be tested to be sure.

3.3.6 Proteins and Other Charged Biological Polymers Migrate in an Electric Field

Multiply charged macromolecules such as proteins and nucleic acids migrate through a medium in response to an electric field (Figure SC- BT-TRB-28).

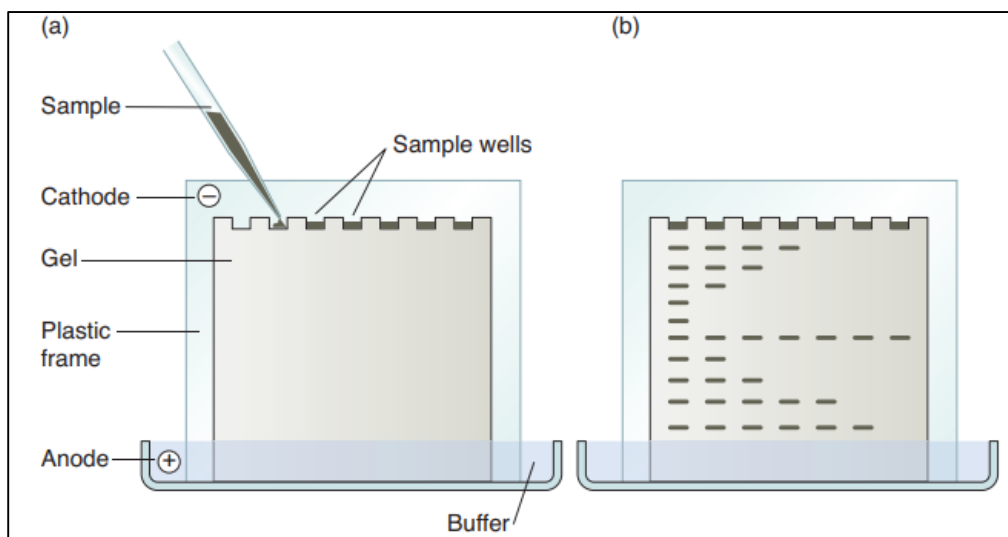


Figure SC-BT-TRB- 28: Gel Electrophoresis. Samples are applied to slots in a porous matrix such as polyacrylamide gel, and an electric field is generated. The buffer pH is adjusted to ensure that the proteins have a net negative charge. Proteins that migrate through the gel at the fastest rate tend to have the greatest net negative charge, the most compact shape, and the smallest size.

A porous matrix, such as a polyacrylamide gel saturated with buffer solution, is used in **Protein Electrophoresis**. The protein sample is put to one end of the gel, and the electric field is created by connecting a power supply to either end of the gel's electrodes. Proteins with the highest net charge, the most compact structure, and the smallest size move through the gel at the fastest rate. Changing the pH of the medium can change the net charge. **Isoelectric pH** refers to the pH at which a protein has no net charge and hence will not migrate in an electric field. Protein bands can be seen by staining them with dyes. A contaminant-free protein will appear as a single band. As a result, electrophoresis is a particularly helpful tool for checking protein purity.

SAMPLE TOPIC ACTIVITIES

Activity 1: Test Exercises

- 1. A purified protein sample contains 10 μg of protein and has an enzyme activity of 1 m mole of ATP synthesized/sec (1 unit). What is the specific activity of the final purified sample?**
 - a) 1,000 units/mg
 - b) 10,000 units/mg.
 - c) 100,000 units/mg
 - d) 1,000,000 units/mg
- 2. Protein's separation can be carried out on the basis of**
 - a) net charge
 - b) solubility in salt solutions
 - c) size or mass
 - d) all of these
- 3. The best way to determine the location of protein in the purification scheme is to measure the**
 - a) rate of ATP synthesis
 - b) changes in the refractive index
 - c) UV absorption
 - d) mass spectroscopy of the protein
- 4. During successful purification scheme, this may be expected that the**
 - a) specific activity increases
 - b) specific activity decreases
 - c) number of proteins in the sample decreases
 - d) both (1) and (3)
- 5. In ion-exchange chromatography**
 - a) proteins are separated on the basis of their net charge
 - b) proteins are separated on the basis of their size
 - c) proteins are separated on the basis of their shape
 - d) either (2) or (3)
- 6. Which of the following may be added to stabilize the protein after yeast cells disruption?**
 - a) NaCl
 - b) Protease inhibitor
 - c) AMP
 - d) All of these
- 7. Gel-filtration chromatography separates on the basis of**
 - a) size and shape using porous beads packed in a column
 - b) size using porous beads packed in a column
 - c) shape using porous beads packed in a column
 - d) none of the above
- 8. Affinity chromatography deals with the**
 - a) specific binding of a protein constituents for another molecule
 - b) protein – protein interaction
 - c) protein – carbohydrate interaction
 - d) none of the above

SAMPLE VIDEO LINKS

Below are some sample videos links that you may use to understand the content.

1. Protein Purification (Link: <https://youtu.be/PVvpEKeOzEM>)
2. Chromatography (Link: <https://youtu.be/ByJ6lzD2Vbq>)
3. Column Chromatography (Link: <https://youtu.be/yiq3QCfBTzc>)
4. Ion-Exchange Chromatography (Link: <https://youtu.be/VOSkyj1dtbc>)
5. Reverse Phase Chromatography (Link: <https://youtu.be/9NFTqUgNU5U>)
6. Gel Filtration (Link: <https://youtu.be/ZA6rgm8wnXE>)
7. Affinity Chromatography (Link: https://youtu.be/8_7cdfNO7OY)
8. Protein Electrophoresis (Link: https://youtu.be/io_hbMQ6kdE)
9. Analysis of Protein Purification (Link 1: <https://youtu.be/7h0XrF1BleM>; Link 2: <https://youtu.be/95PAEel-Fy8>)

Topic 4: Protein Sequencing

4.1 What is Protein Sequencing?

To understand what protein sequencing is, we need to know more about what proteins are.

Amino acids are the building blocks of proteins. The primary structure of the protein is formed by stringing together these amino acids in a specific order. Higher stages of folding transform this primary structure into secondary, tertiary, and higher levels of structure. As a result, the first step in deciphering the structure of a protein is to comprehend its fundamental structure, which is the sequence of amino acids. As a result, the procedure is known as protein sequencing.

4.2 Recall: How do Amino Acids Form a Protein?

Amino acids are frequently linked together in polypeptides, which are lengthy chains of amino acids. This could be comparable to a string of brightly colored beads, each color representing a distinct amino acid.

All of the amino acids in the middle are joined together by one amino acid on each side. The front and back amino acids, on the other hand, are free because they do not form a loop. In scientific terms, these free amino acids are referred to as N-terminal and C-terminal amino acids. Each of these has one free end. When it comes to sequencing, this is extremely useful.

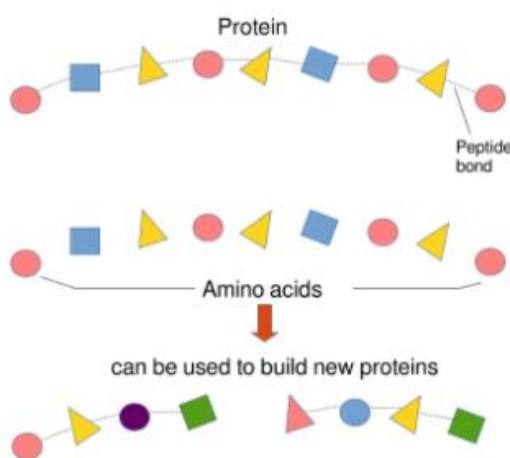


Figure SC-BT-TRB- 29: Amino acids and proteins illustration with beads

4.3 Sanger's Method: Finding the Terminals

Frederick Sanger is a well-known biochemist who won two Nobel Prizes for his contributions to the field. In the late 1940s and early 1950s, his work on protein sequencing was groundbreaking.

Protein structure was unknown to the scientific community at the time, as we now know. There was no known method for reliably identifying the proteins' ends ("terminals") – sequencing was a pipe dream.

Sanger, along with other scientists, studied how amino acids may build proteins under Albert Chibnall's supervision. Insulin, primarily bovine (cow) insulin, was the protein of interest to him.

In the late 1940s, his early research focused on discovering compounds that could aid in the identification of proteins' N-terminals. He discovered that FDNB (fluorodinitrobenzene) could adhere to the N-terminals of proteins and thereby identify them after several failed searches.

The entire approach involves the use of FDNB, followed by the breaking of the polypeptide and the identification of the pieces using separation methods such as chromatography. Because of the method's success, Sanger was able to focus his efforts on deciphering the protein's full sequence.

2.3.1 Unravelling the Protein

Sanger realized that some enzymes could be used to break the protein only at specific places. Let us use the string of **beads analogy** to understand this.

Consider that beads of five colors – red (R), blue (B), green (G), yellow (Y), and pink (P) – form the string. Let us represent the string as **GRPBGYBPBRPBGBBRYP**.

Sanger proceeded to cut the protein at different places using different enzymes. As each enzyme had different spots to cut, this gave various fragments.

Using the beads analogy again, consider enzyme 1 cuts the string only between R and P in that order.

This would give the pieces:

GR PGYPBR PBGBBRYP

Now suppose enzyme 2 cuts the string only between P and B in that order. This gives:

GRPGYP BRP BGBBRYP

These steps are done for many enzymes, so different kinds of pieces are obtained. Now, the pieces are compared using chromatography. As the ends (terminals) are already identified using the previous method, one can find what the overlapping sections are.

For example, in the above case, **PGYPBR** and **BRP** are overlapping sections from different enzyme cuts.

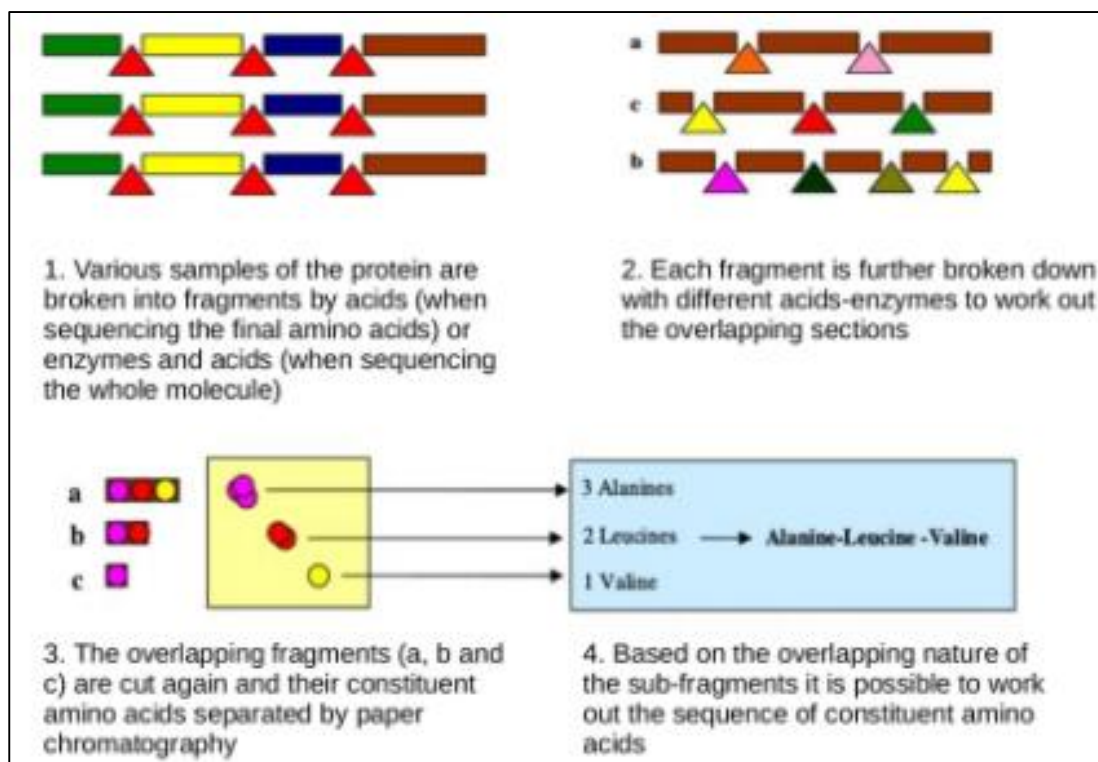


Figure SC-BT-TRB- 30: Sanger's Degradation Procedure for Sequencing Insulin

Each overlapping piece is studied further by breaking it down and doing further chromatography. The information gathered can then be utilized to reconstruct each piece's structure.

Finally, knowing which piece overlaps with which aids in determining the parts' overall order. As a result, we can determine the complete protein sequence.

In the early 1950s, Sanger utilized this method to successfully extract the sequence of bovine insulin.

4.4 Arriving at Different Solutions: Edman Degradation

Pehr Edman was a biochemist from Sweden. In the same era (early to late 1950s), he devised a method for sequencing proteins. Chemicals were utilized instead of enzymes in this process.

His approach consisted of merely attaching a **chemical to the protein's N-terminal amino acid**. Then the right circumstances were created so that only one terminal amino acid was detached, leaving the rest of the protein intact. This amino acid has now been found independently.

Meanwhile, to replace the amino acid that was removed, a new N-terminal has been added to the newly reduced protein. The procedure was repeated, but this time each new N-terminal amino acid was removed one at a time. As a result, the protein was sequenced one amino acid at a time.

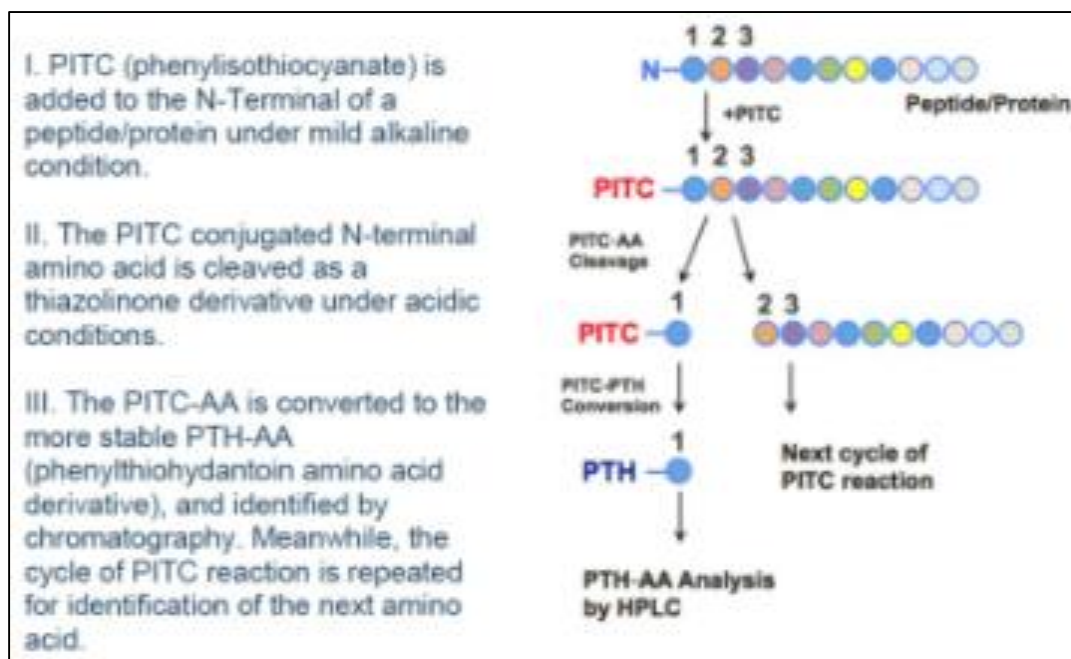


Figure SC-BT-TRB- 31: Sanger Method Sample Illustration

While Sanger's method involved using enzymes to break the protein into various parts and then deducing the original sequence, Edman's method was more straightforward. There were no multiple methods involved; instead, there was just one step that was repeated over and over. Because the technique included starting at one end and breaking the protein step by step, it also gave the sequence directly.

Because of this efficiency, scientists like Stanford Moore and William Stein developed **automatic protein sequencers** (also known as **sequenators**) that could perform this procedure automatically.

One disadvantage of this approach is that it is unreliable for polypeptides (protein fragments) longer than 50 amino acids. As a result, large proteins are frequently broken down into tiny bits first (as in Sanger's approach) and subsequently degraded using Edman's method.

4.5 More Advanced Methods

2.5.1 Mass Spectrometry

Protein sequencing became a reality thanks to these early technologies. They allowed researchers to investigate the primary structures of proteins of various sizes. New strategies for dealing with bigger proteins appeared as technology and scientific knowledge advanced.

Mass spectrometry is one way. In this procedure, the protein is broken down into small fragments, which are then analyzed using a mass spectrometer. The method is capable of dealing with extremely big proteins while also providing speedy and precise findings. This has recently been paired with the usage of the internet to provide massive databases of protein sequences that allow us to quickly arrive at the correct sequences.

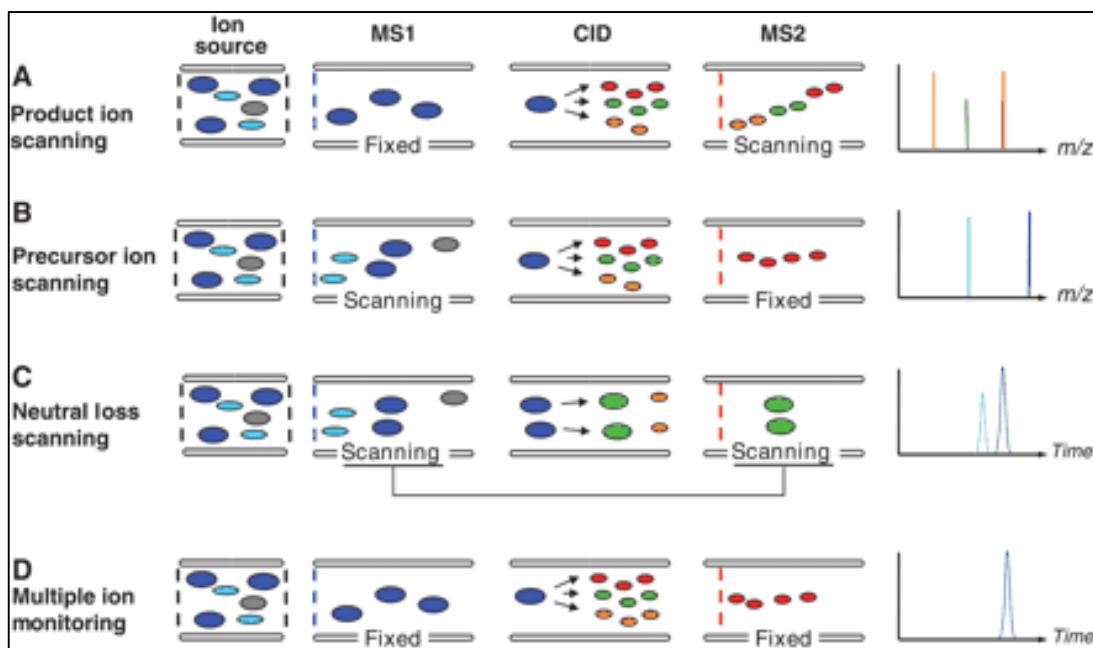


Figure SC-BT-TRB- 32: Mass Spectrometry sample illustration

The **relationship between DNA/RNA** and proteins is another technique. DNA directs the development of RNA, which then gives rise to proteins, according to a fundamental principle of molecular biology. We can predict the output protein and so sequence it if we know the DNA sequence and a few other parameters.

This process frequently entails doing Edman degradation on a portion of the protein, identifying the sequence, and then comparing it to DNA to determine the rest of the sequence. Sanger was also responsible for the development of the first DNA sequencing methods.

Protein sequencing is an important tool for biochemists since it allows them to study the protein's fundamental structure. This prompts them to investigate the protein's higher-level structure and function, which is an attractive topic of study.⁴

2.5.2 Dansyl Chloride Method

Your protein's N-terminal analysis will reveal the distinct types of subunits. Because each polypeptide chain has an N-terminus (also known as a protein's "end group"), a fluorescent chemical like 5-dimethylamino-1-naphthalenesulfonyl chloride (also known as "dansyl chloride") can be used to cleave to the main amine (N-terminal of a protein is the amine group).

So, What Does Dansyl Chloride Do?

Dansyl chloride fluoresces an amino acid by cleaving or attaching itself to the n-terminus. The dansylated amino acid is subsequently separated from the free amino acids using acid hydrolysis, as demonstrated below:

⁴ [Protein Sequencing Explained | AtomsTalk](#)

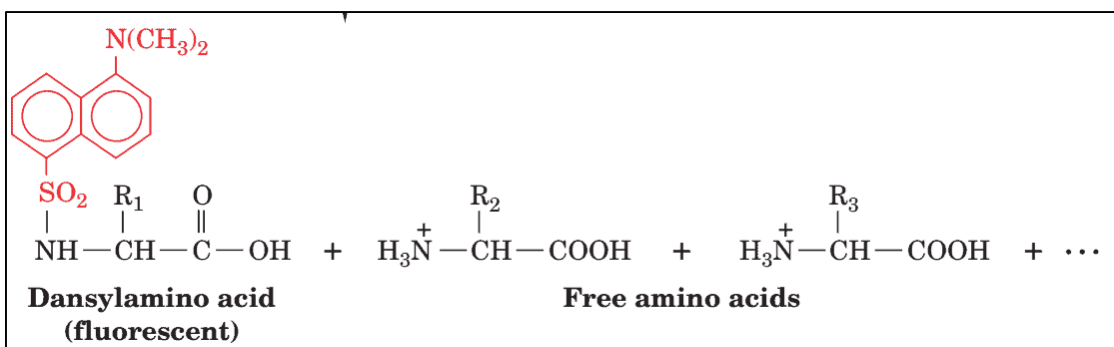


Figure SC-BT-TRB- 33: Acid Hydrolysis of Dansylamino Acid

Chromatography is used to compare the fluoresced amino acid to a known standard to determine the amino acid.

SAMPLE TOPIC ACTIVITIES

Activity 1: Exercise

1. During denaturation, proteins will not lose their structure, with regard to:
 - E. Primary structure
 - F. Secondary structure
 - G. Tertiary structure
 - H. Quaternary structure
2. The characteristic features of the peptide bond include all the following, except:
 - E. Does not allow freedom of rotation
 - F. It is a partial double bond
 - G. Always has cis configuration
 - H. Absorbs UV light at 280 nm
3. The force maintaining the primary structure of a protein:
 - E. Peptide bonds
 - F. Hydrophobic forces
 - G. Hydrogen bonds
 - H. Electrostatic (ionic) bonds
4. The forces maintaining the secondary, tertiary and quaternary structures of a protein are the following, except:
 - E. Electrostatic (ionic) bonds
 - F. Hydrophobic forces
 - G. Van der Waals forces
 - H. Peptide bonds
5. The amino acid which did not allow formation of alpha-helix is:
 - E. Glutamate
 - F. Proline
 - G. Tyrosine
 - H. Histidine
6. Tertiary structure of a protein describes:
 - E. The sequence of amino acids
 - F. Location of disulphide bonds
 - G. Amino terminal end amino acid
 - H. The nature of protein folding
7. One of the following proteins does not have a quaternary structure:
 - E. Albumin
 - F. Hemoglobin
 - G. Lactate dehydrogenase
 - H. Immunoglobulin G
8. All the following reagents are used for identifying the first amino acid in a protein, except:
 - E. Cyanogen bromide
 - F. Fluorodinitrobenzene
 - G. Dansyl chloride
 - H. Phenyl isothiocyanate

- 9.** Proteins can be precipitated by the following methods, except:
- E. Adding alcohol and acetone
 - F. Saturating with ammonium sulphate
 - G. Using salts of heavy metals
 - H. Shifting the pH away from the iso electric point
- 10.** Denatured proteins:
- E. Are soluble
 - F. Are difficult to digest
 - G. Are biologically inactive
 - H. Peptide bonds are broken
- 11.** Which of the following is a simple protein?
- E. Casein
 - F. Insulin
 - G. Hemoglobin
 - H. Tyrosinase
- 12.** In glycoproteins, the carbohydrate chains are combined through glycosidic linkages with:
- E. Hydroxyl groups of serine or threonine residues of proteins
 - F. Epsilon amino nitrogen of lysine residues of proteins
 - G. Guanidium group of arginine residues of proteins
 - H. Phenol group of tyrosine residues of proteins
- 13.** The protein which does not answer the aldehyde test is:
- E. Hemoglobin
 - F. Albumin
 - G. Casein
 - H. Gelatin
- 14.** Proteins may be estimated by the following methods, except:
- E. Biuret method
 - F. Heat coagulation
 - G. Kjeldahl's digestion
 - H. Nephelometry
- 15.** All the following are examples of tertiary structure of proteins, except:
- E. Alpha helix
 - F. Beta pleated sheet
 - G. Triple stranded helix
 - H. Peptide bonds
- 16.** Tertiary structure of a protein describes:
- E. Sequence of amino acids
 - F. Location of disulphide bonds
 - G. Amino terminal end amino acid
 - H. The nature of protein folding
- 17.** Proteins may be denatured irreversibly by:
- E. Adding urea
 - F. Bringing to iso electric pH
 - G. Heat coagulation
 - H. Reduction with mercaptoethanol

- 18.** Lectins are:
- E. Animal proteins having specific amino acid binding site
 - F. Antibody molecules acting against cells
 - G. Plant proteins having specific carbohydrate binding site
 - H. Blood proteins having a lecithin group
- 19.** Ultraviolet light at 280 nm is absorbed by which component of proteins?
- E. Peptide bonds
 - F. Sulfhydryl group of cysteine
 - G. Indole ring of tryptophan
 - H. Imidazole ring of proline
- 20.** The nature of the bond linking amino acids to each other is:
- E. Covalent
 - F. Co-ordinate
 - G. Ionic
 - H. Hydrophobic
- 21.** How many peptide bonds are present in glutathione?
- E. 1
 - F. 2
 - G. 3
 - H. 4
- 22.** Basic difference between two polypeptides is in the —
- E. Structural conformation
 - F. Primary sequence of amino acids
 - G. Number of side chains
 - H. Number of hydrophobic bonds
- 23.** Human insulin differs from bovine insulin in:
- E. Biological activity
 - F. Number of amino acids
 - G. Position of disulfide bonds
 - H. Sequence of amino acids
- 24.** A covalent bond between the alpha carboxyl group of one amino acid and alpha amino group of the neighboring amino acid is called
- E. Cis double bond
 - F. Isopeptide bond
 - G. Pseudopeptide bond
 - H. D. Peptide bond
- 25.** Study of linear sequence of amino acids is done by all techniques listed except:
- E. End group analysis
 - F. Hydrolysis by proteolytic enzymes
 - G. Analyzing the content of each amino acid
 - H. Denaturing the protein
- 26.** Different polypeptide chains are held together by:
- E. Peptide bonds
 - F. Disulphide bonds

- G. Glycosidic bonds
- H. Ester bonds

27. Primary structure decides:

- E. Rate of synthesis of protein
- F. Biological activity of the protein
- G. Rate of degradation of the protein
- H. Effect of proteolytic enzymes on protein

28. Secondary and tertiary levels of protein structure are dependent on:

- E. Presence of disulfide bonds
- F. Primary structure
- G. PH of the medium
- H. PK value of component amino acids

29. The protein having predominantly alpha helical structure is

- E. Collagen
- F. Keratin
- G. Fibroin
- H. Myoglobin

30. Which of the following is NOT true regarding the tertiary structure of proteins?

- E. It is a random coil structure
- F. Disulfide bonds are formed between any two cysteine residues
- G. Position of disulfide bonds are predetermined and fixed
- H. Denaturation using reducing agents does not affect the disulfide bonds

31. Protein having a large number of disulfide bonds is:

- E. collagen
- F. keratin
- G. hemoglobin
- H. albumin

32. Which of the following proteins does not possess a quaternary structure?

- E. Myoglobin
- F. Lactate dehydrogenase
- G. Hemoglobin
- H. Immunoglobulin M

33. Which of the following is NOT true regarding hemoglobin?

- E. Has 4 independent subunits
- F. Each subunit has one heme residue
- G. Each subunit can bind one molecule of oxygen
- H. All four subunits are similar

Activity 2: Self-Test Exercises

1. How proteins are made up of?
2. What is a peptide bond?
3. What is a dipeptide?
4. How many peptide bonds are present in a tripeptide?
5. What is a polypeptide?
6. What are the levels of organizations of proteins?
7. What is meant by primary structure of a protein?
8. What force maintains the primary structure?
9. How are the end amino acids of proteins called?
10. What is a pseudopeptide?
11. Can you give an example of a pseudopeptide?
12. What are the salient structural features of insulin?
13. What is proinsulin?
14. What is a mutation?
15. Can you give an example?
16. What are the forces that maintain the secondary, tertiary and quaternary structures of a protein?
17. What are the salient features of alpha structure of proteins?
18. What is secondary structure of a protein?
19. What is meant by tertiary structure of a protein?
20. What is quaternary structure of a protein?
21. Name proteins having quaternary structure.
22. What are the reagents that are used for identifying the first amino acid in a protein?
23. What is isoelectric point of a protein?
24. What are the features of isoelectric point?
25. What is the isoelectric pH of human albumin?
26. How proteins are precipitated from solution?
27. How is albumin precipitated?
28. What about globulins?
29. Give example of isoelectric precipitation.
30. What is the isoelectric pH of casein?
31. What are the features of denaturation?
32. What is heat coagulation?
33. Give examples of proteins that coagulate.

SAMPLE VIDEO LINKS

Below are some sample videos links that you may use to understand the content.

1. Protein Sequencing Steps (Link 1: <https://youtu.be/qFTDOT9Kojc>; Link 2: https://youtu.be/HM_eBO1QbO0)
2. Sanger's Degradation Method (Link: <https://youtu.be/OeJqXWChfgk>)
3. Edman Degradation Method (Link 1: <https://youtu.be/GUhZoNTCCPo>; Link 2: <https://youtu.be/cRjwWAqzyDY>)
4. Mass Spectrometry (Link: <https://youtu.be/myoIF-h1kKI>)

THEMA: HEALTH AND MEDICINE

Introduction

Medicine is the art, science and practice of caring for a patient and managing the diagnosis, prognosis, prevention, treatment or palliative care of their injury or diseases. Medicine encompasses health care practices evolved to maintain and restore health by the prevention and treatment of illness. The practice of medicine has significantly changed with the advent of molecular biology with its enormous implications in biological sciences (sequencing of human genome), sophisticated new imaging techniques and advances in bioinformatics and information technology. That has sparked an explosion of scientific information that fundamentally influenced and changed the way in which patient care and management has been practiced.

The knowledge gleaned from the science of medicine has already improved and undoubtedly will further improve our understanding of complex disease processes and provide new approaches to disease treatment and prevention. Though the practice of medicine requires a scientific basis, the combination of medical knowledge, intuition, experience and judgment defines the art of medicine.

The study of the human and body, how this function, and how they interact, not only with each other but also with their environment, has been of utmost importance in ensuring human well-being. Research on potential treatments and preventive medicine has expended greatly with the development of modern medicine, and a network of disciplines, including such fields as genetics, psychology, and nutrition, aims to facilitate the betterment of our health.

MODULE 12.1: DRUG DEVELOPMENT PROCESS

4-5 Weeks

Papua New Guinea (PNG) has the potential to solve some of the world's health-related problems regarding therapy and treatments drugs. Why? Because PNG has majorly unexplored rainforests and large biodiversity. Think about it! PNG has a rich tradition of using herbal medicine, and about 50% of our rural population relies on traditional herbal medicine for therapy and treatment. This should imply to us that PNG is a potential source for drug discovery⁵.

We (Students) need to be encouraged to exploit the scientific opportunities that our country has. If we choose to investigate that comparative advantage, we may likely stumble upon what may be new medical technologies or drugs that may not only solve PNG's health-related problems, but the worlds.

Medicinal Chemistry is an interdisciplinary study that combines aspects of chemistry, physical chemistry, pharmacology, microbiology, biochemistry, and computational chemistry. Medicinal Chemistry to the micromolecular level is a very interesting yet complex topic. However, for this Module, the focus will be on the process of discovery and development of drugs at a bird-eye view. By the end of this Module, the students should be sufficiently equipped with the concept of drug discovery, design and development should they choose to conduct further research on drug development as well as pursue a career in the field of pharmaceutical biotechnology.

⁵ [2327-5162.S1.011-006.pdf \(hilarispublisher.com\)](https://www.hilarispublisher.com/p/2327-5162.S1.011-006.pdf)

Topic 1: Introduction to Drugs and their Actions

Key Points

- **Lead compounds** are the substances on which a development is founded, while **analogues** are synthetic compounds created from a lead.

Review

Before we study the main concepts of this topic, let us do a short Review of what we have covered from Grade 11/Thema: Health and Medicine/ Module 11.1: Biochemistry in Medicine:

- Drugs are chemicals that are taken into the body to affect the normal function of the body
- If the therapeutic effect of the drug helps the body, it is medicinal.
- If the therapeutic effect of the drug harms the body, it is poison.
- The US Food and Drug Administration (UFDA) does a lot of certifications for USA but its process of certification is robust and credible that once approved in USA the medicine can be approved for distribution in Papua New Guinea.
- World Health Organization (WHO) is another authority that approves distribution of drugs for global use.
- Drugs are categorized into three main categories based on their Modes of Action (MoA): (1) Chemotherapeutic Agents; (2) Pharmacodynamic Agents; and (3) Miscellaneous Agents.
- Chemotherapeutic Agents are used to cure infectious disease and cancer (e.g., antibiotics)
- Pharmacodynamic Agents are used to treat non-infectious diseases (e.g., Sedatives)
- Drugs are used for diagnosis, prevention, treatment of symptoms and cure.
- Drugs work by binding to the target receptor site.
- A drug receptor is a specialized target macromolecule that binds a drug and mediates its pharmacological action (e.g., enzyme, nucleic acid, etc.)
- Efficacy is the maximal response a drug can produce.
- Potency is a measure of the dose that is required to produce a response.

Understanding how the drug works in the subject's body is a huge part of studying drugs. Therefore, students must conduct review study on Pharmacokinetics and Pharmacodynamics for added background knowledge as they study the process of drug discovery.

1.1 Drugs: What are They?

Drugs are basically chemicals (except food) that are taken into the body to affect the normal function of the body. When entered into the body, they exert a therapeutical effect (known as **Activity**) that may be poisonous (harms the body) or medicinal (helps the body).

One example of drug acting as poison is overdosing, which ultimately leads to discomfort, pain, unconsciousness, and death. Some drugs that are medicinal (helpful) can cause **Side Effects**. For example, Aspirin's therapeutic effect is to relieve Headaches and side effects is stomach discomfort and bleeding.

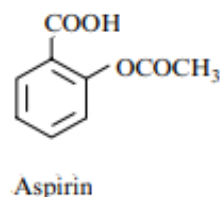


Figure SC-H&M-TRB - 1:
Aspirins and Paracetamol

1.2 Drugs: Why do we need new ones?

We are constantly in need of new drugs because there is always needed improvement in the treatment of existing diseases, treatment of newly discovered diseases, and the manufacture of safer pharmaceuticals by reducing or eliminating negative side effects.

1.3 Drug Discovery and Design, a historical outline

People all across the world have employed a variety of natural ingredients for medical purposes since ancient times. These compounds, which came from animal, vegetable, and mineral sources, could be quite helpful in some cases. Many of the items, however, were quite hazardous. Users didn't have easy access to information about these ancient treatments until the printing press was invented in the 15th century.

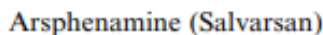
Herbals and pharmacopoeias were widely published and circulated as a result of this development. As a result, the use and misuse of herbal and other medicines has skyrocketed. In the 18th and 19th centuries, however, improved communication amongst practitioners led in the gradual removal of useless or dangerous concoctions from herbals and pharmacopoeias. It also resulted in more reasonable drug development. Initially, natural products isolated from plant and animal material were utilized as the starting point for drug development, but as knowledge grew, a wider spectrum of pharmaceutically active molecules were used as the starting point. **Lead compounds** are the substances on which a development is founded, while **analogues** are synthetic compounds created from a lead.

The discovery of new lead compounds with specialized medical qualities is the focus of the medicinal chemist's job. It entails the creation of more effective and safer analogues from both novel and old lead compounds. This normally entails manufacturing and testing hundreds of chemicals before settling on one that is satisfactory. One out of every 10,000 chemicals produced is thought to be appropriate for medical application.

Paul Ehrlich and Sacachiro Hata were the first to rationally develop synthetic medications, producing the antiprotozoal arsphenamine in 1910 by combining synthesis with reliable biological screening and evaluation processes. Ehrlich realized at the turn of the twentieth century that a drug's helpful and harmful qualities were both significant in determining its effectiveness. He realized that the more effective medications were more selective for the target bacteria than for the host microorganism. As a result, in order to assess the efficacy of different chemicals, he characterized a drug's selectivity, and hence its efficacy, in terms of its chemotherapeutic index, which he described as:

$$\text{Chemotherapeutic Index} = \frac{\text{minimum curative dose}}{\text{maximum tolerated dose}}$$

The chemotherapeutic index of the 600 chemicals Ehrlich and Hata produced was determined and cataloged, allowing them to find arsphenamine in 1909. (Salvarsan). This medicine was extremely poisonous, but it was far less so than Atoxyl, which was widely used at the time. It was used until the mid-1940s, when penicillin took its place.



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STEM GRADE 12 Chemistry Students Resource Book

knowledge. It has allowed medicinal chemists to put groups in the molecular structure of a medication that alter absorption, bio-system stability, distribution, metabolism, and excretion. The addition of a sulphonic acid group to a drug's structure, for example, will increase its water solubility. This may improve absorption and/or excretion rates from the body. However, due to the complexity of biological systems, there is always some ambiguity in estimating the effect of structural changes on a drug's activity. As a result, significant testing is always required to determine the repercussions of altering a structure. Furthermore, modifying or introducing a group can alter the nature of the compound's activity. For example, the change of the ester group in procaine to an amide (procainamide) changes the activity from a local anaesthetic to anti-rhythmic.

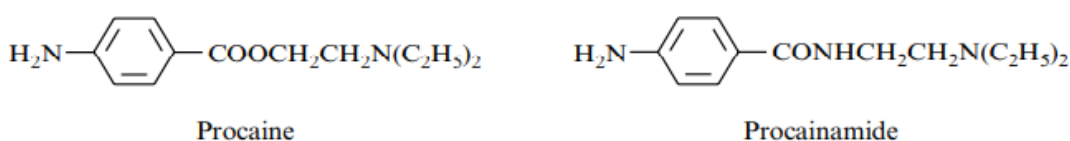
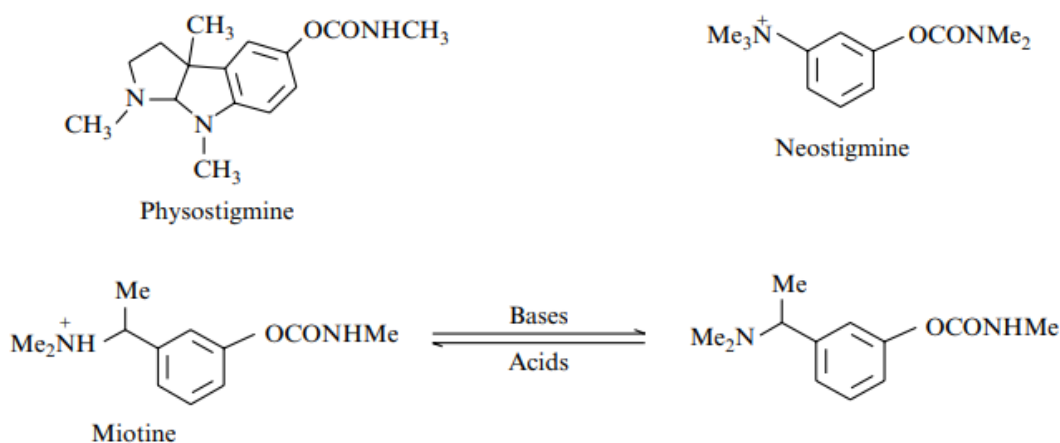


Figure SC-H&M-TRB - 4: Procaine and Procainamide

The addition or removal of charged groups or groups capable of forming ions into or out of a structure can have a significant impact on pharmacological action. This is because most medications must pass through nonpolar lipid membrane barriers to reach their target. When a result, as the drug's polarity grows, it becomes more difficult for it to get through these barriers. In order to restrict the flow of a medication through a membrane, quaternary ammonium salts, which are irreversibly charged, can be utilized as an alternative to an amine in a structure. A quaternary ammonium group in the structure of the anticholinesterase neostigmine, which was derived from physostigmine, prevents the molecule from crossing the blood–brain barrier. This stops unwelcome CNS activity. Its counterpart miotine, on the other hand, can be used as a free basis. As a result, it can pass lipid membranes, potentially causing negative CNS effects.



The development team must choose the suitable beginning point for both SAR and QSAR research. Inevitably, serendipity plays a big role in the selection of that spot. However, modern approaches introduced in the 1970s and 1990s, such as computer modeling and combinatorial chemistry, are expected to diminish the number of intuitive discoveries.

The necessity to synthesize every analogue of a lead chemical has been minimized thanks to computer modeling. It's also frequently utilized in retrospect to verify information gleaned from other sources. Combinatorial chemistry, which began in the discipline of peptide chemistry, has now evolved to encompass a wide range of topics. The term refers to a series of similar approaches for producing huge numbers of chemicals at the same time for biological testing. As a result, it's used to investigate structure-action relationships and find novel lead compounds. It's possible that the operations will be automated.

1.4 Sources of Drugs and Lead Compounds

The discovery of a novel medication is a combination of luck and meticulous research. Drugs and lead compounds originated from natural sources, such as animals, plants, trees, and microorganisms, were the first to be developed. Until the mid-twentieth century, marine sources were rarely used. Natural sources of lead are still essential today, although the majority of lead compounds are manufactured in laboratories. The biochemistry of the pathogenic state is used to determine the nature of these synthetic chemicals at first.

Many modern discoveries begin with pharmacologists conducting biological tests (bioassays or screening programs) on possible sources to establish the nature of their pharmacological action and potencies. These screening programs might be broad or narrow in scope. Random screening procedures test all available drugs and compounds, regardless of their structures. The antibiotics streptomycin and tetracycline, as well as many other lead compounds, were discovered by random screening of soil samples. While random screening is still utilized, more specialized screening approaches that assess certain structural types are becoming more common.

Following the identification of compounds with pharmacological activity of interest through a screening method, the chemical responsible for that activity is separated and employed as a lead compound for the manufacture of related analogues. Further screening procedures are performed on these compounds. The most promising of these molecules are synthesized into analogues, which are then exposed to the screening process. This process of selective screening and analogue synthesis could be done several times before a potentially useful medicine is discovered. Frequently, the sequence must be abandoned because it is either ineffective or too costly.

1.4.1 Natural Sources

Natural sources of lead compounds and novel drugs are still essential. However, because of the vast diversity of potential natural sources around the planet, random screening is a hit-or-miss operation. Ethnopharmacology (the study of local folk remedies) provides the foundation for a more scientific approach. Many important therapeutic compounds have been discovered as a result of this, including the antimalarial quinine from cinchona bark, the cardiac stimulant digitalis from fox gloves (Figure SC-H&M-TRB-5), and the antidepressant reserpine derived from *Rauwolfia serpentina*.

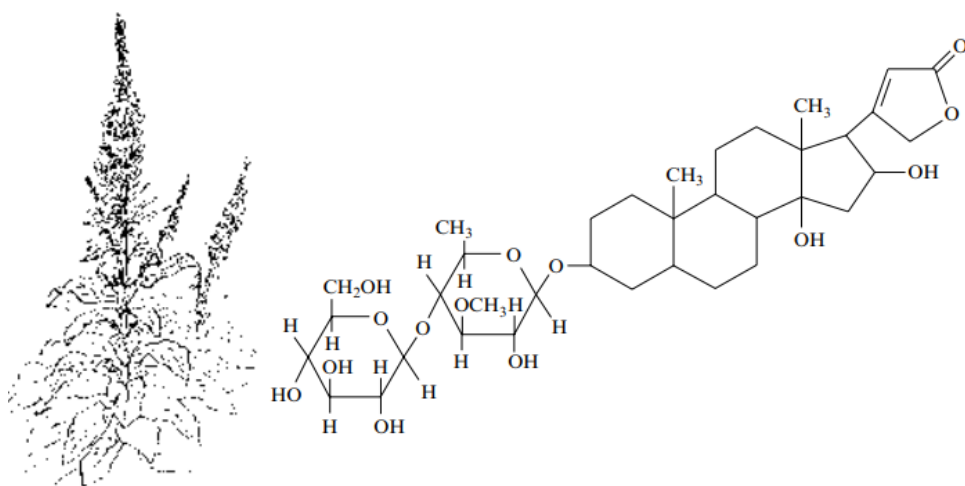


Figure SC-H&M-TRB - 6: Common Foxglove and the structure of digitalis

Once a material containing an active molecule has been identified by screening, the problem of extraction, purification, and pharmacological activity assessment must be addressed. However, isolating effective quantities of a medication from its land or sea sources can have negative environmental consequences. For example, the promising anticancer drug Taxol (Figure SC-H&M-TRB-6) was discovered in the bark of the Pacific yew tree. However, producing substantial amounts of Taxol from this source would result in the complete extinction of the tree, which is an environmentally unsustainable situation. It is critical to conserve the world's plant, shrub, tree, and marine sources from further erosion, as there is no doubt that they will continue to supply beneficial medicinal molecules in the future.

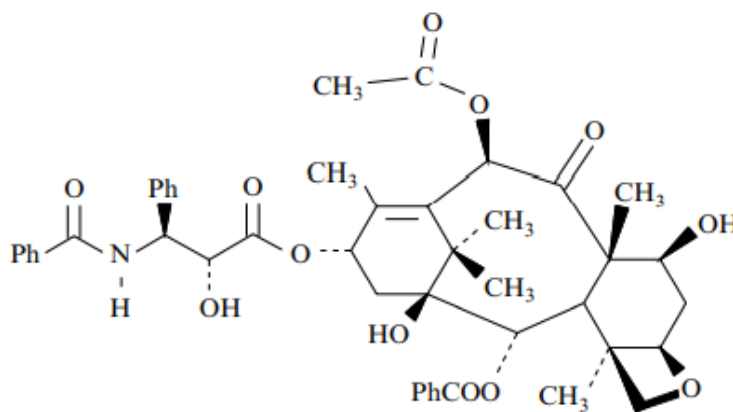


Figure SC-H&M-TRB - 7: Taxol, isolated from the bark of the Pacific yew

1.4.2 Drug Synthesis

Starting with the pathophysiology of the diseased state and determining the point where intervention is most likely to be beneficial is the most common method to drug design by synthesis. This allows the medicinal chemist to make recommendations for potential lead molecules. These substances are created in order to test their pharmacological effects. Once a viable active lead has been identified, structural analogues of that lead are created and screened in the hopes of finally obtaining a molecule appropriate for clinical application. Obviously, this method is time consuming, and the likelihood of a successful outcome is highly dependent on luck. To lessen the element of luck, various modifications to this strategy have been made.

1.4.3 Market forces and 'me-too drugs'

The cost of bringing a new medication to market is quite costly, and it is continuing to rise. Before a new drug is put on the market, it must be absolutely certain that it will be profitable. As a result, rather than ethical or medical concerns, the board of directors' decision to market or not market a drug is largely based on facts provided by the accounting department. Companies can save costs by producing medications with comparable actions and molecular structures as their competitors. These medicines are referred to as 'me-too drugs.' They are useful in that they provide the practitioner with a selection of medications with similar mechanisms of action. This option is useful in a variety of scenarios, such as when a patient has an unpleasant reaction to a prescription drug or when a drug is withdrawn from the market on a rare occurrence.

1.5 Classification of drugs

Drugs are classed in a variety of ways depending on where they are used and how they are utilized. Chemical structure and pharmacological action, which comprises the site of action and target system, are the methodologies that medicinal chemists are most interested in. Unfortunately, classifying medications by their chemical structure type has the drawback of resulting in highly varied forms of pharmacological activity among members of the same structural group. Steroids can serve as hormones (testosterone), diuretics (spironolactone), and antibacterial agents (fusidic acid), among other things. Drugs whose active form is created by enzyme or chemical action at or near their site of action are referred to as prodrugs. Other classifications, such as the type of the ailment and the bodily system on which the medicine operates (physiological classification), are also employed in medicinal chemistry and other domains, depending on the information's intended application.

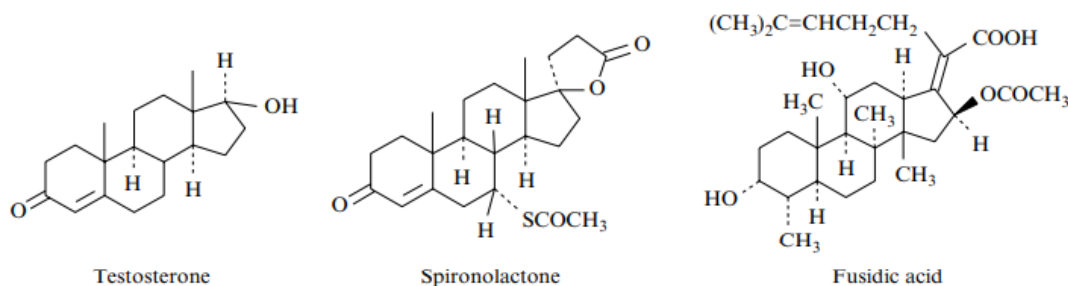


Figure SC-H&M-TRB - 8: Testosterone, Spironolactone, and Fusidic Acid

1.6 Routs of administration (Pharmaceutical Phase

The dose form refers to the physical form in which a medicine is given. The active component is usually combined with other chemicals known as excipients in dosage forms. Fillers (bulk supplying agents), lubricants, binders, preservatives, and antioxidants are all examples of excipients. A change in the excipients' nature can have a big impact on the active ingredient's stability and release from the dosage form. Similarly, changes in the active principle's manufacture, such as using a different solvent for purification, might impact its bioavailability and, as a result, its efficacy as a medicine. This emphasizes the significance of quality control procedures for all pharmaceuticals, particularly those in the manufacturing stage.

The pharmaceutical technologist is responsible for dosage form design, although the medicinal chemist should consider it when generating a medicine from a lead molecule. It's pointless to have a wonder medicine if it can't be packaged in a way that makes it physiologically available and patient-acceptable.

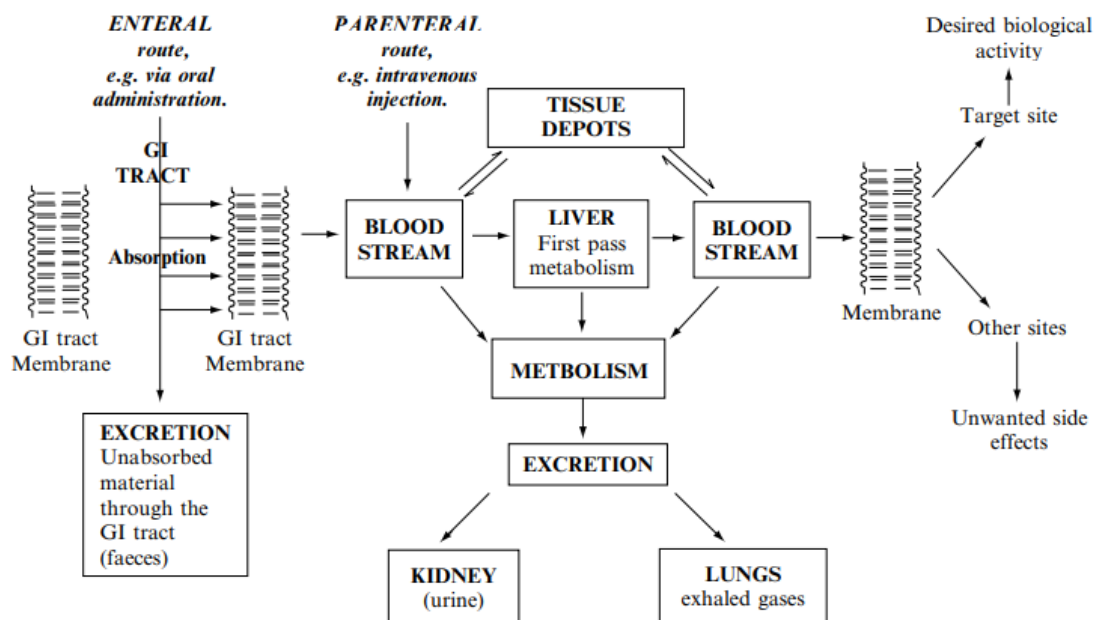


Figure SC-H&M-TRB - 9: The main routes of drug administration and distribution in the body. The distribution of a drug is also modified by metabolism, which can occur at any point in the system

Typically, drugs are used topically or systemically. The routes are divided into two categories: parenteral and enteral (Figure SC- H&M-TRB-8). Parenteral approaches avoid the gastrointestinal tract (GI. tract), with intramuscular injection being the most common method (IM). Drugs are absorbed by the enteral route from the alimentary canal (PO per oral), rectal, and sublingual routes. The chemical stability of a drug will determine the route of administration, both when it is transported through a membrane (absorption) and in transit to the site of action (distribution). It will also be influenced by the patients' age, as well as their physical and mental abilities. For example, due to age-related metabolic changes, elderly individuals may require lower pharmacological dosages to get the desired clinical effect. Schizophrenics and individuals with chronic medical disorders are especially vulnerable to either overdosing or underdosing. In these circumstances, a slow-release intramuscular injection, administered once every two to four weeks rather than on a daily basis, may be the most effective treatment option.

As a result, the design of a medicine should take into account the nature of its target groups at an early stage in its development.

Once a drug enters the bloodstream, it is transported throughout the body, and a part of it is either excreted as other products or attached to biological locations other than its target site by excretion metabolism. As a result, the dose given is invariably larger than what would be required if all of the medicine reached its target biological site. The dose of a medicine given to a patient is the amount needed to achieve and maintain the concentration required to produce a positive reaction at the biological action site. A dose that is too high creates undesirable side effects, whereas a dose that is too low causes the therapy to fail. The therapeutic window is the range during which a drug is

effective as a therapeutic agent (Figure SC- H&M-TRB-9.). Because of the amount of drug that plasma can hold combined with processes that permanently remove the drug from its site of action, the drug concentration reaches a plateau. A dose that is too high will result in a plateau outside of the therapeutic window, as well as hazardous side effects. A dose that is too low will cause a plateau below the therapeutic window, resulting in unsuccessful treatment.

The dosing regimen refers to a drug's dose and how it is administered. Dosage regimens can range from a single dose to relieve a headache to regular daily doses to treat epilepsy and diabetes, as well as continuous intravenous infusions for very ill patients. Regimens are created to keep the drug concentration at the site of action inside the therapeutic window for the amount of time required for therapeutic success. The regimen's design is determined by the nature of the medical problem and the physician. The latter necessitates not only an understanding of a drug's biological effects, but also its pharmacokinetic qualities, or the rate at which it is absorbed, distributed, processed, and eliminated from the body.

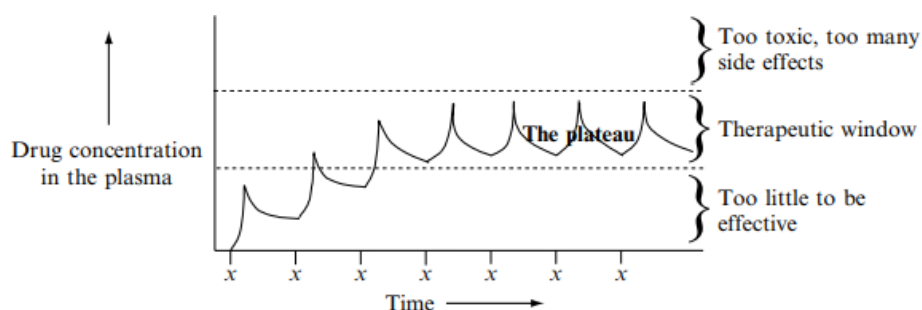


Figure 2.4 A simulation of the therapeutic window for a drug given in fixed doses at fixed time intervals X

Figure SC-H&M-TRB - 10: A simulation of the therapeutic window for a drug given in fixed doses at fixed time intervals X

1.7 Drug Action

The interaction of a medicine with endogenous and external substrate molecules in the body is thought to be the cause of its activity. When one or more active drug molecules attach to the target endogenous and exogenous molecules, the biological activity of these molecules is changed or inhibited. The stability of the drug–substrate complex determines how effective a drug is at bringing about these changes, whereas the medical success of a pharmacological intervention is determined by whether enough drug molecules bind to enough substrate molecules to have a significant impact on the disease's progress.

The concentration of the drug in the aqueous medium in contact with the substrate molecules determines the degree of drug action. The pharmacokinetic and pharmacodynamic phases of drug action can be used to classify the elements that influence this concentration in a biological system. The pharmacokinetic phase is concerned with the investigation of the parameters that regulate the drug's trip from administration to activity. The pharmacodynamic phase is concerned with the chemical nature of the drug-target relationship: in other words, the drug's effect on the body.

1.7.1 Pharmacokinetic phase

The drug's Absorption, Distribution, Metabolism, and Elimination (ADME) phase is part of the pharmacokinetic phase of action. Many of the elements that determine drug activity are present throughout the pharmacokinetic phase. A drug's solubility, for example, is a key element in its absorption, distribution, and elimination. Furthermore, when a solid drug is supplied by enteral routes, as a solid or suspension, the rate of drug dissolution, or the rate at which a solid drug dissolves in the aqueous medium, influences its activity.

1.7.1.1 Absorption

After enteral administration, absorption refers to the transfer of the medication from the site of administration into the plasma. It entails the medication passing through the proper membranes. A medication molecule's polar (hydrophilic) and nonpolar (hydrophobic) groups must be balanced in order for it to be absorbed effectively. Drugs that are too polar will likely to stay in the bloodstream, whereas drugs that are too nonpolar will be absorbed into the lipid core of the membranes and remain there. In both circumstances, the medicine is likely to be ineffective, depending on the target. Partition coefficient, solubility, pKa, excipients, and particle size are all factors that influence absorption. The acids released by the parietal cells in the stomach lining, for example, hinder the ionization of the painkiller aspirin in the stomach. As a result, it is absorbed in large quantities into the bloodstream through the stomach membrane in its unionized and so uncharged form.

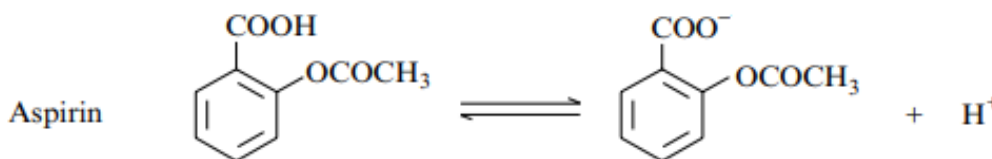
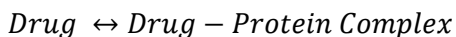


Figure SC-H&M-TRB - 11: Aspirin

1.7.1.2 Distribution

The movement of a medicine from its initial location of administration or absorption to its site of action is known as distribution. The circulatory system is the primary route of distribution, but the lymphatic system also plays a role. In the first situation, once the medicine is absorbed, it is promptly transported throughout all of the body's blood-supplied locations. Drugs are delivered either in a 'free form' or reversibly bound to plasma proteins, dissolved in the aqueous medium of the blood.



Until drug molecules attached to plasma proteins are released from those proteins, they have no pharmacological impact. However, if a drug forms a more stable compound with a protein, it may be able to displace another drug from that protein. This could generate undesired side effects, which could complicate the design of medication regimens including multiple drugs. Furthermore, in some disorders, such as rheumatoid arthritis, decreased plasma protein concentrations can impact drug distribution.

The solubility and bioavailability of medications in the biological milieu of the blood are major determinants of distribution. Water-soluble substances may be deposited sparingly in blood arteries, resulting in blood flow limitation. Because serum proteins can act as enzymes that catalyze the breakdown of drugs, drug stability is especially important. Decompositions like these

can lead to the requirement for a greater dose of the drug to have the desired pharmacological action, increasing the patient's risk of harmful side effects. Decomposition of the administered drug, on the other hand, produces the active form of some drugs. Prodrugs are pharmaceuticals that work in this way. The bacteriacide prontosil, for example, was discovered in 1935 but is not active because it is converted in situ to the antibiotic sulphanilamide.

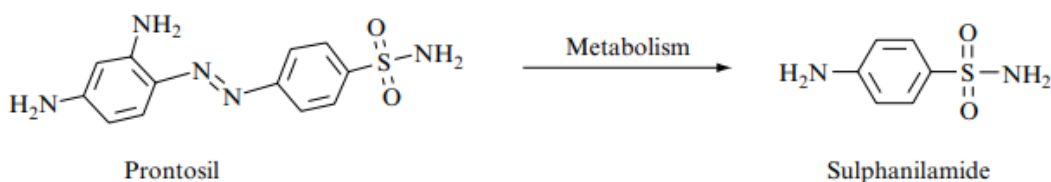


Figure SC-H&M-TRB - 12: Prontosil and Sulphanilamide

1.7.1.3 Metabolism

The biotransformation of a medicine into other chemicals known as metabolites is referred to as drug metabolism. These biotransformations are mostly found in the liver, but they can also be found in the blood and other organs like the brain, lungs, and kidneys. The pharmacological activity and toxic effects of a medication are usually reduced or completely suppressed when it is metabolized, resulting in a lower or total suppression of the drug's pharmacological action and toxic effects in the systemic circulation. Prodrugs, such as prontosil, are exceptions since metabolism produces the active version of the drug.

Metabolism typically comprises multiple pathways and leads in the creation of a series of metabolites (Figure SC- H&M-TRB-12). Each of these metabolites could have a distinct or similar effect on the parent medication. As a result, all of a drug's metabolites must be evaluated throughout the creation of a new medicine. Because metabolites are usually more water soluble than their parent medication, they are eliminated in the urine.

1.7.1.4 Elimination

The term "elimination" refers to the metabolic and excretion processes that remove a drug from the body irreversibly during its journey to its site of action. It decreases the drug's medicinal effect by lowering its concentration at the point of action. The build-up of drug concentration in the body might be caused by a sluggish elimination process. This could benefit the patient since the dose needed to sustain the therapeutic effect can be lowered, lowering the risk of undesired side effects. Rapid drug elimination, on the other hand, necessitates either higher doses, which involves a higher risk of harmful side effects, or more frequent doses, which carries a higher danger of under- or over-dosing. Drugs and their metabolites are primarily excreted in solution in the urine through the kidney. A considerable proportion of medications and their metabolic metabolites, on the other hand, are eliminated in the feces via the bowel.

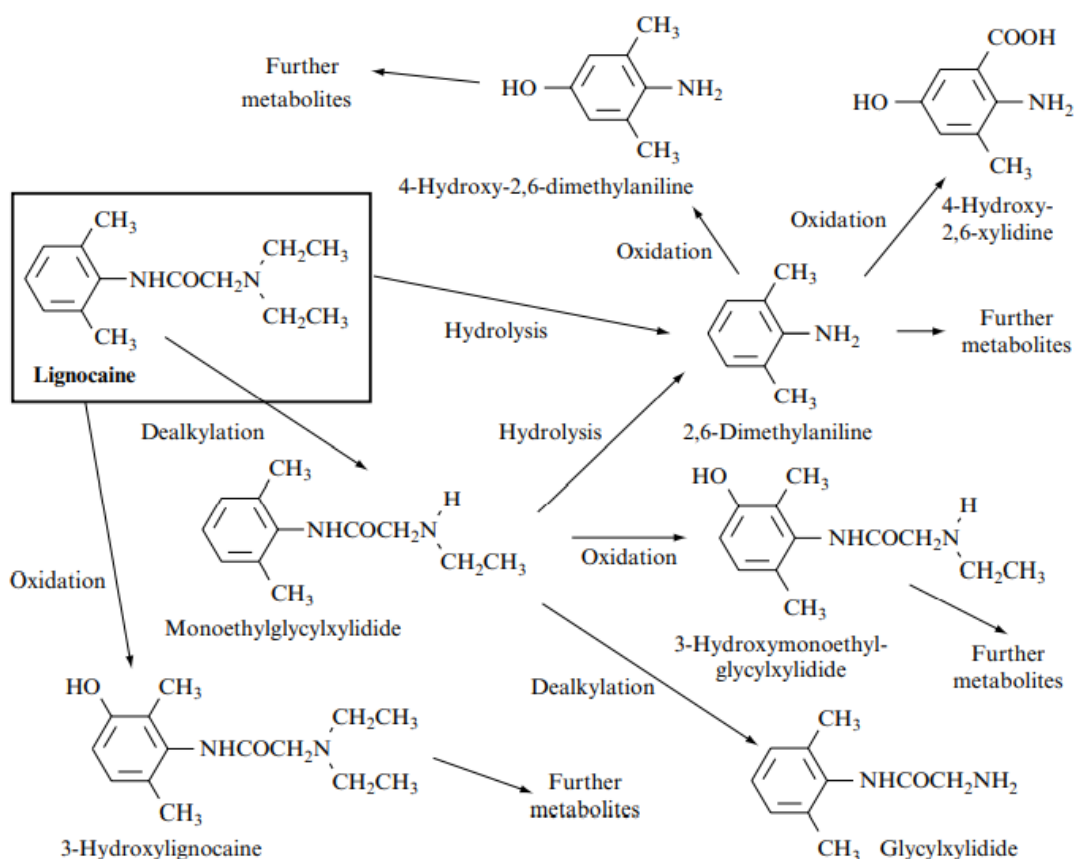


Figure 2.5 An outline of the known metabolic pathways of the local anaesthetic lignocaine

Figure SC-H&M-TRB - 13: An outline of the known metabolic pathways of the local anaesthetic lignocaine

Drugs are removed from the body by glomerular filtration or tubular secretion in the kidneys. However, tubular reabsorption, a recycling mechanism, reclaims part of the species lost as a result of these processes. Tubular reabsorption is a mechanism that returns vital bodily chemicals including water, amino acids, salts, and glucose from the urine to the circulatory system, but it can also return drug molecules. Because charged molecules are not easily transported across membranes, reabsorption of acidic and basic medicines is limited when the pH promotes salt production.

Biliary clearance occurs in the liver, where big molecules are converted into smaller compounds before being expelled. Some of the ejected medications, however, are reabsorbed during the enterohepatic cycle. The addition of appropriate compounds in the dose form can minimize this reabsorption; for example, the ion exchange resin cholestyramine is used to lower cholesterol levels by blocking its reabsorption.

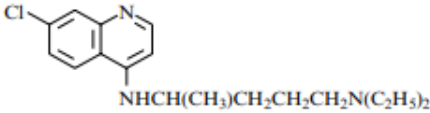
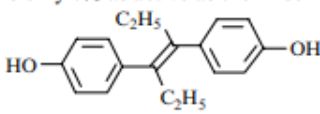
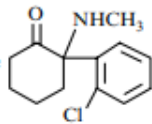
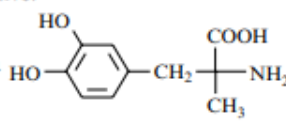
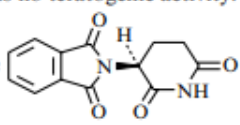
1.7.1.5 Bioactivity of a Drug

A drug's bioavailability is defined as the proportion of a drug's dose that is found in general circulation. ADME, for example, has an impact on it. Bioavailability is not continuous, but fluctuates according to the physiological state of the organism.

1.7.2 Pharmacodynamic Phase

Pharmacodynamics is the study of what happens when a drug interacts with the body at its site of action, or what the drug does to the body. It is now well understood that a drug's shape and electron distribution, or stereoelectronic structure, is most effective when it is complimentary to the active site or receptor's stereoelectronic structure. The medicinal chemist's job is to design and synthesize a drug structure with the most beneficial effects and the fewest toxic side effects possible. The stereoelectronic features of the target active or receptor site, as well as the drug's stability in situ, polarity, and relative solubilities in aqueous media and lipids, must all be considered in this design. The drug's stereochemistry is crucial since stereoisomers often have diverse biological effects, ranging from inert to severely hazardous (see Table 3).

Table 3: Variations in Biological Activities of Stereoisomers

First stereoisomer	Second stereoisomer	Example
Active	Activity of same type and potency	The R and S isomers of the antimalarial chloroquine have equal potencies.  <chem>Clc1ccc2nc(NCCN(CC)CC)ccc2c1</chem>
Active	Activity the same type but weaker	The E isomer of diethylstilbesterol, an estrogen is only 7% as active as the Z isomer  <chem>CC/C=C/C1=CC=C(C=C1)C(=C2C=CC(=C2)O)O</chem>
Active	Activity of a different type	S-Ketamine is an anaesthetic R-Ketamine has little anaesthetic action but is a psychotic  <chem>CN1CCCC12C(=O)C=CC=C2C1</chem>
Active	No activity	S-α-Methyldopa is a hypertensive drug but the R isomer is inactive.  <chem>C[C@H](N)C(=O)OCC1=CC=C(C=C1)O</chem>
Active	Active but different side effects	Thalidomide: the S isomer is a sedative and has teratogenic side effects. The R isomer is also a sedative but has no teratogenic activity.  <chem>O=C1NC(=O)CC[C@H]1C2=CC=CC=C2C3=CC(=O)N3</chem>

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover background on the drug, brief discovery history of the drug, what the drug targets, and its drug action (pharmacokinetics and pharmacodynamics)
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 2: Practical (Optional)

Title: Evaluating Drug Action in Humans

Note: This can be done on a test subject other than human (e.g., rat).

Objectives:

- To understand the importance of informed consent in the conduct of experiments on humans.
- To investigate the variability in baseline parameters in a group of normal healthy volunteers.
- To investigate the differences between β -adrenoceptor antagonists on cardiovascular and respiratory parameters using a semi-automated blood pressure monitor, peak flow meter and skin temperature thermistor.
- To assess the validity of indirect measures to determine drug effects in humans.
- To gain an understanding of the use of abstracts as a form of scientific communication.

Background Reading

- Current place of Beta-Blockers in the Treatment of Hypertension
- Current Vascular Pharmacology
- Hypertension

Introduction

Clinical trials in humans are an essential step in the development of new therapeutic agents. These range from initial Phase 1 trials in small groups of volunteers to confirm safety indicated in the preclinical animal studies to large-scale phase 3 trials in the intended treatment cohort. Even when drugs have been approved, they continue to be monitored for their long-term effects. Phase 4 studies can reveal unwanted side effects such as the identification of increased cardiovascular risk associated with the COX-2 inhibitors and the potentially fatal prolongation of QT interval by terfenadine. Conversely, they may lead to expanded uses of a drug. Thus, ACE inhibitors originally used for hypertension are now also used for treatment of heart failure. Likewise, β -adrenoceptor antagonists are used for a variety of conditions including hypertension, arrhythmias, angina and anxiety. Of note is that while they were initially contraindicated for use in heart failure, they have now been shown to be effective in managing the condition.

Studies involving humans have specific ethical requirements to maximize safety of volunteers and ensure that they can provide informed consent to participate in a study. The National Statement on Ethical Conduct in Human Research (2007, updated in 2018) provides guidelines around human research in accordance with the National Health and Medical Research Council Act 1992. More than

that, the National Statement discusses the ethos or attitude that should be adopted by those involved in any aspect of human research. Further information can be found here: <https://www.nhmrc.gov.au/about-us/publications/national-statement-ethical-conduct-human-research-2007-updated-2018>.

Aim

There are two aims of this practical.

1. To compare the reliability and reproducibility of heart rate measurements obtained via smartphone- based apps and pulse palpation.
2. To study the effects of two orally administered β -adrenoceptor antagonists on blood pressure, heart rate, lung function and peripheral blood flow. The β -adrenoceptor antagonists which will be used are: atenolol, which does not have intrinsic sympathomimetic activity or membrane stabilising activity and which is β_1 -adrenoceptor selective, and pindolol, which has intrinsic sympathomimetic activity, no membrane stabilising activity and which does not show selectivity for β_1 - or β_2 -adrenoceptors 1). Their effects will be studied under two conditions: at rest, and during a period of exercise.

Table 1: Properties of some β -adrenoceptor antagonists

	Approximate relative potency*	Antagonistic selectivity	Agonist activity**	Membrane stabilising activity
Pindolol	5	non-selective	+	0
Propranolol	1	non-selective	0	+++
Timolol	6	non-selective	0	+
Atenolol	1	β_1	0	0
Metoprolol	1	β_1	0	+
Practolol	0.3	β_1	+	0

*relative to propranolol, which = 1; ** intrinsic sympathomimetic activity

Approach

The effects of the β -adrenoceptor antagonists will be studied under two conditions: at rest, and during a period of exercise. A period of baseline measurements will be taken, and a therapeutically effective dose of each drug will be administered. The trial will be conducted "single blind" with a placebo (vitamin B6) control and a no treatment control, as approved by the University of Melbourne Human Ethics Committee (1646840.1). The identity of the drug / placebo will be revealed to you at the end of the class.

Important Notes

- As you will be exercising during this practical, you will need to wear appropriate clothing and have something to eat prior to coming to class.
- Any student having a history of asthma, wheezy bronchitis, diabetes, or heart disease must not participate as a subject in this experiment.
- If in any doubt about your ability to participate, see the physician in attendance. All participating students will be asked to complete a consent form.

- Once you have taken the drug, do not leave the class without notifying a demonstrator.

Protocol

Read the practical notes carefully before starting the experiment. Pretreatment observations will be taken at rest (at times 0, 15, 30, 45 and 60 min) and following exercise (at times 30 and 60 min) for each student. After taking the 60 min reading, students should take their unsigned consent form to the dispensing bench. Laboratory staff will then witness your signature giving your informed consent, and then dispense either: Placebo, OR Atenolol (100 mg) OR Pindolol (10 mg). Subjects will be required to take their allocated treatment immediately in the presence of laboratory staff and continue taking measurements at rest (every 15 min from times 75 to 150 min) and post-exercise (at times 90, 120 and 150 min) according to the following protocol.

Measurement at Rest

The subject should be seated and rested for a few minutes before resting readings are taken every 15 min from time 0 to 150 min. Remember - exercise periods occur after the resting measurements at 30 and 60 min. The placebo/drug must be taken after the second exercise period at 60 min.

Resting heart rate

Measure resting heart rate (RHR, beats per minute) by palpation of the radial or carotid pulse. To measure your radial pulse, lightly press the first two or three fingers of one hand against the hollow next to the tendon on the thumb side of your other hand. Count your pulse for 15 s and multiply this figure by four to obtain beats per minute.

Blood Pressure

Systolic (SBP) and diastolic (DBP) arterial blood pressure will be measured using an electronic sphygmomanometer.

Peripheral Blood Flow

The infrared thermometer should be aimed at the cheek and time allowed for the temperature to equilibrate before recording the temperature (FT) reading. The same position on the face should be used for subsequent measurements. The principle behind this measurement is that the temperature recorded is proportional to blood flow in the vascular bed.

Peak expiratory flow rate

One member of the team should record lung function with a peak flow meter. Correct use of the air flow meter will be demonstrated at the commencement of the class. Measure the peak expiratory flow rate (PEFR) reading (L/min). All other team members should record their respiratory rate by counting the number of respirations in 15 s and multiplying by 4.

COVIDSafe Rules

In addition to the usual rules in place to ensure a COVIDSafe environment, the following rules should be observed by all students and staff.

- Exercise to be performed while maintaining 1.5 m distancing
- Gloves are to be worn when using equipment
- Shared equipment (electronic sphygmomanometer and infrared thermometer) must be sanitised between each use using the supplied alcohol-based wipes

- Only one member of each team will measure lung function with the peak flow meter. Use of the peak flow meter must be performed in the designated area for each team.

Measurements during and post-exercise

After taking resting measurements at times 30, 60, 90, 120 and 150 min, each subject will exercise at an intensity level suited to an estimate of their own fitness (see table below). Resting heart rate can be used as an index of fitness.

Unfit	Fit
30 s of star jumps* 20 s rest 30 s of star jumps	40 s of star jumps 10 s rest 30 s of stationary jogging 10 s rest 40 s of star jumps

Exercise Heart Rate

Exercise heart rate (EHR) should be measured immediately after your last start jump.

Respiratory Rate

Respiratory rate should be measured immediately after the heart rate measurement

Fatigue

The following scale should also be used to record the level of fatigue felt immediately after exercise.

1	2	3	4	5
Not at all	Slight	Moderate	Out of breath and difficulty talking	Never been more tired

Recording of Observations

Record the listed parameters at the times shown in the table overleaf. Before leaving the class, find out from the posted list what drug you were given and have your results screened by a demonstrator. You should record your data electronically immediately after each measurement. In the quieter times between exercising and between measurements, transfer your data into the Class Data spreadsheet (LabArchives → Class data → Prac 9 → In Vivo Human Class Data S1 2021).

Example results table:

Treatment number: _____

Name: _____

Gender: _____

Fitness level: _____

Measurements at rest:

resting heart rate (RHR)

systolic blood pressure (SBP) and diastolic blood pressure (DBP)

peak expiratory flow rate (PEFR) OR respiration rate (RR)

facial temperature (FT)

Measurements during/post exercise:

heart rate during exercise (EHR)

level of fatigue

PRE-TREATMENT READINGS

TIME (min)	REST					EXERCISE from t = 30 min at 30 min intervals	
	RHR (beats/min)	SBP (mmHg)	DBP (mmHg)	PEFR or RR (L/min or rpm)	FT (°C)	EHR (beats/min)	FATIGUE (1 – 5)
0						NO EXERCISE	
15						NO EXERCISE	
30							
45						NO EXERCISE	
60							

After recording the results at t = 60 min (at rest and during exercise), IMMEDIATELY take your unsigned consent form to the prep room window. After your signature is witnessed by staff, take your allocated treatment in their presence.

POST-TREATMENT READINGS

75						NO EXERCISE	
90							
105						NO EXERCISE	
120							
135						NO EXERCISE	
150							

Fatigue

1	2	3	4	5
Not at all	Slight	Moderate	Out of breath and difficulty talking	Never been more tired

Activity 3: Homework Exercises

- (1) Explain the meaning of the terms:
 - a. lead compound
 - b. excipient
 - c. parenteral administration
 - d. pharmacophore
 - e. prodrug
- (2) State the general factors that need to be considered when designing a drug.
- (3) Distinguish between parenteral and enteral routes of administration.
- (4) Define the terms pharmacokinetic phase and pharmacodynamic phase in the context of drug action. List the main general factors that affect these phases.
- (5) The drug amphetamine ($\text{PhCH}_2\text{CH}(\text{NH}_2)\text{CH}_3$) binds to the protein albumin in the blood stream. Predict how a reduction in pH would be expected to influence this binding. Albumin is negatively charged at pH 7.4 and electrically neutral at pH 5.0.
- (6) Discuss the general effects that stereoisomers could have on the activity of a drug

⁶ [Practical 9 Manual F2F - Practical 6: Toxicity Testing – All Drugs are Poisons BLOCK 4: In vivo Drug - StuDocu](#)

Topic 2: Drug Discovery Process

In this topic, students will be learning about the general approach to drug discovery. In addition, students will learn about stereochemical and water solubility consideration that are necessary when choosing a lead compound structure.

2.1 What is Drug Discovery

Drug discovery is a process by which new medicinal candidates are discovered. It is basically a combination of chance and research. Drug discovery is typically done by a group of people that are specialized or experienced in fields such as medicine, biochemistry, chemistry, computerized molecular modeling, pharmaceuticals, pharmacology, microbiology, toxicology, physiology, and pathology.

2.1.1 Brief History

Drug Discovery (or Research) started When chemistry had advanced to the point that its concepts and methods could be applied to problems other than chemistry, and pharmacology had established itself as a well-defined scientific area in its own right.

Shown below are some essential foundations of chemical theory that were laid that helped start and propel Drug Discovery.

- 1856 – August Kekulé formulated his pioneering theory on the structure of aromatic organic molecules
- 1870 – some essential foundations of chemical theory had be laid such as the confirmation of Avogadro’s Atomic Hypothesis, establishment of the Periodic Table, and a set of theories for Acids and Bases.
- 20th Century – Drug research began to take shape and was aided by a number of new technologies that improved the drug discovery process. In many aspects, biochemistry had a huge impact on medication development. The idea of using enzymes as therapeutic targets arose, which led to the development of enzyme substrates that worked as inhibitors or demonstrated their activity by altering various feedback pathways.

Table 1 shows some important discoveries in the field of medicine, right from 19th century to 21st century.

Table 4: Important Discoveries in Medicine

Sr. no	Year of Discovery	Drug Name	Category
1.	1806	Morphine	Hypnotic agent
2.	1899	Aspirin	Analgesic and Anti-pyretic agent
3.	1922	Insulin	Anti-Diabetic agent
4.	1928	Penicillin	Antibiotic
5.	1960	Chlordiazepoxide	Tranquillizer
6.	1971	L-dopa	Anti-Parkinson agent
7.	1987	Artemisinin	Anti-malarial agent
8.	1998	Sildenafil	Erectile Dysfunctioning treatment
9.	1999	Celecoxib, Rofecoxib	Selective COX-2 inhibitors
10.	1999	Zanamavir, Oseltamivir	Anti-influenza agents
11.	2001	Imatinib	Leukemia treatment

An Approach to Drug Design

The approach or method that you want to use to design your drug typically depends upon the goal(s) of the drug design team. Before embarking on a drug design investigation, these goals will almost always necessitate a thorough examination of the disease's pathophysiology and, in some circumstances, basic biochemical studies. Figure SC-TRB-46 shows an approach to drug design.

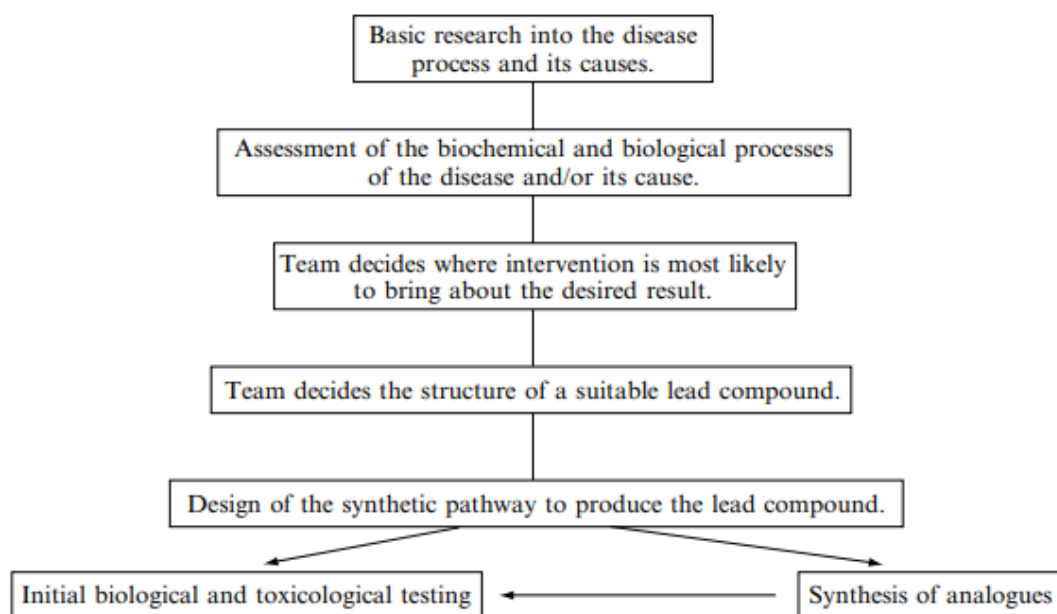


Figure SC-H&M-TRB - 14: The general steps in the design of a new drug

The team uses the information gathered to determine which intervention is most likely to achieve the intended outcome. The team must then propose a structure for a lead chemical that could conceivably bring about the desired alteration after the point of intervention has been determined.

This typically necessitates a thorough literature and database search to uncover substances contained in the organism (**endogenous compounds**) as well as compounds not found in the organism (**exogenous compounds**) that have biological effects at the intervention location.

To assist the team in reaching a choice, molecular modeling approaches are occasionally used. Many structures are determined to be appropriate, but due to the high cost of developing medications, the team must choose only one or two of these compounds to act as the lead or to serve as inspiration for the lead molecule. The team's experience will determine the final selection.

Combinatorial chemistry is a more chance-based technique to finding a lead. To generate a large number of potential leads, a simultaneous multiple synthesis technique is used. To find the most active lead compounds, these prospective leads are subjected to quick high throughput biological screening. These lead compounds are then subjected to further development once they have been found.

Once the potential lead's structure has been agreed upon, the medicinal chemist is responsible for devising a synthetic method and preparing a sample of the chemical for testing. The molecule is first tested for pharmacological and toxicological effects when it is produced. Because the lead molecule is unlikely to be acceptable for use as a medicine, the results of these experiments allow the team to determine if it is profitable to continue development by developing analogues. Preparing a number of analogues, measuring their activity, and comparing the results to determine the structure with the best activity is the standard procedure. To discover the nature of this structure, this analysis may use SARs, QSARs, computational chemistry, and combinatorial chemistry.

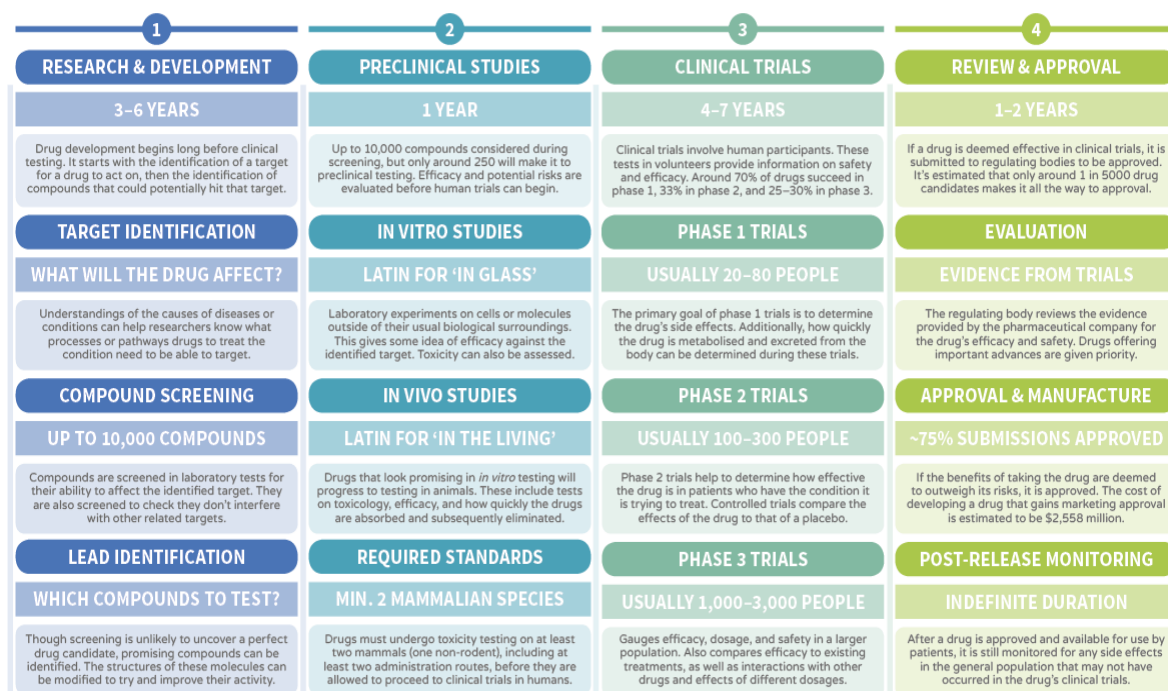
At the commencement of a research, the selection of a lead molecule and the construction of a synthetic pathway for its manufacture is not the only factor to consider. In addition, researchers must develop appropriate in vivo and in vitro assays to evaluate the efficacy and toxicity of the chemicals created. It's pointless to spend money on a synthetic procedure if the product can't be tested at the conclusion of the process.

2.3 Drug Discovery Process Overview

Drug discovery is a lengthy process that might take up to 13 years to complete. Screening for potentially active compounds is usually the first step in the Early Drug Discovery process. These molecules must have a therapeutic impact on the condition in question, and after they've been identified, testing for safety and efficacy begins.

Only one out of every 5,000 medications reach the level of commercial approval. Furthermore, only 250 medication candidates out of 5,000 to 10,000 make it to preclinical testing.

On the next page is an oversimplified diagram (Figure SC-H&M-TRB-15)describing the process of discovering, testing, and eventually gaining approval for selling a drug. It shows that this process is a very long and arduous one. Figure ## also shows the different stages involved, and the approximate length of time that each stage takes, to eventually arrive at an approved drug that can be given to patients.



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Figure SC-H&M-TRB - 16: Understanding the Drug Discovery Process.

2.3.1 STAGE 1

- **Research & Development:** Drug development begins long before clinical testing. It starts with the identification of a target for a drug to act on, then the identification of compounds that could potentially hit that target.
- **Target Identification:** (What will the drug affect?) Understandings of the causes of disease or conditions can help researchers know what processes or pathways drugs to treat the conditions need to be able to target.
- **Compound Screening:** Up to 10,000 compounds are screened in laboratory tests for their ability to affect the identified target. They are also screened to check they don't interfere with other related targets.
- **Lead Identification:** (Which Compounds to Test?) Though screening is unlikely to uncover a perfect drug candidate, promising compounds can be identified. The structures of these molecules can be modified to try and improve their activity.

2.3.2 STAGE 2

- **Preclinical Studies:** Up to 10,000 compounds considered during screening, but only around 250 will make it to preclinical testing. Efficacy and Potential risks are evaluated before human trials can begin.
- **In Vitro Studies:** (Latin for 'In Glass') Laboratory experiments on cells or molecules outside of their usual biological surroundings. This gives some idea of efficacy against the identified target. Toxicity can also be assessed.
- **In Vivo Studies:** (Latin for 'In the Living') Drugs that look promising in *in vitro* testing will progress to testing in animals. These include tests on toxicology, efficacy, and how quickly the drugs are absorbed and subsequently eliminated.
- **Required Standards:** Minimum of 2 mammalian species. Drugs must undergo toxicity testing on at least two mammals (on non-rodent), including at least two administration routes, before they are allowed to proceed to clinical trials in humans.

2.3.3 STAGE 3

- **Clinical Trials:** This involves human participants. These tests in volunteers provide information on safety and efficacy. Around 70% of drugs succeed in phase 1, 33% in phase 2, and 25-30% in phase 3.
- **Phase 1 Trials:** Usually 20-80 people. The primary goal of phase 1 trials is to determine the drug's side effects. Additionally, how quickly the drug is metabolized and excreted from the body can be determined during these trials.
- **Phase 2 Trials:** Usually 100-300 people. Phase 2 trials help to determine how effective the drug is in patients who have the condition it is trying to treat. Controlled trials compare the effects of the drug to that of a placebo.
- **Phase 3 Trials:** Usually 1,000-3,000 people. Gauges efficacy, dosage, and safety in a larger population. Also compares efficacy to existing treatments, as well as interactions with other drugs and effects of different dosages.

2.3.4 STAGE 4

- **Review & Approval:** If a drug is deemed effective in clinical trials, it is submitted to regulating bodies to be approved. It is estimated that only around 1 in 5000 drug candidates makes it all the way to approval.
- **Evaluation:** Evidence from trials – the regulatory body reviews the evidence provided by the pharmaceutical company for the drug's efficacy and safety. Drugs offering important advances are given priority.

- **Approval & Manufacture:** about 75% submission approved. If the benefits of taking the drug are deemed to outweigh its risks, it is approved. The cost of developing a drug that gains marketing approval is a given estimated total.
- **Post-Release Monitoring:** Indefinite duration – after a drug is approved and available for use by patients, it is still monitored for any side effects in the general population that may not have occurred in the drug’s clinical trials.

2.4 Four Stages of Drug Discovery

There are several phases and actions involved in the drug discovery process. Early Drug Discovery, Pre-Clinical Phase, Clinical Phases, and Regulatory Approval are the four primary stages of the process. Let's look at the key steps involved in each of these stages of drug development.

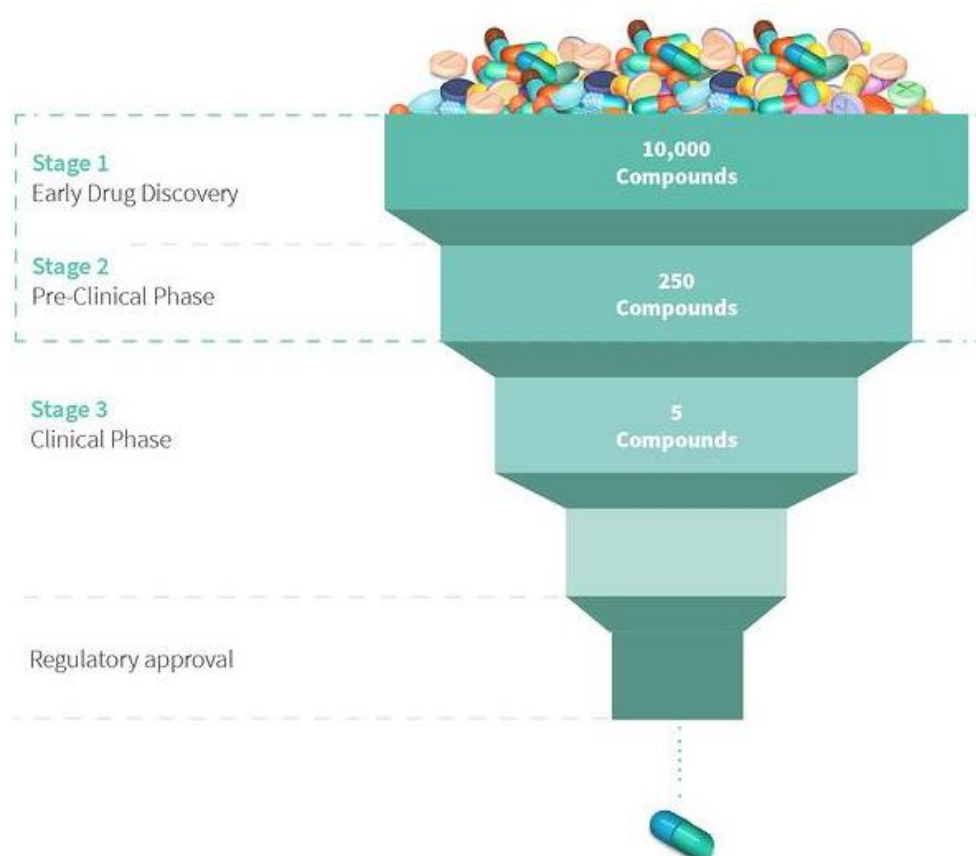


Figure SC-H&M-TRB - 17: Four Phase of Drug Discovery Process (Source: [The Drug Discovery Process: What Is It and Its Major Steps \(biobide.com\)](https://www.biobide.com/))

2.4.1 Early Drug Discovery

The Early Drug Discovery Process involves many different actions and testing. Researchers work together to find and optimize possible leads for a certain target. In order to treat a disease, the leads must elicit a desired impact on a specific biological target involved in the condition. At present time, research is carried out in the lab employing in silico platforms, biochemical tests, cell cultures, and a variety of animal models. Target Identification and Validation, High Throughput Screening or High Content Screening, Hit Identification, Assay Development and Screening, Hit-

To-Lead (H2L), Lead Generation and Optimization, and In vivo and In vitro Assays are the sub-processes that flow through this stage.

2.4.2 Pre-Clinical Phase

The second step of the Drug Discovery process is the Pre-Clinical Phase, which follows the four key phases of the process. The chemicals discovered during Early Drug Discovery are improved, optimized, and extensively evaluated in a laboratory and in animal or alternative models during the Pre-Clinical Phase. Before Clinical Trials in people can begin, adequate proof of safety and efficacy must be provided, and once this point is established, it is also important to determine the optimum dosages to test in humans. It must be assured that the new chemical will be available in adequate amounts during the clinical trials before they begin.

Because only modest amounts were required earlier, manufacturing must now be adjusted to meet the much-increased demand in the Clinical Phase.

Preclinical investigations are required by regulatory agencies before any investigational new drug application may proceed to the Clinical Phases.

2.4.3 Clinical Phases

Clinical trials are divided into four stages: Phase I, Phase II, Phase III, and Phase IV. We'll go through each of these phases in greater depth later. Nonetheless, the tolerance and safety of the drug candidate will be examined in a small sample of healthy people, generally 20 to 80, in the first stage. Phase I is intended to provide answers to the following questions:

- a. Is the Drug Safe?
- b. What dose can be given without side effects?
- c. How does the substance behave in the body?

The active component must initially be synthesized under GMP conditions for these trials. That is, it is tightly regulated in accordance with "Good Manufacturing Practice" norms.

Phases IIa and IIb assess the efficacy, tolerability, and dose in a larger population after tolerance and effectiveness have been examined in a small sample. The dosage form is initially created for this.

The therapeutic idea is principally verified (proof of concept) in phase IIa trials; the goal of phase IIb research is to discover the correct dose. In most Phase II trials, 100 to 500 adult patients are included.

Doctors test the treatment on thousands of patients in the last phase before it can be approved as a drug to check whether its effectiveness and safety can be validated in a wide range of individuals. Other medication interactions are also investigated. Phase II and phase III trials are sometimes referred to as "controlled studies," in which some patients get the new medicine while others receive the old standard of care.

2.4.4 Regulatory Approval

After an active substance's Clinical Trials are completed, the data is gathered and examined. It can then be submitted for evaluation to the proper authorities. A national regulatory authority or centralized mechanism must approve a medication or vaccination before it may be commercialized. Only one molecule out of a big number examined makes it through the clinical study phases and regulatory testing in the end. As a result, just one molecule has been licensed for use as a medicine or vaccine.

2.4.5 Post-Market Monitoring

After getting marketing authorization from the authorities, Phase IV studies (also known as Post-Marketing Surveillance Trials) are conducted. More information about the effectiveness and safety of the new medicine can be acquired. A higher number of patients using the medicine offers more data and allows for comparisons with existing therapies. These studies are used to evaluate a drug's long-term effects. Adverse incidents can be recorded and avoided this way.

2.5 The Early Drug Discovery and Pre-Clinical Phase (Step-by-Step)

Let's take a closer look at the many procedures that take place during the Early Drug Discovery stage of drug development. Target identification, high-throughput and high-content screening, hit identification, assay development, hit-to-lead, lead generation, and in vivo and/or in vitro tests will all be discussed.

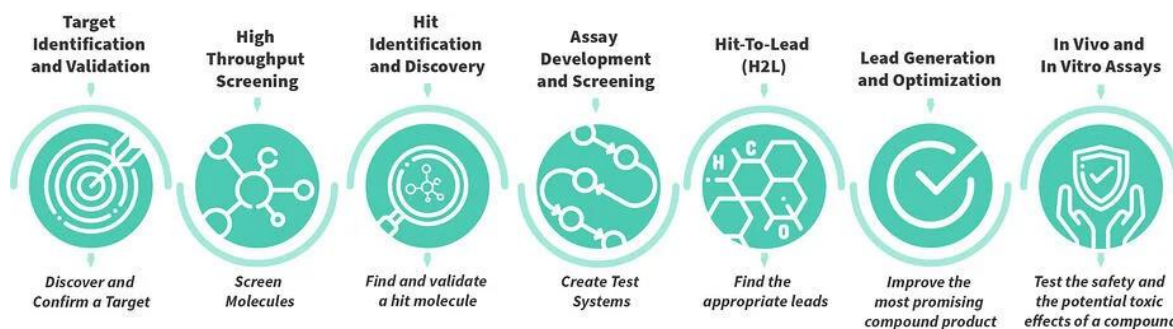


Figure SC-H&M-TRB - 18: Procedures that take place during Early Drug Discovery Stage

2.5.1 Target Identification and Validation

Understanding the pathophysiology of a disease is one of the most important aspects of creating a successful medicine. When a therapeutic molecule, known as a "hit," may affect the biological activity of an appropriate biological target, it is said to be "druggable." By scouring public databases and conducting research, potential targets can be identified. Once a target has been discovered, researchers must determine if it is suitable for drug development before moving on to the screening phase to find hits.

2.5.2 High Throughput Screening

The most crucial processes in the discovery of a medicine are frequently high throughput screening (HTS) and high content screening (HCS). The beginning point can be determined once a possible target has been identified and verified by screening a large number of compounds that can interact with it. An ad hoc test must be established, which means data on the molecule's efficacy and selectivity must be provided. It must also be able to endure the strict testing standards. HCS can provide more data than HTS.

2.5.3 Hit Identification and Discovery

Various strategies can be employed to find a hit molecule. A hit is a chemical that must interact with the previously described target in order to provide the intended therapeutic effect. A great number of techniques are employed to find hits, including high content screening, phenotypic screening, fragment-based screening, structure-based screening, and virtual screening.

2.5.4 Assay Development and Screening

High Content Screening (HCS) is a unique technology that uses high throughput microscopy to take a huge number of pictures. In vitro cell models, such as image-based cellular assays and even zebrafish embryos (classified as in vitro), play a critical role in the use of HCS in the development of novel active components. HCS is utilized in preclinical research for secondary substance screening, lead optimization, substance profiling, discovery and validation of effective targets, ADME (absorption, distribution, metabolism, and excretion) investigations, and toxicity studies.

2.5.5 Hit-To-Lead (H2L)

A hit is a phrase used to describe a chemical product that has a desired therapeutic impact on a recognized target molecule, as we described before. Similarly, a lead is a result of the screening process that can be employed in later stages.

One of the most crucial stages in early drug development is the Hit-to-Lead (H2L) stage. The H2L's major goal is to locate the right leads to advance along the route to a final clinically active medication. Different screening approaches, such as High-Throughput Screening, affinity selection of vast chemical libraries, fragment-based procedures, and target-focused libraries, are employed to optimize the original compounds. Using in vitro and computer-based methodologies, the screening procedure seeks to decrease the number of these compounds into fewer and more qualified leads. To test their effectiveness in any in vivo models, the lead characteristics must be appropriate.

2.5.6 Lead Generation and Optimization

The lead optimization procedure will begin when the lead is produced in the Hit-To-Lead process. The goal of lead optimization is to improve the most promising chemical products in order to improve their efficacy, reduce toxicity, or boost absorption.

2.5.7 In Vivo and In Vitro Assays

The generated candidate compounds must be examined under circumstances comparable to those found in living cells at the conclusion of the early Drug Discovery phases. Here comes the role of in vivo or in vitro testing. Animal models, alternative models, such as Zebrafish, or cell cultures are employed in these testing tests to determine the safety and potential harmful effects of a chemical.

2.6 Clinical Phase and Regulatory Approval

Following the Early Drug Discovery and Pre-Clinical Phases, the Clinical Phases begin. Let's take a look at the four major procedures that take place throughout the Clinical Trials and Regulatory Approval Phase.

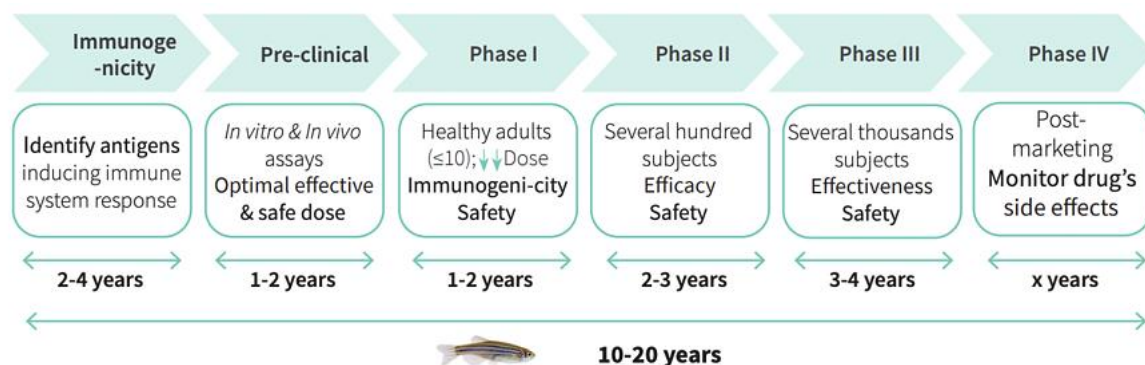


Figure SC-H&M-TRB - 19: Clinical Trials that Vaccines go through

2.6.1 Phases I-III Trials

Clinical research comprises evaluating people in order to discover new ways to prevent and treat diseases. In these studies, vaccines and drugs are tested for effectiveness and acceptability. For active substances, three clinical phases are required.

2.6.1.1 Phase I

The impact is assessed on patients who are suffering from a specific ailment in Phase II clinical investigations. The Phase II research will include up to 500 participants. Not only should the drug's efficacy and acceptability be verified during this period, but the proper dose should also be determined.

2.6.1.2 Phase II

The impact is assessed on patients who are suffering from a specific ailment in Phase II clinical investigations. The Phase II research will include up to 500 participants. Not only should the drug's efficacy and acceptability be verified during this period, but the proper dose should also be determined.

2.6.1.3 Phase III

If the medicine passes Phase II, it will be studied in a wider group of patients in Phase III clinical trials. The goal is to demonstrate that the medicine has the intended impact in a large number of patients. During this stage, rare side effects may be detected.

If Phase III is successful, the firm can file for drug approval, which is the formal license to commercialize the medicine.

2.6.2 Pharmacokinetics and Pharmacodynamics Analysis

Drug dynamics data within the body will be generated and explained following the administration of a drug in an in vivo experiment. Pharmacokinetics (PK) and pharmacodynamics (PD) analysis are terms used to describe this type of analysis. This procedure is crucial because it allows doses and regimens to be determined.

2.6.3 Bioanalytical Method Validation

Quantitative assessments of the matching analyte in biological matrices such as urine, saliva, blood, and so on are used to validate bioanalytical procedures. The effectiveness of pre-clinical and later clinical pharmacological investigations depends on the confirmation of such quantitative assessments of the analyte, such as medications in development or their metabolic products or biomarkers. Validated procedures give crucial information on the safety and efficacy of the medicine under development.

2.6.4 Regulatory Approval and Post Approval

When the development of a drug is almost completed, the approval documentation is drawn up and submitted to the competent authority. This procedure is very time-consuming because all results of the drug testing have to be presented in extensive documents. The approval authority checks all data and decides on approval. The essential prerequisites for approval of a drug are adequate pharmaceutical quality, therapeutic effectiveness, and safety. To do this, the drug must have a favorable risk-benefit ratio, which means that the desired drug effect should be accompanied by as few and harmless side effects as possible. The risk-benefit ratio is the most important approval criterion that is weighed up by the responsible authorities. Even after approval has been granted, the risk-benefit ratio should be continuously monitored. If it changes, this has an impact on the approval. In the worst-case scenario, the drug even has to be withdrawn from the market.

In the course of the authorization process, a very important document is also produced: the “Summary of Product Characteristics.” This document contains all of the important information for patients about the area of application, contraindications, dosage, interactions, and side effects.⁷

2.7 Conclusion

You've probably gained a better understanding of the entire Drug Discovery process by now. The Drug Discovery process is a long, but vital procedure that serves to improve lives and progress research. It starts with early drug discovery and continues with pre-clinical trials, clinical trials, and regulatory approval. Fortunately, emerging alternative models can make this process more cost-effective and shorter while also conforming to higher ethical standards.

⁷ (Biobide, 2021)

2.8 Stereochemistry and Drug Design

The shape of a molecule is one of the most critical elements influencing pharmacological activity. As a result, when developing an analogue, the overall shape of the molecule's structure is a significant consideration. Some structural elements impose a high level of stiffness on a structure, whilst others make it more flexible.

Stereoisomers are formed by combining two or more structures, and they might have varied potencies, types of activity, and undesired side effects (see Table 3). This necessitates pharmacological evaluation of particular stereoisomers and racemates. As a result, when proposing structures for prospective leads and analogues, all of these stereochemical properties must be considered. The extent to which these structural traits may be exploited, however, will be determined by our understanding of the target biological system's structure and biochemistry.

2.8.1 Structurally rigid groups

Unsaturated groups of all kinds, as well as saturated ring systems, are structurally stiff (Figure SC-TRB-47). Esters and amides, as well as aliphatic conjugated systems, aromatic and heteroaromatic ring systems, fall into the first category. The binding of these rigid structures to a target site can reveal details about the shape of the site as well as the nature of the ligand-site interaction. Furthermore, because the structure is rigid, it can be swapped with other rigid structures of comparable size and shape to create analogues with different binding characteristics and, as a result, varying activity or potency.

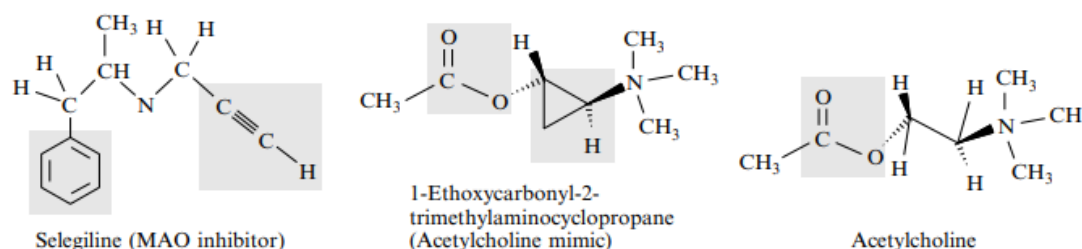


Figure SC-H&M-TRB - 20: Examples of structural groups that impose a rigid shape on sections of a molecule. The shaded areas represent the rigid sections of the molecule.

2.8.2 Conformation

Schueler and Archer proposed in the 1950s and early 1960s that the flexibility of the architecture of both ligands and receptors explained why the same ligand may bind to distinct sub-types of a receptor. Archer also came to the conclusion that when a ligand binds to distinct sub-types of a receptor, it appears to take on different conformations. Acetylcholine, for example, has both muscarinic and nicotinic properties. Archer et al. proposed that the anti or staggered conformation was responsible for muscarinic activity, while the syn or eclipsed form was responsible for nicotinic activity (Figure SC-TRB-48).

Bulky substituents, unsaturated structures, and ring systems are the most common ways to introduce conformational limitations. The most prevalent option is a ring system (Figure SC-TRB-49). In all circumstances, the structures utilized must be carefully chosen, as there is always the risk of steric hindrance preventing the analogue from binding to the target. However, if enough

information is available, molecular modeling can be very useful in determining which structures to use.

2.8.3 Configuration

Configurational centers provide the portions of the molecule where they occur a rigid shape. Their existence, however, causes geometric and optical isomerism. Because the geometries of these stereoisomers differ, biologically active stereoisomers will often have variable potencies and/or activity (Table 3). When a chiral center is positioned in a crucial position in the structure of a molecule, pharmacological variants are more likely. As a result of these discrepancies, it is now required to produce and test all of a drug's different stereoisomers independently.

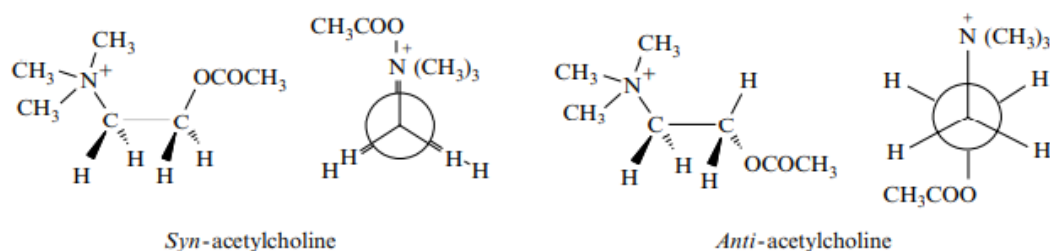


Figure SC-TRB-1: *Syn* and *Anti* Conformers of Acetylcholine and 2-Tropanyl Ethanoate Methiodides

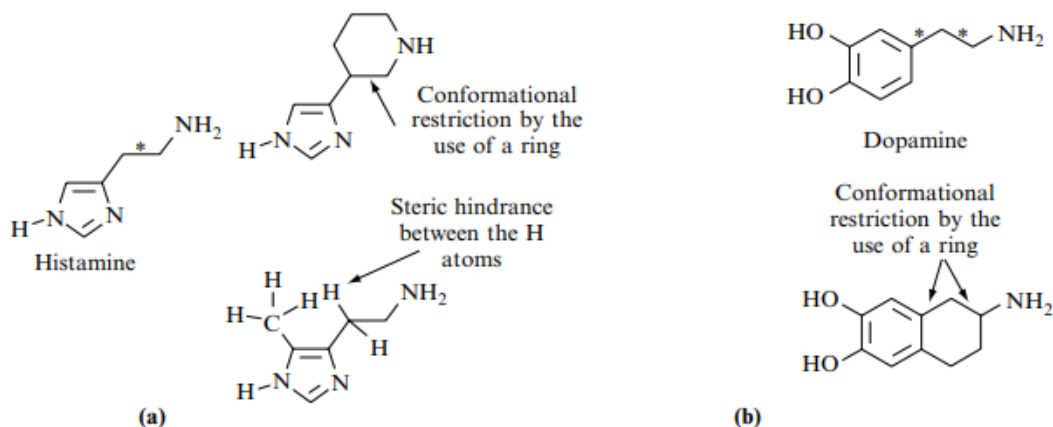


Figure 3.4 Examples of the use of conformational restrictions to produce analogues of (a) histamine and (b) dopamine. Bonds marked * can exhibit free rotation and form numerous conformers

Figure SC-H&M-TRB - 21: Examples of the use of conformational restrictions to produce analogues of (a) histamine and (b) dopamine. Bonds marked * can exhibit free rotation and form numerous conformers

Varied stereoisomers will have different physiochemical properties, including as absorption, metabolism, and elimination, in addition to an effect on activity. (-)-norgestrel, for example, is absorbed twice as quickly as (p-+)-norgestrel across the buccal and vaginal membranes. S-indacrinone has a plasma half-life of 2–5 hours, while the R isomer has a half-life of 10–12 hours.

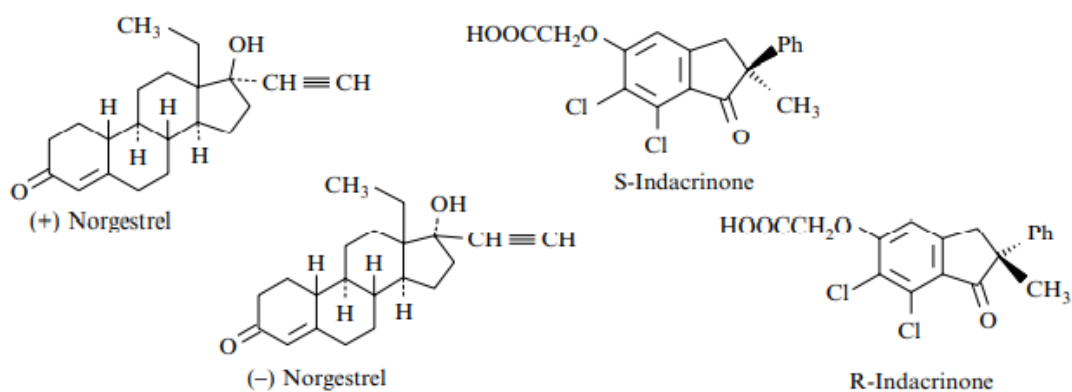


Figure SC-H&M-TRB - 22: (-) Norgestrel and R-Indacrinone

2.8.4 Solubility and Drug Design

The relative solubilities of medicines in the body's aqueous media and lipid tissues have a significant role in their absorption and transport to their sites of action. A medicine must usually have a suitable amount of water and lipid solubility to pass through a membrane. Water solubility that is appropriate will often improve drug distribution and action inside the circulatory system.

2.8.4.1 The importance of water solubility

Because our bodies' cells contain roughly 65 percent water, a drug's solubility and behavior in water is extremely essential. Water serves as an inert solvent, a dispersion medium for colloidal solutions, and a nucleophilic reagent in a variety of biological reactions in living matter. Furthermore, the conformations of biological macromolecules are influenced by hydrogen bonding and hydrophobic interactions in water, which impacts their biological behavior. It also simplifies medication toxicity testing, bioavailability assessment, and clinical use. This means that early in the creation of a novel drug, it is usually necessary to design a suitable degree of water solubility into the structure of that molecule.

Orally administered solid or suspension drugs must dissolve in the aqueous stomach fluid (**dissolution**) before being absorbed and delivered to their site of action via the systemic circulation. The rate and extent of a drug's dissolution are important factors in determining how well it is absorbed. This is because the concentration of the medication in the fluid in the gut lumen is one of the most important parameters influencing drug transfer across the gastrointestinal tract's membranes (see Appendix 5). (GI tract). The rate of dissolution is determined by the solid's surface area, which is determined by the physical nature of the drug's dosage form as well as its chemical structure. The extent of dissolution, on the other hand, is solely determined by the drug's solubility, which is determined by the chemical structure of the drug. The dosage form is a formulation challenge that is generally beyond the scope of the medicinal chemist's expertise, but the design of the structure of lead compounds in terms of solubility is.

The drug's water solubility will influence its ease of transport to the bodily compartments available to it once it has reached the circulatory system, either through absorption or direct administration. Drugs that are only slightly soluble in water may be deposited en route to their target, clogging blood vessels and causing organ damage. Many sulphonamides, such as sulphamethoxazole, crystallize in the kidney, potentially causing catastrophic liver and kidney damage. Drug transport through membranes is also influenced by water solubility.

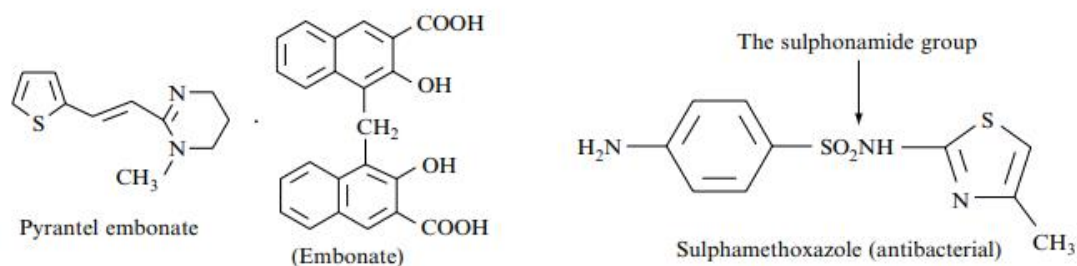


Figure SC-H&M-TRB - 23: Pyrantel Embonate, Embonate, Sulphamethoxazole (antibacterial)

Orally administered solid or suspension drugs must dissolve in the aqueous stomach fluid (dissolution) before being absorbed and delivered to their site of action via the systemic circulation. The rate and extent of a drug's dissolution are important factors in determining how well it is absorbed. This is because the concentration of the medication in the fluid in the gut lumen is one of the most important parameters influencing drug transfer across the gastrointestinal tract's membranes (GI tract). Because taste is dependent on the chemical producing an aqueous solution, a drug's low water solubility can be utilized to create drug depots, chewable dosage forms, and hide bitter-tasting pharmaceuticals.

Because water solubility is so important in drug action, one of a medicinal chemist's research goals for a novel medicine is to create analogues with the requisite level of water solubility.

2.8.5 Solubility and Drug Structure

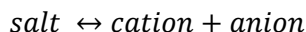
The degree of solvation of a chemical in the solvent determines its solubility. A more soluble solute will have structural characteristics in the molecule that boost the degree of solvation. As a result, an organic compound's water solubility is determined by the amount and kind of polar groups in its structure, as well as the size of its carbon–hydrogen skeleton. In general, the larger the proportion of polar groups to total carbon atoms in a structure, the more water soluble it is. Additionally, aromatic molecules are less soluble than their nonaromatic counterparts. It is possible to examine the relative solubilities of compounds with similar carbon–hydrogen skeletons using these general criteria. However, as the structure becomes more complex, these assessments become less accurate.

A lead compound's water solubility can be increased in three ways: salt production, incorporation of water solubilizing groups into its structure, particularly those that can hydrogen bond with water, and the employment of specific dosage forms, the latter of which is outside the focus of this article. The drug's activity is usually unaffected during salt production, however its potency may differ. When new structural groups are integrated into the structure of a drug, however, the drug's activity may be altered. As a result, a comprehensive trial program on the new analogue will be required. Both of these alterations can be expensive if they have to be done at a late stage in the medication development process. Specialized dosage forms do not normally necessitate major changes to the trials program, but these formulation methods are only appropriate for certain medications.

The structural elements that affect a compound's lipid solubility are diametrically opposed to those that control its water solubility. As a result, substituting polar groups with nonpolar structures or groups that are much less polar in nature may increase lipid solubility.

2.8.5.1 Salt Formation

Salt formation usually improves the water solubility of acidic and basic drugs because the salts of these drugs dissociate in water to produce hydrated ions:



If hydrogen and hydroxide ions mix with the right cation or anion to generate less soluble acids or bases, they can disrupt this equilibrium. As a result, the pH of a biological fluid can influence a drug's solubility and, as a result, its action. Increasing the salt's hydrophilic character should boost its water solubility in general. This generalization, however, has many exceptions, and each salt should be judged on its own merits.

Acidic medications are frequently transformed to metallic or amino salts, whereas basic drugs are usually converted to organic acid salts. (Table 4).

Table 5: Examples of Acids and Bases used to form salts of drugs

Anions and anion sources	Cations and cation sources
Ethanoic acid – ethanoate (CH_3COO^-)	Sodium – sodium ion (Na^+)
Citric acid – citrate (see Figure 3.5)	Calcium – calcium ion (Ca^{2+})
Lactic acid – lactate (see Figure 3.5)	Zinc – zinc ion (Zn^{2+})
Tartaric acid – tartrate (see Figure 3.5)	Diethanolamine – R_2NH_2^+ (see Figure 3.5)
Hydrochloric acid – chloride (Cl^-)	N-Methylglucamine – $\text{RNH}_2^+\text{CH}_3$ (see Figure 3.5)
Sulphuric acid – sulphate (SO_4^{2-})	2-Aminoethanol ($\text{HN}_2\text{CH}_2\text{CH}_2\text{OH}$) – RNH_3^+
Sulphuric acid – hydrogen sulphate (HSO_4^-)	

The structure of the acid or base used to make the salt determines its degree of water solubility. Acids and bases containing water solubilizing groups in their structures, for example, will create salts that are more water soluble than compounds without these groups (Figure SC-TRB-52).

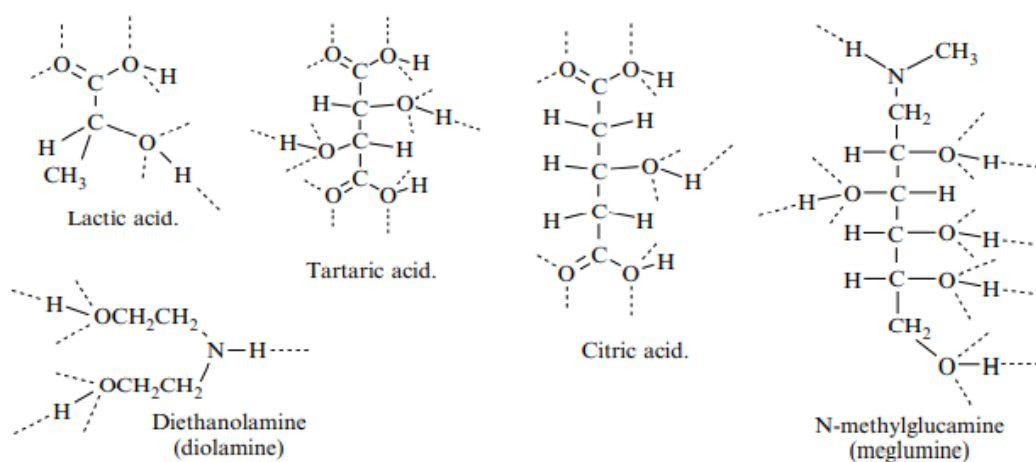


Figure SC-H&M-TRB - 24: Diethanolamine (diolamine), Lactic Acid, Tartaric Acid, Citric Acid, and N-methylglucamine (meglumine)

However, if a medicine is too water soluble, it will not dissolve in lipids and thus will not be transported easily through lipid membranes. This usually results in a decrease in activity or an increase in the time it takes to take effect. Because of the common ion effect, the presence of a high concentration of chloride ions in the stomach will lower the solubility of sparingly soluble chloride salts.

2.8.6 The incorporation of water solubilizing groups in a Structure

The incorporation of water solubilizing groups into the structure of a lead compound can be neatly divided into four general categories::

1. the type of group introduced;
2. whether the introduction is reversible or irreversible;
3. the position of incorporation and
4. the chemical route of introduction.

2.8.6.1 Type of group

Alcohol, amine, amide, carboxylic acid, sulphonic acid, and phosphorus oxyacid groups, which either ionize or have very high intermolecular forces of attraction with water (hydrogen bonding), will usually result in counterparts with increased water solubility. Acidic and basic groups are especially helpful because they can be utilized to make salts, giving the final product a greater choice of dosing forms. However, introducing an acid group into a structure containing a base or a base group into a structure containing an acid group can limit water solubility by forming zwitter-ions. Weakly polar groups, such as carboxylic acid esters, aryl halides, and alkyl halides, do not considerably improve water solubility but do increase lipid solubility. In any case, the degree of solubility obtained by inclusion cannot be predicted precisely because it is dependent on a variety of other parameters. As a result, the type of group that is introduced is usually chosen based on previous experience.

Acidic residues in a lead structure are less likely to influence the type of activity, although they can cause the analogue to exhibit haemolytic features. Furthermore, aromatic acid groups usually have anti-inflammatory properties, whereas carboxylic acids with an alpha functional group might operate as chelating agents. Because bases frequently interact with neurotransmitters and biological processes involving amines, basic water solubilizing groups have a tendency to modify their mode of action. However, because of their inclusion, the analogue can be produced as a variety of acid salts. Acidic and basic groups have drawbacks that nonionizable groups do not.

2.8.6.2 Reversibly and irreversibly attached groups

The type of group chosen is also determined by the needed level of permanence. Less reactive C–C, C–O, and C–N bonds are likely to permanently attach groups to the carbon skeleton of the lead complex. As the analogue is transported from its point of administration to its site of action, groups attached to the lead via ester, amide, phosphate, sulphate, and glycosidic linkages are more likely to be metabolized from the resultant analogue to rebuild the parent lead molecule. As prodrugs, compounds having this type of solubilizing group are more likely to have the same activity as the parent lead molecule. The rate of loss of the solubilizing group, on the other hand, will be determined by the type of the transfer pathway, which could alter the drug's action.

2.8.6.3 The position of the water solubilizing group

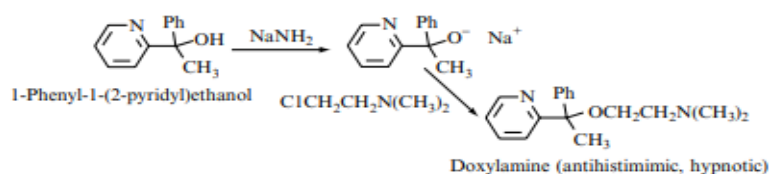
The water solubilizing group should be linked to a region of the structure that is not engaged in the drug–receptor interaction in order to preserve the sort of activity displayed by the lead molecule. As a result, the pathway taken to introduce a new water solubilizing group and its location in the lead structure will be determined by the pharmacophore's and the remainder of the molecule's relative reactivities. The chemicals used to add the new water solubilizing group should be chosen so that they do not react with the pharmacophore or are in close proximity to it. This reduces the chance that the new group will have an impact on the relevant drug–receptor interactions.

2.8.6.4 Methods of Introduction

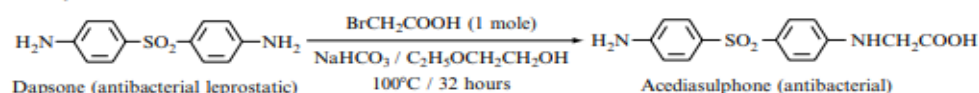
Water solubilizing groups are best added in the start of a drug's production, but they can be added at any point. The problem of a later introduction affecting the kind and/or character of the drug–receptor interaction is avoided by starting at the beginning. To introduce a water solubilizing group, a variety of approaches can be utilized; the one chosen will depend on the type of group being introduced and the chemical nature of the target structure (Figures SC-TRB-53 and SC-TRB-54). To prevent undesired interactions of either the water solubilizing group or the lead structure, several of these methods require the addition of protective agents.

ACID and BASIC GROUPS

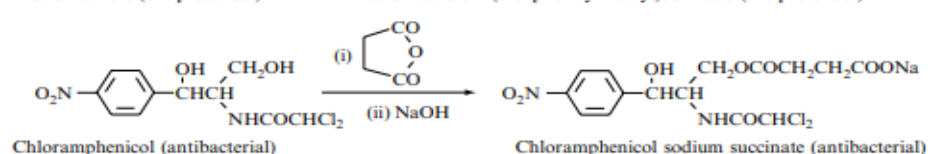
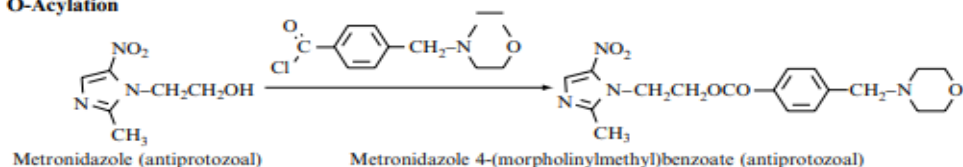
O-Alkylation



N-Alkylation



O-Acylation



N-Acylation

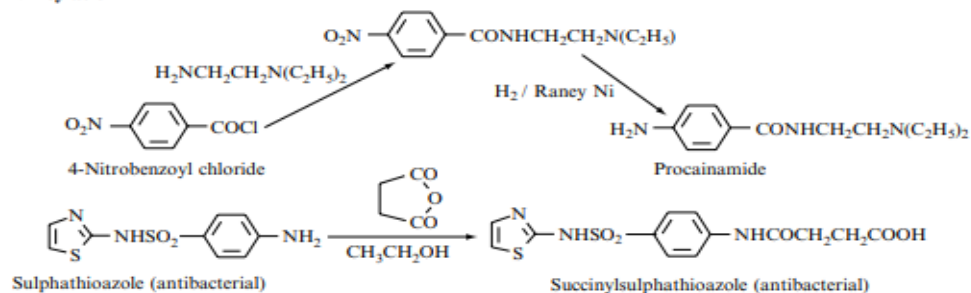
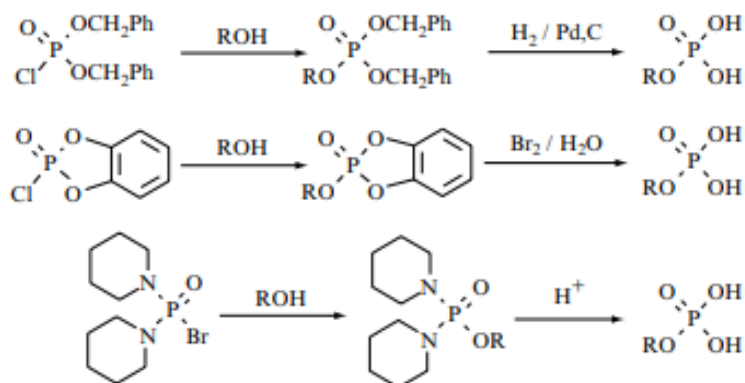
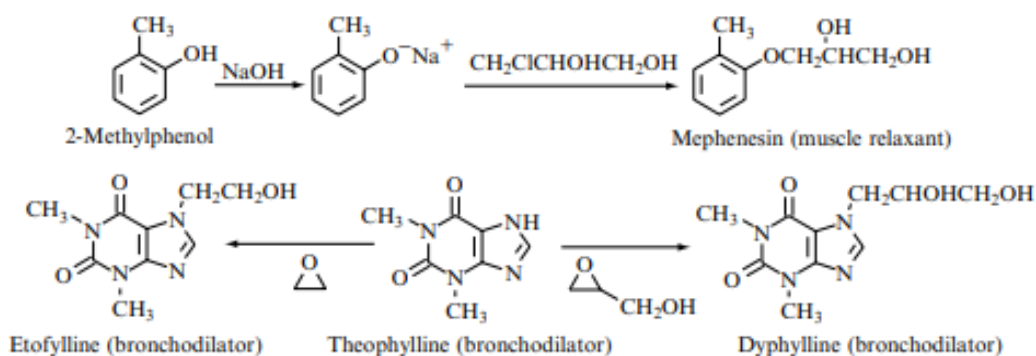


Figure SC-H&M-TRB - 25: Examples of water solubilizing structures and the routes used to introduce them into the lead structures. O-alkylation, N-alkylation, O-acylation and N-acylation reactions are used to introduce both acidic and basic groups. Acetylation methods use both the appropriate acid chloride and anhydride.

PHOSPHATE GROUPS



HYDROXY GROUPS



SULPHONIC ACID GROUPS

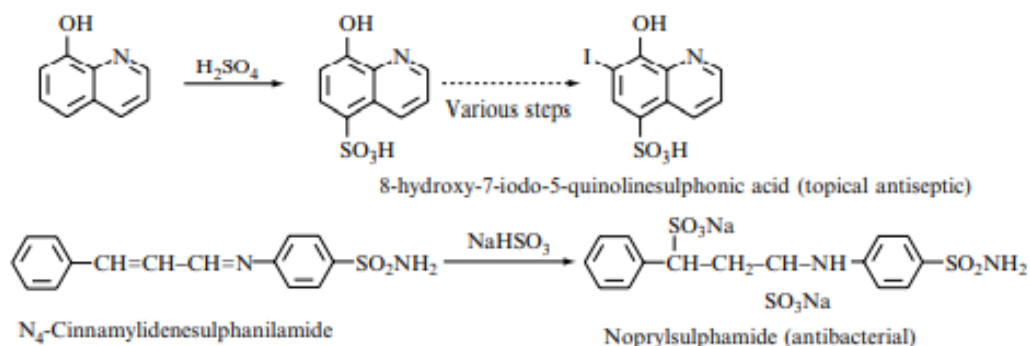


Figure SC-H&M-TRB - 26: Examples of water solubilizing structures and the routes used to introduce them into lead structures. Phosphate acid halides have been used to introduce phosphate groups into lead structures. The hydroxy groups of the acid halide must normally be protected group. These protecting groups are removed in the final stage of the synthesis to reveal the water solubilizing phosphate ester. Structures containing hydroxy groups have been introduced by reaction of the corresponding monochlorinated hydrin and the use of suitable epoxides amongst other methods. Sulphonic acid groups may be introduced by either direct sulphonation or the addition of bisulphite to reactive C=C bonds amongst other methods.

SAMPLE TOPIC ACTIVITIES

Activity 1: Group Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover the drug discovery phases.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 2: Practical

Title: Synthesizing Oil of Wintergreen from Aspirin.

Source: PharmaChemistry in the Classroom: A Drug-Discovery Experiment for the High School Chemistry or Biotechnology Classroom (URL: [ed300336c.pdf](#))

Experiment Information

In the experiment presented here, students execute many roles of scientists within the drug-discovery pipeline: identifying a medical problem, performing a chemical synthesis to make a new molecule, characterizing the new molecule, and performing a biological assay comparing their starting material to their product. More specifically, this experiment involves the search for a “new” antibacterial agent. As an introduction to the experiment, students are taught about pathogenic bacteria and the potential health threat it poses. Students are then directed to use the natural product acetylsalicylic acid (aspirin) as a lead molecule. Next, students are guided through a chemical synthesis where they transform the lead molecule aspirin into methyl salicylate, more commonly known as oil of wintergreen. Unlike previously published articles on the synthesis of oil of wintergreen from aspirin,^{6,7} this experiment further investigates the starting material and product. On the basis of observations of physical properties before and after the synthesis, students see evidence that their reaction was successful and that their molecule has changed (solid to oil, bitter smell to minty smell, white color to colorless). The students then grow bacteria in the presence of the two different molecules and observe the effect each molecule has on bacterial survival by measuring the area around the molecule where bacterial growth is halted, also known as a zone of inhibition. Following the experiment, students convene to discuss their results and talk about the importance of those results in light of the drug-discovery process.

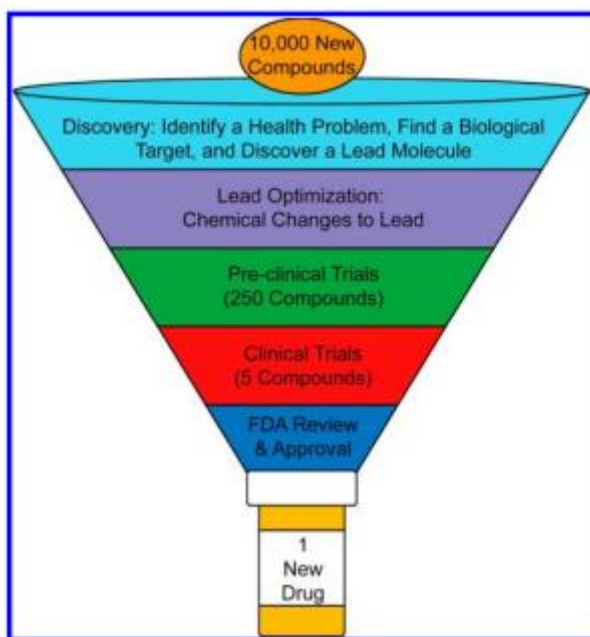


Figure 1: Drug Discovery Process steps

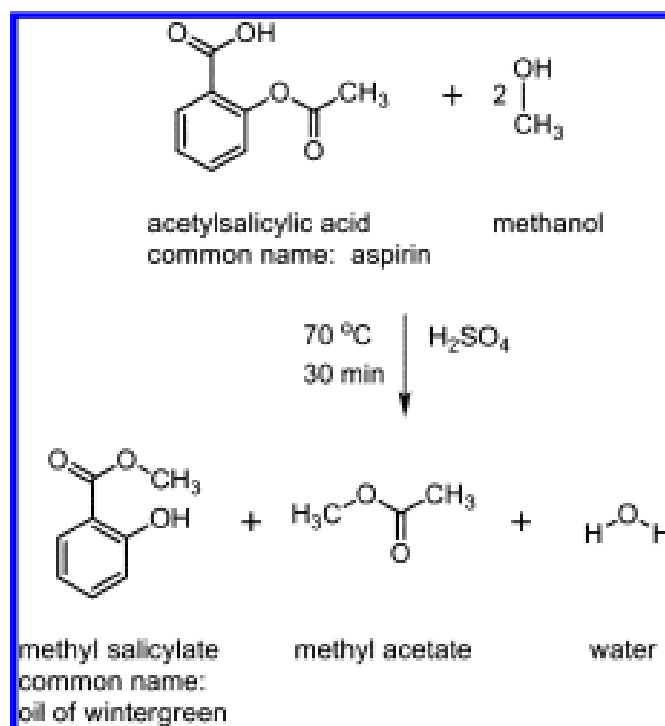
Materials

Methanol, sulfuric acid, salicylic acid, and ethanol were purchased from Sigma-Aldrich. Filter paper, large (90 cm diameter) Petri dishes, agar, and 6-well plates were purchased from Flinn Scientific. Aspirin (regular strength 325 mg tablets, noncoated, nonbuffered), cotton swabs, and bleach were purchased from a local grocery store. Gloves were purchased from the local hardware store. Mortar and pestles were purchased from Flinn Scientific. Bacteria stab used for making preplated lawns was provided by ScienceBridge.⁸ Paper discs were made from standard filter paper using a 1/2 in. hole punch. Sterile water was purified using a mega-pure water purification system from Flinn Scientific. The experiments require a fume hood, water bath, and a culture incubator.

Synthesis of Oil of Wintergreen from Aspirin Tablets

Students should be provided with two tablets of 325 mg regular strength aspirin. After grinding the tablets using a mortar and pestle, students analyzed the physical properties of the powder, which would be compared with the product to verify a successful reaction. Next, students dissolved the active ingredient, acetylsalicylic acid, in methanol and filtered off the inactive ingredients. In the fume hood, students added two drops of the catalyst, concentrated sulfuric acid, to the test tube containing their sample, and placed it in a water bath set to 70 °C. After 30 min the transesterification, or Fisher esterification, reaction was complete and the sample was removed from the water bath (Scheme 1). The physical properties of this sample were evaluated as the product. To fit into the time constraints, equipment availability, and approved chemical list of the school district, the lab was adapted accordingly from previously published procedures.⁹

Scheme 1. Synthesis of Methyl Salicylate from Acetylsalicylic Acid



Antibacterial Testing

Students were provided with agar plates and preplated colonies to create their own bacterial lawn, as well as solutions of aspirin and salicylic acid for preparation of paper discs for antibacterial testing. Students prepared their samples by placing 2–3 drops of these solutions onto the filter paper discs.¹⁰ While allowing the discs to dry, students created a lawn of bacteria on large agar plates using a cotton swab and spreading a colony of pregrown *Escherichia coli* bacteria onto their sample plate. After allowing 5 min for the paper discs to dry, students placed the paper discs onto appropriately labeled quadrants and placed them in an incubator set to 37 °C overnight. On day two, students took the agar plates out of the incubator and used a ruler to measure the zone of inhibition of samples from the paper discs.

HAZARDS

Sulfuric acid is corrosive and must be handled with care. Methanol is extremely flammable and toxic by inhalation, ingestion, or absorption through the skin. Therefore, gloves should be used. Acetylsalicylic acid and methyl salicylate are irritants. Bacteria strains used should be harmless, and students should be instructed to wash hands thoroughly after handling. Heating of methanol results in evaporation and should be carried out in a fume hood. All procedures should be carried out wearing safety glasses.

RESULTS AND DISCUSSION - Comparing Physical Properties between Starting Material and Product

The reaction of aspirin (acetylsalicylic acid) to form oil of wintergreen (methyl salicylate) is a convenient synthesis utilizing relatively common chemical reagents, heat, and a sulfuric acid catalyst. By analyzing the physical properties before and after the reaction, students were able to observe notable physical changes between starting material and product (Table 1). From class discussions

and pre- and postlab questions, students studied the synthesis and were able to observe that small changes in a molecule can lead to substantial differences in physical nature.

Table 1. Sample Student Observations of Physical Properties before and after Chemical Synthesis

Physical Property To Observe	Observations of Starting Material: Acetylsalicylic Acid (aspirin)	Observations of Product: Methyl Salicylate
Smell ^a	Bitter	Minty
Color	White	Colorless
State	Solid	Liquid

^aThe example of smell as a physical property is provided for the students, and they are responsible to fill in the others.

As students observed, acetylsalicylic acid was a white solid exhibiting a bitter odor and methyl salicylate was a colorless oil with a minty odor, indicating a successful synthesis. The synthetic procedure allowed students to play the role of a medicinal chemist in the search for new potential drugs.

Testing Oil of Wintergreen for Antibacterial Activity

The biological assay was performed by plating the starting material (acetylsalicylic acid or aspirin), the product (methyl salicylate or oil of wintergreen), a positive control (salicylic acid, a known antibacterial substance), and a negative control (evaporated ethanol exhibiting no antibacterial effect) to show a comparison between antibacterial and nonantibacterial substances.⁹ After incubation, the students were able to observe whether a substance was antibacterial through the absence or presence of a zone of inhibition, or a clear area where bacterial growth is inhibited around the impregnated paper discs.⁴ Aspirin is not antibacterial and will therefore exhibit no zone of inhibition. Oil of wintergreen, however, is antibacterial and will exhibit a clear ring around the paper disc soaked with the compound. The diameter of the zone of inhibition can be compared to known antibiotics and will be proportional to the strength of antibacterial activity. In comparison with the positive and negative controls, students observed that oil of wintergreen killed bacteria, whereas aspirin did not (Figure 2). Students also compared data between groups to conclude that they created a new molecule with different medicinal properties. In completing their observations and participating in useful discussions, students came full circle in mirroring the processes used in drug discovery. Through further discussion, students were directed to analyze the active ingredients from the labels of hygienic products such as mouthwash and topical analgesics and were able to observe and discuss standard uses for methyl salicylate (Figure 3).¹¹

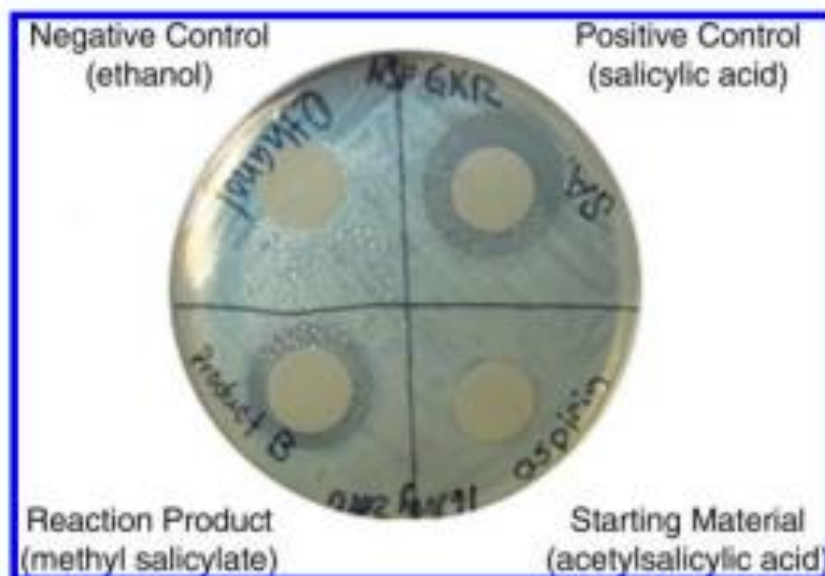


Figure 2: The plate should illustrate the antibacterial effects of the negative control, evaporated ethanol (upper left quadrant), the positive control, salicylic acid (upper right quadrant), the starting material, acetylsalicylic acid (lower right quadrant), and the reaction product, methyl salicylate (lower left quadrant).

Active Ingredients of Mouthwash	Purpose
Eucalyptol	Antiplaque/antigingivitis
Menthol	Antiplaque/antigingivitis
Methyl salicylate	Antiplaque/antigingivitis
Thymol	Antiplaque/antigingivitis

Figure 3: Discuss the use of methyl salicylate (oil of wintergreen) in common products. One example used is looking at the active ingredients of antiseptic mouthwash.

Activity 2: Exercises

- (1) Outline the general steps in the discovery of a new drug.
- (2) Outline by means of suitable examples the significance of:
 - (i) structurally rigid groups,
 - (ii) conformations and
 - (iii) configuration on the design of new drugs.
- (3) Explain why water solubility is an important factor in drug design.
- (4) List the structural features that would indicate whether a compound is likely to be reasonably water soluble. Illustrate the answer by reference to suitable examples.
- (5) Suggest general methods by which the water solubility of a compound could be improved without affecting its type of biological action.
- (6) Suggest, by means of chemical equations, one route for the introduction of each of:
 - (i) an acid residue and
 - (ii) a basic residue into the structure of 4-hydroxybenzenesulphonamide when the sulphonamide group acts as the pharmacophore in this compound.

Topic 3: Drug Development and Production

3.1 Introduction

Drug development is the process of turning a physiologically active chemical into a safe, commercial product. It's a multidisciplinary process that needs collaboration between teams of employees from many professions. Its success is dependent on their judgment and ability. The work that these groups undertook at various stages of the development process will be described in this topic. Many of these tasks are interdependent, implying that they should be completed in the order listed or at the same time. As a result, given the necessity of speed in all aspects of growth, these jobs will require meticulous preparation and organization.

It normally takes seven to 10 years from the commencement of the development process to the commercialization of the drug. Furthermore, only one out of every 400–1000 pharmaceutical candidates that are evaluated for development makes it to market. As a result, development is exceedingly expensive, with the average cost of producing a successful treatment in 1999 estimated to be approximately 350 million pounds. Because development is such a high-risk endeavor, a comprehensive strategy for mitigating both medical and financial risks must be devised. The first step in this method is to figure out if the treatment will achieve its pharmacological and financial goals. As a result, it's vital to figure out whether the new treatment will be able to compete with existing drugs and, if so, what advantages it will have. This review should be undertaken at appropriate points during the process to ensure that the development is still sustainable. This, as well as other aspects of the project, are frequently managed by keeping an eye on its critical path (See Figure SC-H&M-TRB-25).

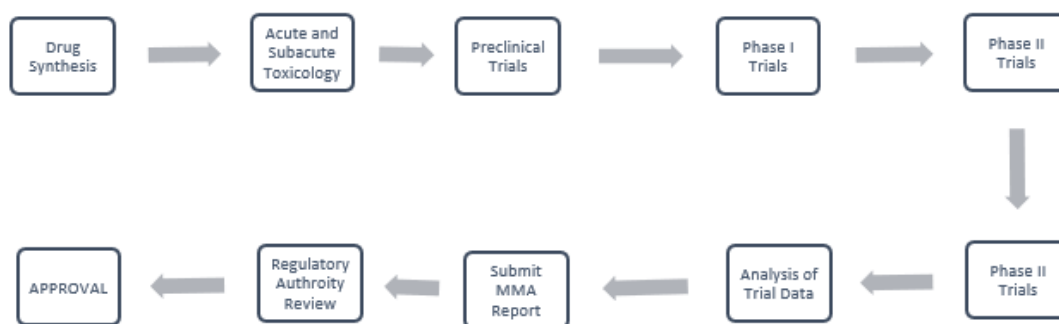


Figure SC-H&M-TRB - 27: An example of the activities that are likely to form the critical path for the development of a drug. MMA is the market authorization application that a company has to submit in Britain in order to produce and market a drug.

This comprises efforts that determine how long a drug takes to register with the appropriate government body. It does not include all of the steps involved in the development and testing of a drug. On the other hand, monitoring the critical route is a useful way to guarantee that the project continues on schedule.

3.2 Chemical Development

The reactions employed by researchers usually use expensive reagents and only yield enough medication for a limited number of biological tests. Furthermore, research methods frequently employ techniques, such as chromatography, that are incompatible with large-scale production procedures. As a result, the first stage in developing chemical synthesis is to find safe and cost-effective alternatives. These alternative reactions should start with inexpensive, easily available components, contain more cost-effective reactions with high yields, and be safe to perform on a large laboratory or factory scale.

They should produce sufficient quantities of lead in acceptable purity for first activity and toxicity testing. Many large-scale synthetic routes will be explored first in order to discover the optimal route. Because they give greater overall yields, convergent techniques are frequently preferred over linear syntheses. Creating a chemically efficient synthesis is often more important in the early stages of research than manufacturing enough lead to conduct extensive testing. Because the results of these first tests will allow company officials to decide if the chemical needs further investigation, this is the case.

Higher quantities of the drug will be required as the drug develops through the development process. In order to address this need, the most promising laboratory synthesis is typically scaled up to pilot plant scale. Converting a synthesis to pilot plant size, which is effectively a mini-manufacturing process, may involve changes to the synthesis to allow for large-scale equipment to be used (see Chemical Engineering Issues section of this Topic). These modifications, waste disposal, and the dangers associated with carrying out the synthesis with larger amounts of reagents should all be considered before performing the conversion (see Chemical Plant, Health and Safety Considerations section of this Topic). The pilot plant should be able to produce enough medication to meet the needs of the regulatory authority. Once a license to manufacture the medicament has been granted, the pilot plant synthesis is transferred to the complete manufacturing process. In this shift, chemical engineering, the environment, waste disposal, safety, quality control, and cost are all important factors to consider.

3.2.1 Chemical Engineering Issues

Chemical engineers must first determine if the reactions can be safely carried out in the existing plant or if a new facility is required. They should also think about if the synthesis can be tweaked to work with existing equipment, as new equipment could be prohibitively expensive. A stainless-steel container is the most common large-scale reaction vessel. A stirring paddle, a heating jacket for managing reaction temperature, openings for loading and unloading solids and liquids, and provisions for distilling or refluxing liquids are all common features. (See Figure SC- H&M-TRB-26)

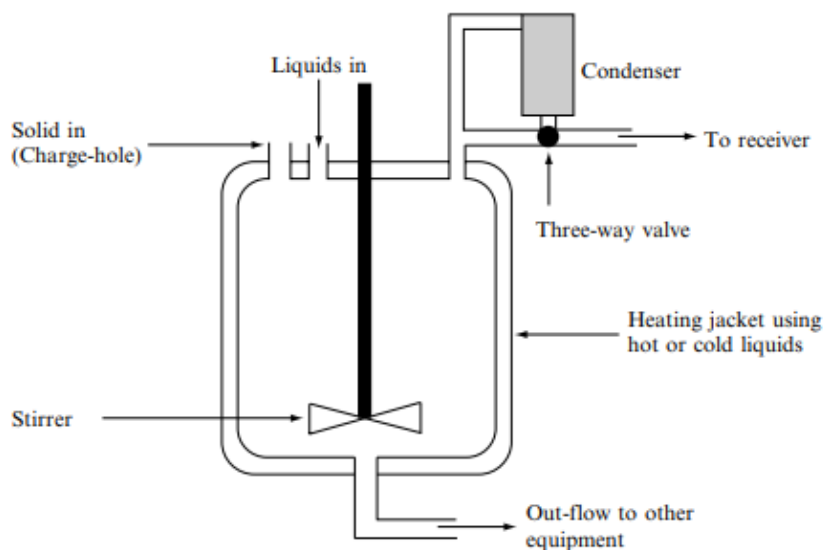


Figure SC-H&M-TRB - 28: A diagrammatic representation of a typical reaction vessel. The three-way valve enables the condenser to be used to either reflux or distil the liquids in the reaction vessel.

Reaction vessels are commonly connected to other pieces of specialized gear via appropriate piping, such as filtration, crystallization, and drying equipment. A variety of methods are used to transmit the product from the reaction vessel to these other pieces of equipment. Among them are pumps, nitrogen gas under pressure, screw, and conveyor systems. The lower liquid phase is generally separated by using the reaction vessel as a separator and allowing it to flow out the bottom. Plant operators can monitor the contents of the vessel and/or piping by installing appropriate sensor systems and small armored glass windows known as sight-glasses, allowing them to regulate the reaction and, if necessary, future operations.

When choosing reactions for pilot and manufacturing processes, plant constraints such as the reaction vessel and supporting equipment required for product isolation and purification must be taken into account. At temperatures ranging from 15 to 140 degrees Celsius, reaction vessels are routinely utilized. As a result, while heating and cooling are both expensive, the reactions chosen should produce a reasonable yield as close to ambient temperature as possible while remaining within the vessel's temperature control constraints. Furthermore, large-scale reactions require longer than laboratory preparations to reach their working temperature, produce a sufficient yield, and cool down, necessitating longer processing times. In the contemporary context, this involves the use of stable reagents, products, solvents, and catalysts.

3.2.2 Chemical Plant, Health and Safety Considerations

Health and safety are critical considerations when selecting reactions for pilot or industrial operations. A full chemical hazard evaluation should be completed before commencing the reaction on a large scale. The chemistry of the process, the safety of the plant's employees, waste disposal, and the environmental impact should all be considered. Chemical and biological hazards of all chemicals and solvents used in the processes should be documented, as should protocols for coping with mishaps, leaks, and spills. Even the most stringently regulated chemical plants will leak chemicals into the environment, such as solvent vapour and chemical dust, therefore monitoring equipment must be suitable for the compounds in use. Only use reactions

involving the handling of extremely toxic substances if there are no other possibilities. A tight protocol governing their use must be created and implemented in this case. Any health and safety protocol must consider the nature of the processes' thermochemistry. The heat production of a large-scale reaction may exceed the cooling capacity of present equipment, resulting in disastrous consequences. As a result, if the reaction temperatures for each of the process steps are unknown, they must be determined and factored into the plant's and workers' safety assessments.

Waste disposal and its environmental impact are becoming increasingly important, and they must now be taken into account as part of the development process. Solid wastes were originally buried at permitted landfill sites, but this is no longer recommended due to the risk of water pollution, unless other solutions, such as incineration, are unavailable. The sewage system, rivers, and the sea are all allowed to accept aqueous liquids. Metals like copper, zinc, mercury, and cadmium, as well as organic halogen compounds like chloroform, which are extremely poisonous to many aquatic organisms, must be removed in large quantities. Distillation is a typical method for recovering solvents, however it is not always possible for some solvents, such as dimethyl sulphoxide and N,N-dimethylformamide. Gases can often be stopped from entering the environment by reacting with the right absorbents in absorbent towers. However, gases that cannot be disposed of this way, such as nitric oxide, require specialized treatment, which can be rather expensive. Dumping highly toxic waste should be avoided at all costs, as it may require costly treatment.

An idea of the impact of a process on the environment may be determined from its effluent load factor (ELF). This is defined as:

$$ELF = \frac{(\text{mass of all ingredients of all the stages} - \text{mass of the produce})}{\text{mass of the product}}$$

The ELF is the amount of waste created by the process per unit mass of the product. It can refer to one or more steps in the synthesis process, as well as the overall procedure. In an ideal situation, where all of the ingredients are converted into the product, the ELF value will be zero. This does not happen in synthetic processes. In pharmacological operations, the ELF is usually in the region of 100. Process design chemists will strive to lower the ELF by selecting processes and operating the facility in order to save money.

3.2.3 Synthesis Quality Control

The ability to identify and assess the chemical purity of the drug, as well as the chemical purity of the intermediate molecules used in each step of the synthesis, will be crucial to drug production efficiency. This usually necessitates the development of sufficient chemical quantitative and qualitative protocols for all of the compounds used in the pilot and manufacturing processes. Physical methods such as HPLC and GC are routinely utilized for these purposes. The licensing authorities also demand the ability to identify impurities, as well as any structurally similar substances or stereoisomers. As a result, much care should be given to how all of these studies could be carried out while creating the production route.

In some synthetic methods, an intermediate product can be used in the next stage of synthesis without being separated and purified. This operation is known as telescoping. It has the

advantage of not requiring hazardous intermediates to be handled. It also makes it easier to work with non-crystalline and oily materials. In order for telescoping to operate, the first reaction must produce a relatively pure result. As a result, when this procedure is used, an analytical approach must be present to ensure that the purity of the product is high enough for the next reaction to be carried out efficiently.

3.2.4 A Case Study

Process Development, Fine Chemicals from Grams to Kilograms, Oxford University Press, 1995. Adapted from S. Lee and G. Robinson, Process Development, Fine Chemicals from Grams to Kilograms, Oxford University Press, 1995. The release of thromboxane A₂ by the body can result in bronchoconstriction, blood vessel constriction, and platelet aggregation, among other things (see Figure SC- H&M-TRB-27a). As a result of this research, ICI discovered a variety of chemicals that can operate as thromboxane antagonists (see Figure SC-TRB-67b).

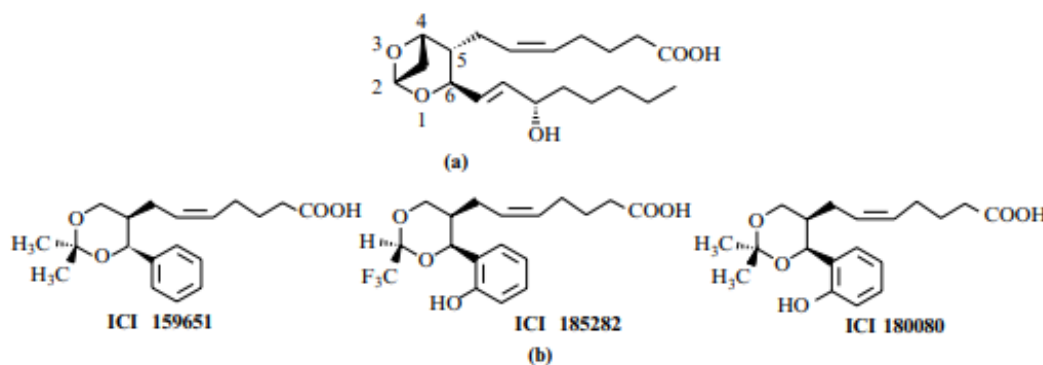


Figure SC-H&M-TRB - 29: (a) Thromboxane A₂ (TXA₂). Note: The locants given are those of the dioxan ring only and not those for the complete molecule. They are used for reference purposes only. (b) Examples of some of the thromboxane antagonists developed by ICI

It was found that significant antagonist activity is obtained when:

1. A substituent cis to the other dioxan ring substituents exists at position 2,
2. The phenolic and alkenoic acid side chains at positions 5 and 6 have a cis orientation, and
3. the alkenoic acid side chain's C=C has a cis configuration.

These active analogues were first synthesized in gram quantities, employing technologies that enabled the manufacture of a large number of analogues for early biological testing (see Figure SC-TRB-68).

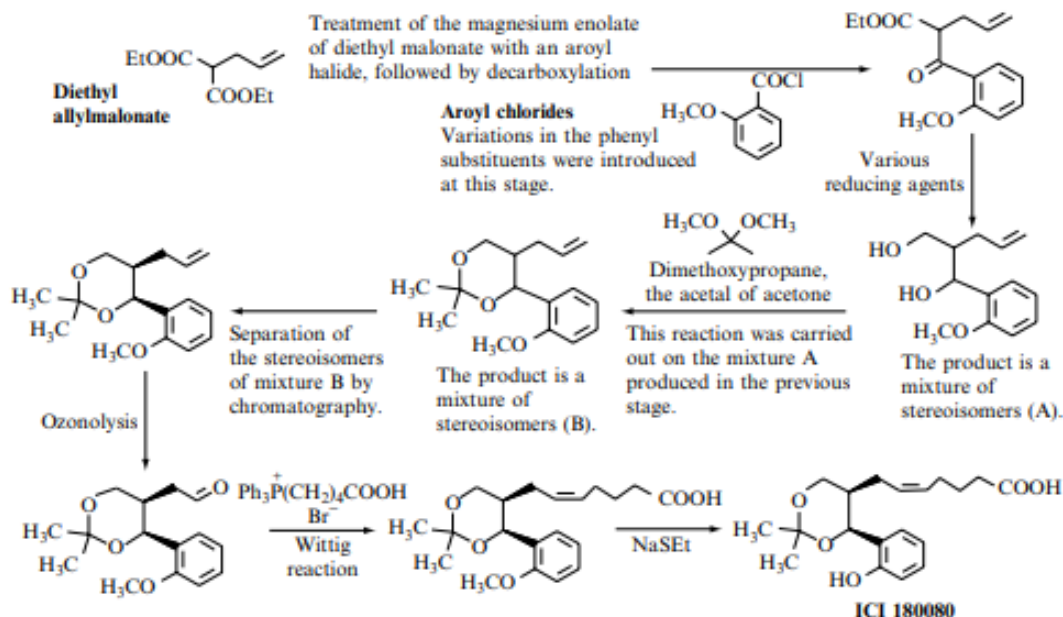


Figure SC-H&M-TRB - 30: An outline of the initial research route for the synthesis of ICI 180080 and other analogues. The intermediates and products were produced as racemates. Geometric stereoisomers were synthesized as shown in the reaction scheme.

The proposed manufacturing strategy for the active thromboxane antagonists made utilizing the small-scale process shown in Figure SC- H&M-TRB-28 must include a means for controlling the product's stereochemistry while still delivering a single enantiomer. Alternatives to chromatographic separation, which is not feasible on a large scale, ozonolysis, which is costly and requires specialized equipment, and sodium ethanethiolate, which would necessitate specific containment due to its volatile nature and terrible odor, must also be considered.

The initial production scale strategy (Figure SC- H&M-TRB-29), which used a Perkin reaction to make an adequately substituted 1,4-butrolactone, protected any phenolic hydroxyl groups by producing their methyl ethers.

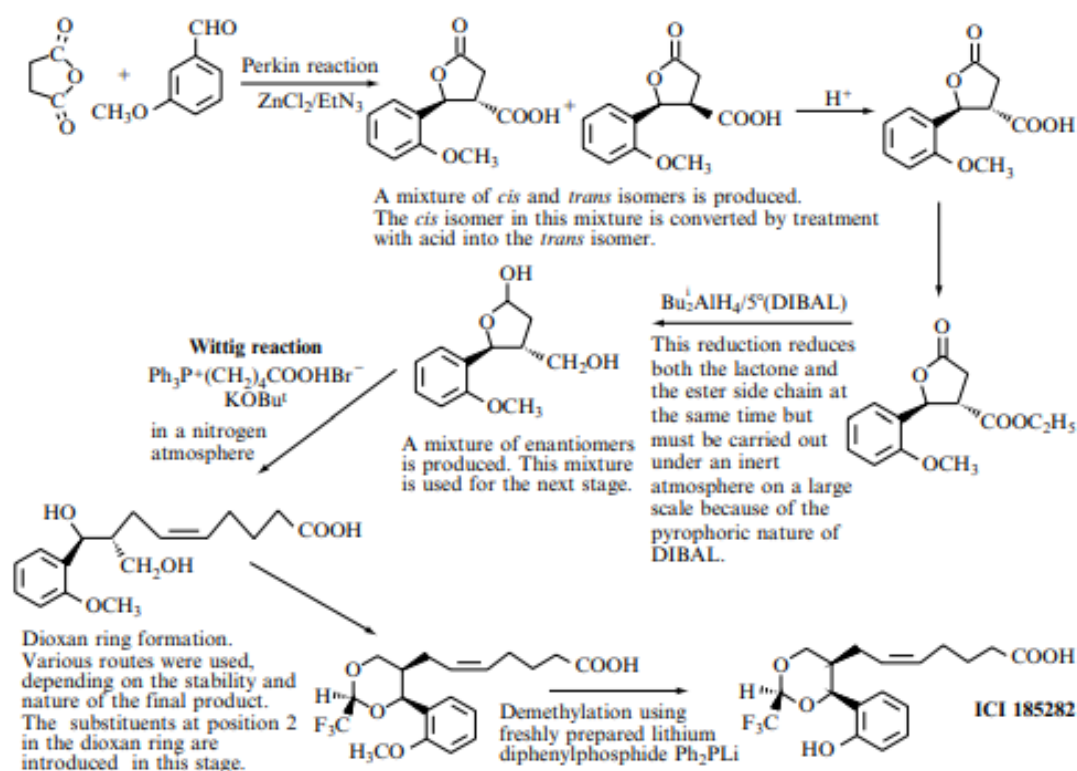


Figure SC-H&M-TRB - 31: An outline of the manufacturing route developed for the production of ICI 185282 and its analogues. Both succinic anhydride and the appropriate aromatic aldehydes are readily available.

To make a final product with the proper *cis* orientation in the dioxan ring, the researchers observed that the lactone ring's 3,4-substituents should be *trans* to each other. After reducing the lactone to lactol, the team was able to perform a Wittig reaction to add the *cis* alkenoic acid side chain to the molecule. Despite the fact that this reaction produces a mixture of isomers, it was quite easy to isolate a relatively pure *cis* product. The Wittig reaction also eliminates the requirement for ozonolysis and the accompanying equipment expenditures. The dioxan ring was created by cyclizing the Wittig reaction product, which may be done in a variety of methods depending on the final product's nature and stability. Finally, if possible, lithium diphenylphosphide rather than sodium ethanethiolate was used to remove the methyl protecting groups of any phenolic hydroxy groups. Demethylation was achieved simply by combining methyl ether with freshly created lithium diphenylphosphide, making this reaction suited for large-scale manufacturing. On an industrial scale, goods comprising less than 1% stereoisomers were reported to be generated utilizing the approach detailed in **Figure SC- H&M-TRB-29**.

3.3 Pharmacological and Toxicological Testing

A novel pharmaceutical must undergo thorough pharmacological and toxicological testing before it can be sold. These research are carried out in two stages: preclinical and clinical trials. These studies look at the risks that come with using a new medicine. They may also provide crucial information regarding the drug's pharmacokinetic properties, which might be useful during the development process. Details on the trials are included in the MAA application, which is submitted to the appropriate government entity (see Regulation section of this Topic). The trials program will be approved or modified by this body. In order to create and market the drug, the producer

must adhere to all of the terms of the MAA, which has replaced the previous drug license. It's a scientific assessment of the new product's safety, efficacy, and quality.

Preclinical trials are essentially toxicity and other biological tests carried out by microbiologists on bacteria and pharmacologists on tissue samples, animals, and sometimes organ cultures to determine whether it is safe to test the drug on humans. The effects of the drug on diverse body systems, such as the respiratory, neurological, and cardiovascular systems, are studied in animal testing. They are performed both in vivo (in a living organism) and in vitro (in a laboratory) (in a controlled environment). The drug's pharmacokinetic properties and interactions with other pharmaceuticals and over-the-counter remedies are also revealed in these preliminary research. Any of these interactions that improve or decrease the drug's activity should be investigated further during clinical trials, and the results should be included in the product labeling and literature.

Preclinical studies should be used to evaluate whether or not the drug is safe to use in humans, as well as the harmful dose. The researchers will be able to calculate a dose level at which Phase I clinical trials can begin. Dose-ranging pharmacokinetic investigations and bioavailability via various administration methods will be among them. Transferring animal experiments to people, on the other hand, is difficult, and the results can only be considered acceptable if the dose–organ toxicity findings have a large safety margin.

The use of animals in drug testing has sparked debate in both the pharmaceutical industry and the general public. The majority of countries are committed to decreasing the number of animals used in this way to a bare minimum and are actively researching the use of chemical and other alternatives. Some of these strategies include avoiding duplicate trials by having different countries centralize their validation procedures, employing human cell in vitro tests rather than animal in vivo studies, and removing methodologies that are not applicable to people. However, it appears unlikely that all animal testing will be abolished. When a medication passes preclinical studies, clinical trials in humans begin. These trials must first be approved by the appropriate legal and ethical committees because they may raise legal and ethical difficulties. In most countries, this approval requires a certificate or license from the appropriate medicine regulatory organization (see Regulation section of this Topic).

In healthy human trials, half of the participants are routinely given an inert drug in a form that cannot be discriminated from the test substance in order to accurately assess the outcomes of a clinical experiment.

To accurately examine the results of a clinical trial, they must be compared to the natural situation; as a result, in healthy human trials, half of the participants are usually given an inert drug in a form that cannot be distinguished from the test substance. The term "placebo" refers to a dosage form that has no active substances. In addition, the results of a trial must be reliable and unaffected by the experiment's conductor or the medication recipient. As a result, conducting a double-blind research, in which both the drug's administrator and the recipient are unaware whether they're dealing with the genuine thing or a placebo, is now normal practice. In addition, patients are randomly randomized to either the placebo or the medicine.

In clinical trials, healthy persons found no evidence of the new drug's favorable effects. To determine its efficacy, double-blind trials on healthy people are required. Using a placebo with

sick patients, on the other hand, presents moral and ethical considerations. Placebos may still be utilized if the discontinuation of treatment has no long-term consequences for the patients. If this isn't possible, the effect of the new drug is compared to a previously given treatment for the same medical condition. This is a reference drug that should be selected with caution. It should not be chosen in a way that exaggerates the new drug's potency, giving the manufacturer an unfair commercial advantage while giving the patient a false impression of the medicine's efficacy. Conducting cross-over trials is a third option. Halfway through the study, patients who are receiving the treatment are switched to the placebo or reference drug, and those who are receiving the placebo or reference drug are given the new drug. Because both groups have been exposed to the benefits of the new medicine, this is more ethical.

Clinical trials are usually conducted on small groups of healthy volunteers who are not minors or the elderly for the first time (Phase I trials). These individuals are subjected to a thorough medical examination prior to the testing, and they are closely monitored throughout the experiment. The research takes place in either in-house clinics or at specialized places outside of the university. The purpose of these healthy people trials is to see how the new treatment works in the body. They also discuss the new medicine's dosage form, absorption, distribution, bioavailability, excretion, and side effects. Due to the link between side effects and specific metabolites, medicinal chemists can create novel analogues that do not produce that metabolite, thus reducing the negative effects. The studies also include data on the drug's blood level following intravenous and oral administration, the rate of excretion in the urine and via the colon, and the effect of gender on these parameters. Throughout the trials, the kidneys and other organs in the body are monitored for adverse effects. Initially, the dose given to volunteers is a fraction of the dose given to animals using the same technique.

Once the drug's safety has been established in healthy volunteers, the testing phase progresses to Phase II trials. However, the company must first obtain a clinical trials certificate from the regulatory agency before conducting these trials in the United Kingdom. A small number of people with the ailment that the drug is meant to treat participate in Phase II studies. They assess the drug's effectiveness in treating the ailment and help determine the drug's dosing level and schedule. Following the success of this phase, the new therapy is tested in Phase III studies on a significant number of patients. In the United Kingdom, phase II studies are only permitted if the study protocol has been reviewed and approved by a local ethics council. In Phase III trials, both placebos and reference standards are used. They're particularly useful for gathering safety and efficacy data to satisfy product licensing authorities. In both Phase II and Phase III trials, a small number of patients will have adverse medication reactions (ADRs). These are unwanted or harmful reactions that occur at the therapeutic levels. They rule out the likelihood of failure of the therapy. These negative effects are documented in the drug's data sheet. However, until a large percentage of people experience the same ADR, the medicine is usually not pulled from usage.

When a new drug is introduced to the market, it is put to the test on large groups of patients in hospitals and general practices. Phase IV trials is the term used to describe this type of monitoring. It offers more details about the drug's safety and efficacy. In addition, smaller specialist groups undertake trials on specific features of medication administration, such as kinetics in the elderly, newborns, neonates, and ethnic groups.

The interpretation of all study outcomes necessitates close collaboration between clinicians and statisticians. Only if the minimum number of patients required for statistical viability is included in

the preliminary research can reliable results be produced. It's not always easy to get a precise measurement of the parameter being examined. As a result, the accuracy of the results is usually expressed as a probability coefficient, with the value of the coefficient representing the precision of the results. Clinical trials, on the other hand, can only provide highly accurate results when a significant number of patients are involved. This is quite uncommon. As a result, producers and regulatory bodies usually choose the statistical compromise that best meets their needs. Because certain adverse effects take years to manifest, it's critical to monitor the drug (Phase IV trials) after it's been approved for general use.

It is usually easier to interpret pharmacological and toxicological data when the ADMEs of the drug and its metabolites are well known (see section 8.4). Single dose studies are typically used to acquire data on tissue distribution, but repeated dose studies should be performed when:

1. the drug's or metabolite's tissue $t_{1/2}$ value is significantly higher than its plasma $t_{1/2}$ value,
2. the drug's or metabolite's C_{ss} is discovered to be substantially greater than that predicted from single dose studies,
3. the drug is targeted to a specific place, and
4. some types of tissue reveal unexpected lesions.

3.4 Drug Metabolism and Pharmacokinetics

Medication metabolism and pharmacokinetic (DMPK) experiments are used to show how drug and metabolite concentrations change with dose and time since the dose was given. Phase I studies are frequently conducted on suitable animal species as well as in people. The findings of animal studies are utilized to establish safe dose levels for human Phase I clinical trials. Animal testing data, on the other hand, is limited in its accuracy because it is extrapolated. Furthermore, in order to fully understand toxicological and pharmacological effects, the dose that just saturates the absorption and elimination systems must be identified.

3.4.1 Formulation Development

A medicine's dosage form refers to how it is given to patients. The physical characteristics of dose forms can be classified as liquid, semisolid, or solid. Liquid formulations include solutions, syrups, suspensions, and emulsions. Creams, ointments, and gels are semisolid formulations, whereas pills, capsules, suppositories, pessaries, and transdermal patches are solid formulations. However, all of these dosage forms include the drug as well as excipients. Excipients include fillers (bulk providing agents), lubricants, binders, preservatives, and antioxidants. The makeup of the excipient can have a significant impact on the drug's release from the dosage form. As a result, before marketing a new dosage form, producers must conduct bioavailability and any other testing necessary by the licensing agency.

The nature of the target and the stage of pharmaceutical development will define the dose form required. Preclinical and early Phase I clinical studies commonly use a simple dosage form, such as an oral solution, because many promising therapeutic candidates fail in the preclinical and Phase I stages. This is done to keep costs as low as possible throughout these high-risk stages of drug development. On the other hand, the manufacturer must use the dosage form of the drug that he wants to use in future clinical trials.

Certain properties, such as stability and pharmaceutical release pattern, must be met by the dosage forms utilized. Stability tests are performed to see if the potency of a dosage form is maintained over time, usually 2–3 years. Its shelf life and ideal storage conditions will be affected as a result of this. Excipients and any slow-release mechanisms used have a direct impact on drug release. In both of these cases, appropriate chemical and biological studies must be performed to gather or validate the relevant data. The findings of these research may lead to adjustments in the dosage form's design. They're usually done in conjunction with clinical trials.

3.5 Production and Quality Control

The marketing authorization (MA) establishes the conditions under which a new medicine must be produced (see Regulation section of this Topic). Due to the rarity of manufacturers dedicating a facility to the production of a single pharmaceutical, it is necessary that the equipment be carefully cleaned and examined for adulterants prior to use. According to many pharmaceutical companies, production line equipment is only used for about 10% of the time to create the product. For the most of the remaining time, it will be torn down, cleaned, and reassembled.

Quality control is required for the safe use of pharmaceuticals and medicines both during and after manufacture. It was only practicable after the mid-nineteenth century, when precise analytical processes were developed. As a result, national pharmacopoeias and other documents were created, outlining the extent and nature of the identification tests and quantitative evaluations required to ensure that the product reaching the general public is fit for its intended use. Pharmaceutical manufacturing, storage, and usage are now covered by these documents. They are constantly scrutinized, yet this does not rule out the possibility of product-related difficulties. The regular update of these documents, on the other hand, minimizes the chances of similar problems appearing in other products. Since the thalidomide disaster, it's comforting to note how few drugs have been removed off the market owing to safety concerns. The development of reliable analytical methods for trials, production quality control, and identification, limit, and assay processes for inclusion in relevant pharmacopeias normally follows the critical path development stages. A full description of these analytical procedures must be included in the product license application.

3.6 Patent Protection

Because of the high cost of drug development and production, it is vital for a company to maximize the value of a new drug. This can only be accomplished by stopping competitors from imitating a new product indefinitely. Patents are used to prevent competitors from producing and selling a product without the creator's permission. On the other hand, many companies advertise the products of other manufacturers under license from the patentee.

Since the early 14th century, patents have been used as a weapon for industrial protection in some form or another. They were created to encourage the creation of new industries and products by granting a developer or manufacturer a monopoly on the use of specific industrial equipment or the creation of specific commodities for a set period of time. This government-enforced monopoly allowed an innovator to be adequately compensated for his labor. In most countries, the issue of a patent prevents third parties from manufacturing and selling the product

without the creator's permission. By disseminating new knowledge, patents, on the other hand, encourage and protect the creation of new ideas.

At first, each country had its own patent laws, but by the mid-eighteenth century, it was clear that patent rights should transcend national borders. The Paris Convention of 1883 was the first international agreement. In 1978, the European Union signed the European Patent Convention (EPC), and in 1970, the Patent Cooperation Treaty (PCT) was signed in Washington. The former, which is solely available to European states, is overseen by the European Patents Office (EPO). The latter is open to all countries in the world and is controlled by the treaty's signatories' national patent offices.

During the research, development, and manufacturing stages of pharmaceutical creation, a patent's protection is crucial. When new compounds are produced and proven to have valuable properties, it is vital to file a patent as quickly as possible. Otherwise, a competitor in the same sector could preempt the patent, rendering a large amount of costly research worthless in terms of corporate profits. In this case, it's vital that the patent covers the relevant field and that no loopholes exist for a competitor to exploit.

From discovery to production, a medicine could take anywhere from 7 to 15 years to produce. Patents normally last 20 years from the date of application in various nations. As a result, a company's time to recoup development expenditures and earn a profit is extremely short, which explains why certain new drugs are so expensive. Furthermore, certain compounds are never developed because the patent-protected production time available to repay the cost of development is too short.

3.7 Regulation

The country's regulatory authorities must approve the introduction of a new medicine onto the market. The Medicines Control Agency (MCA) in the United Kingdom, the European Medicines Evaluation Agency (EMA) in the European Union, and the United States' Food and Drug Administration are all examples of this (FDA). These authorities, which are essentially consumer protection agencies, give a so-called product licence or marketing authorisation when they are satisfied with the production technique, efficacy, safety, and quality of the product (MA). A pharmaceutical company must submit a full dossier covering all essential pharmacological, formulation, and toxicological data, as well as detailed information on all aspects of the manufacturing process and dosage forms, in order to get a product license. A product license allows a pharmaceutical company or the individual who orders the product to produce and sell the new product in the awarding country, subject to specified conditions. Licenses can be revoked if the producer does not follow the agreement's terms and conditions. Any additional changes to the production process, dosage forms, or indications (use) must be approved, and the corporation is responsible for any additional testing required.

SAMPLE TOPIC ACTIVITIES

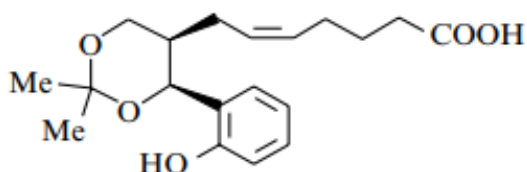
Activity 1: Group Presentation

Instruction to Teachers

1. Group students
2. Task Groups to research and select medicinal drugs of interest. Their research must cover the development phase of the drug and its performance in the pharmaceutical industry/market.
3. Groups to prepare PowerPoint Presentation to show-case to the class their findings

Activity 2: Homework/Assignment Exercise

- (1) (a) What is the critical path in drug development? (b) List the main stages in the critical path of the development of a drug.
- (2) Explain the meaning of the terms (a) dosage form, (b) ELF, (c) telescoping in drug production, (d) excipient and (e) double blind trial.
- (3) Outline the chemical factors that need to be considered when scaling up a research synthesis to pilot plant scale.
- (4) Outline a possible production scale route for the preparation of the thromboxane analogue A.



- (5) Explain the differences between (a) preclinical and clinical trials and (b) Phase I and Phase II trials.
- (6) What is a patent? Why is it necessary to patent drugs?

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